

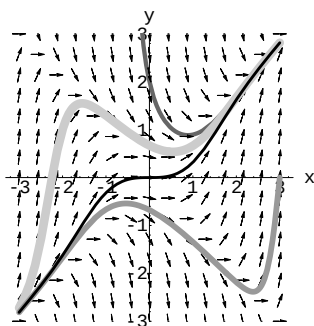
2

First-Order Differential Equations

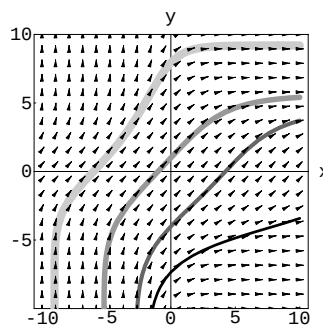
EXERCISES 2.1

Solution Curves Without the Solution

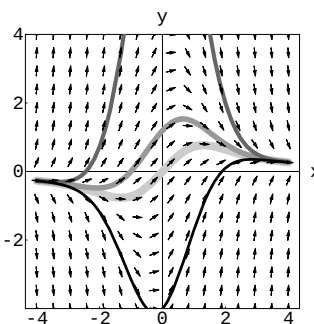
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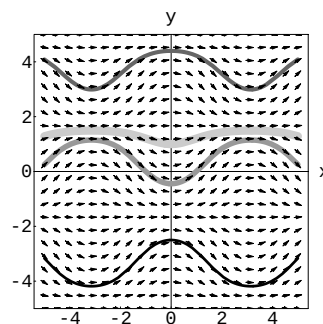
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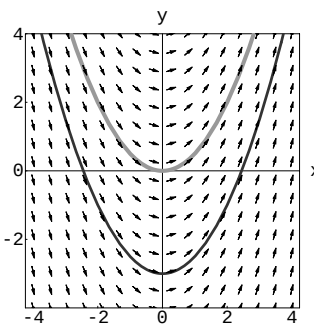
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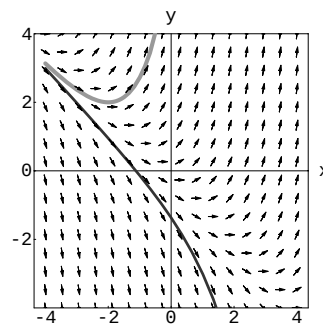
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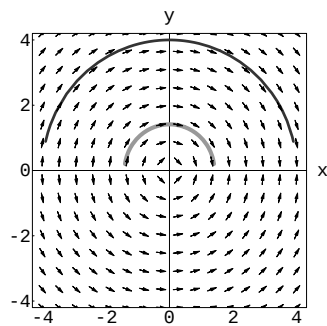


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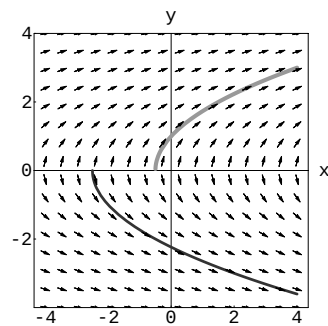


2.1 Solution Curves Without the Solution

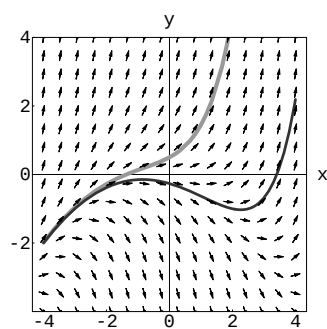
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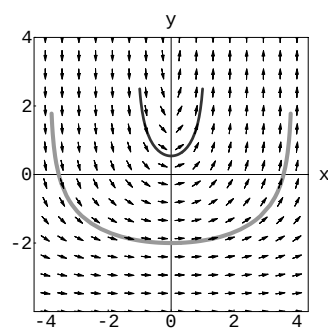
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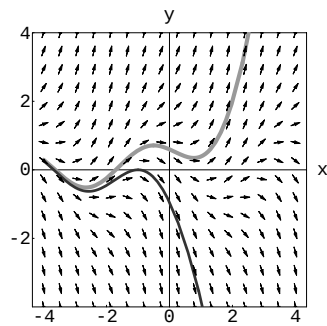
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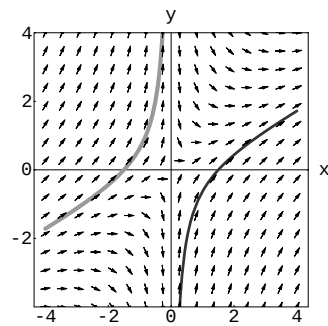
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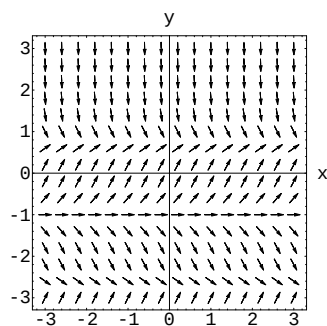
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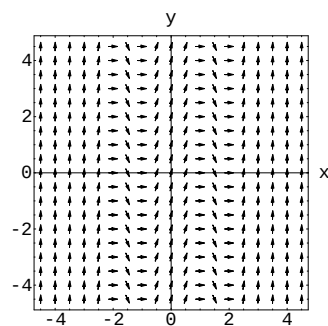
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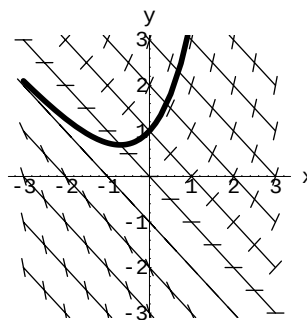


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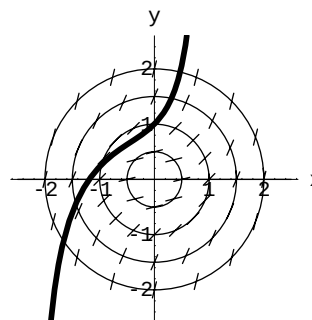


2.1 Solution Curves Without the Solution

15. (a) The isoclines have the form $y = -x + c$, which are straight lines with slope -1 .



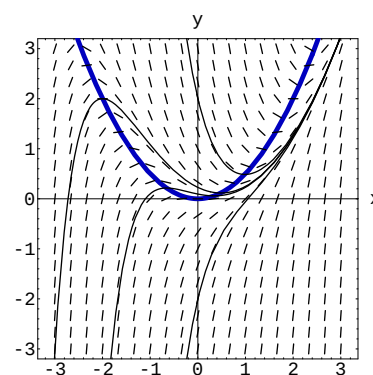
- (b) The isoclines have the form $x^2 + y^2 = c$, which are circles centered at the origin.



16. (a) When $x = 0$ or $y = 4$, $dy/dx = -2$ so the lineal elements have slope -2 . When $y = 3$ or $y = 5$, $dy/dx = x - 2$, so the lineal elements at $(x, 3)$ and $(x, 5)$ have slopes $x - 2$.

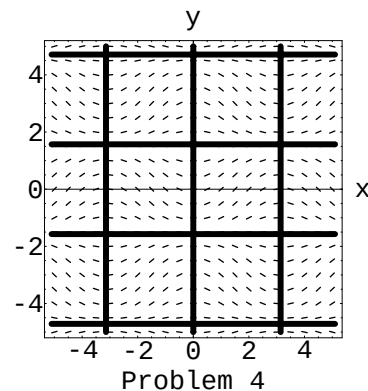
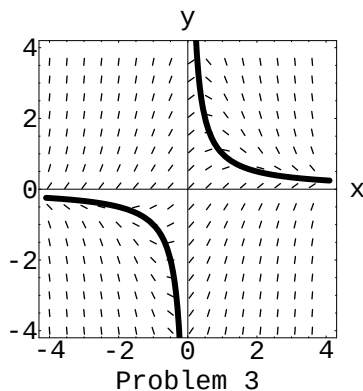
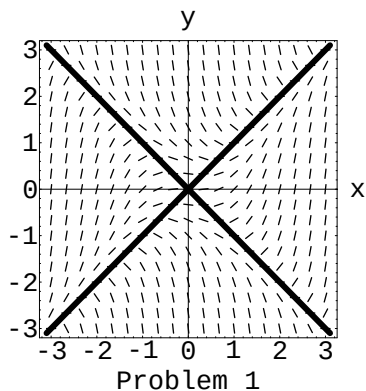
- (b) At $(0, y_0)$ the solution curve is headed down. If $y \rightarrow \infty$ as x increases, the graph must eventually turn around and head up, but while heading up it can never cross $y = 4$ where a tangent line to a solution curve must have slope -2 . Thus, y cannot approach ∞ as x approaches ∞ .

17. When $y < \frac{1}{2}x^2$, $y' = x^2 - 2y$ is positive and the portions of solution curves “outside” the nullcline parabola are increasing. When $y > \frac{1}{2}x^2$, $y' = x^2 - 2y$ is negative and the portions of the solution curves “inside” the nullcline parabola are decreasing.



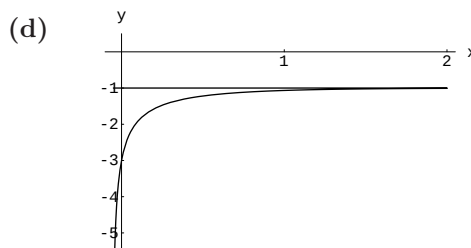
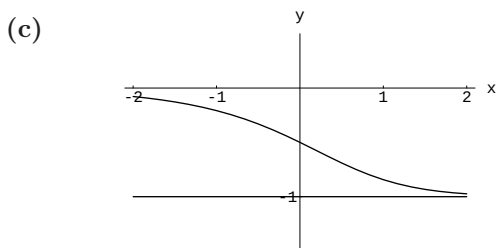
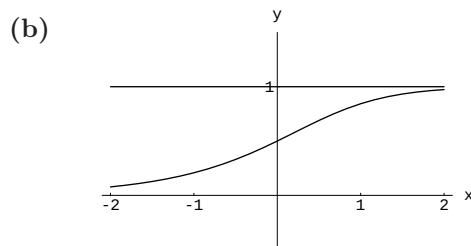
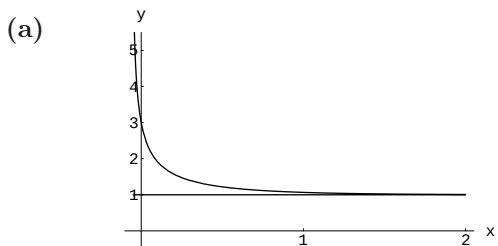
18. (a) Any horizontal lineal element should be at a point on a nullcline. In Problem 1 the nullclines are $x^2 - y^2 = 0$ or $y = \pm x$. In Problem 3 the nullclines are $1 - xy = 0$ or $y = 1/x$. In Problem 4 the nullclines are $(\sin x) \cos y = 0$ or $x = n\pi$ and $y = \pi/2 + n\pi$, where n is an integer. The graphs on the next page show the nullclines for the differential equations in Problems 1, 3, and 4 superimposed on the corresponding direction field.

2.1 Solution Curves Without the Solution

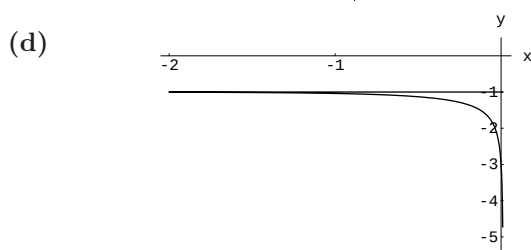
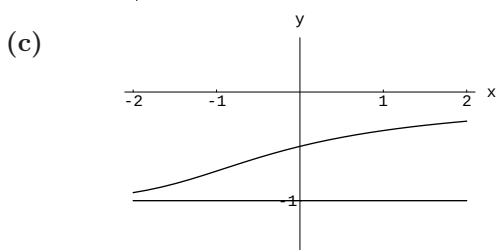
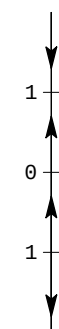
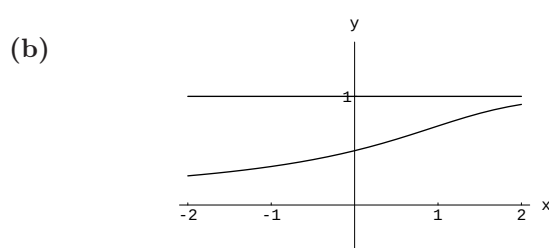
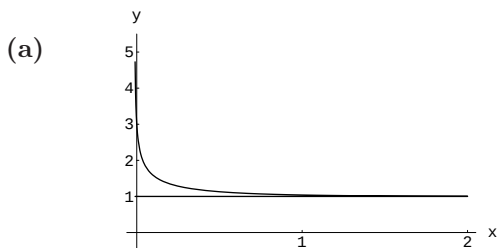


(b) An autonomous first-order differential equation has the form $y' = f(y)$. Nullclines have the form $y = c$ where $f(c) = 0$. These are the graphs of the equilibrium solutions of the differential equation.

19. Writing the differential equation in the form $dy/dx = y(1-y)(1+y)$ we see that critical points are located at $y = -1$, $y = 0$, and $y = 1$. The phase portrait is shown at the right.



20. Writing the differential equation in the form $dy/dx = y^2(1-y)(1+y)$ we see that critical points are located at $y = -1$, $y = 0$, and $y = 1$. The phase portrait is shown at the right.



2.1 Solution Curves Without the Solution

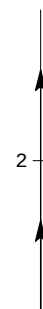
21. Solving $y^2 - 3y = y(y - 3) = 0$ we obtain the critical points 0 and 3. From the phase portrait we see that 0 is asymptotically stable (attractor) and 3 is unstable (repeller).



22. Solving $y^2 - y^3 = y^2(1 - y) = 0$ we obtain the critical points 0 and 1. From the phase portrait we see that 1 is asymptotically stable (attractor) and 0 is semi-stable.



23. Solving $(y - 2)^4 = 0$ we obtain the critical point 2. From the phase portrait we see that 2 is semi-stable.



24. Solving $10 + 3y - y^2 = (5 - y)(2 + y) = 0$ we obtain the critical points -2 and 5 . From the phase portrait we see that 5 is asymptotically stable (attractor) and -2 is unstable (repeller).

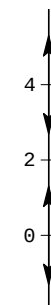


2.1 Solution Curves Without the Solution

25. Solving $y^2(4-y^2) = y^2(2-y)(2+y) = 0$ we obtain the critical points -2 , 0 , and 2 . From the phase portrait we see that 2 is asymptotically stable (attractor), 0 is semi-stable, and -2 is unstable (repeller).



26. Solving $y(2-y)(4-y) = 0$ we obtain the critical points 0 , 2 , and 4 . From the phase portrait we see that 2 is asymptotically stable (attractor) and 0 and 4 are unstable (repellers).



27. Solving $y \ln(y+2) = 0$ we obtain the critical points -1 and 0 . From the phase portrait we see that -1 is asymptotically stable (attractor) and 0 is unstable (repeller).

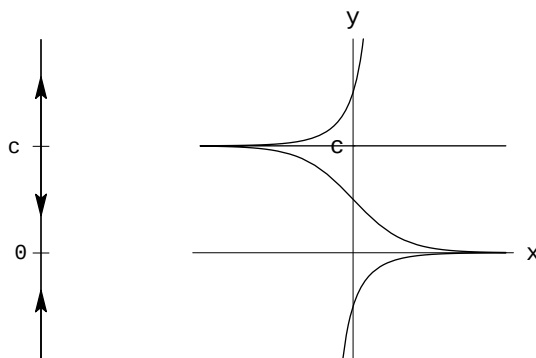


28. Solving $ye^y - 9y = y(e^y - 9) = 0$ we obtain the critical points 0 and $\ln 9$. From the phase portrait we see that 0 is asymptotically stable (attractor) and $\ln 9$ is unstable (repeller).

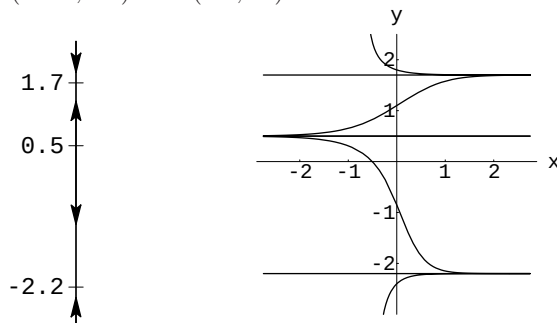


29. The critical points are 0 and c because the graph of $f(y)$ is 0 at these points. Since $f(y) > 0$ for $y < 0$ and $y > c$, the graph of the solution is increasing on $(-\infty, 0)$ and (c, ∞) . Since $f(y) < 0$ for $0 < y < c$, the graph of the solution is decreasing on $(0, c)$.

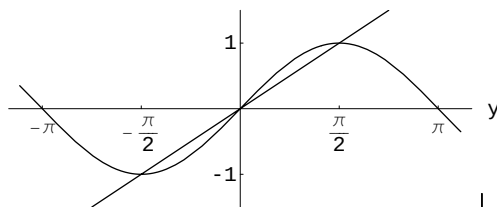
2.1 Solution Curves Without the Solution



30. The critical points are approximately at $-2, 2, 0.5$, and 1.7 . Since $f(y) > 0$ for $y < -2.2$ and $0.5 < y < 1.7$, the graph of the solution is increasing on $(-\infty, -2.2)$ and $(0.5, 1.7)$. Since $f(y) < 0$ for $-2.2 < y < 0.5$ and $y > 1.7$, the graph is decreasing on $(-2.2, 0.5)$ and $(1.7, \infty)$.



31. From the graphs of $z = \pi/2$ and $z = \sin y$ we see that $(\pi/2)y - \sin y = 0$ has only three solutions. By inspection we see that the critical points are $-\pi/2, 0$, and $\pi/2$.



From the graph at the right we see that

$$\frac{2}{\pi}y - \sin y \begin{cases} < 0 & \text{for } y < -\pi/2 \\ > 0 & \text{for } y > \pi/2 \end{cases}$$

$$\frac{2}{\pi}y - \sin y \begin{cases} > 0 & \text{for } -\pi/2 < y < 0 \\ < 0 & \text{for } 0 < y < \pi/2. \end{cases}$$



This enables us to construct the phase portrait shown at the right. From this portrait we see that $\pi/2$ and $-\pi/2$ are unstable (repellers), and 0 is asymptotically stable (attractor).

32. For $dy/dx = 0$ every real number is a critical point, and hence all critical points are nonisolated.
33. Recall that for $dy/dx = f(y)$ we are assuming that f and f' are continuous functions of y on some interval I . Now suppose that the graph of a nonconstant solution of the differential equation crosses the line $y = c$. If the point of intersection is taken as an initial condition we have two distinct solutions of the initial-value problem. This violates uniqueness, so the graph of any nonconstant solution must lie entirely on one side of any equilibrium solution. Since f is continuous it can only change signs at a point where it is 0 . But this is a critical point. Thus, $f(y)$ is completely positive or completely negative in each region R_i . If $y(x)$ is oscillatory

2.1 Solution Curves Without the Solution

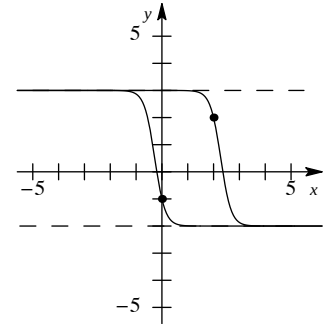
or has a relative extremum, then it must have a horizontal tangent line at some point (x_0, y_0) . In this case y_0 would be a critical point of the differential equation, but we saw above that the graph of a nonconstant solution cannot intersect the graph of the equilibrium solution $y = y_0$.

34. By Problem 33, a solution $y(x)$ of $dy/dx = f(y)$ cannot have relative extrema and hence must be monotone. Since $y'(x) = f(y) > 0$, $y(x)$ is monotone increasing, and since $y(x)$ is bounded above by c_2 , $\lim_{x \rightarrow \infty} y(x) = L$, where $L \leq c_2$. We want to show that $L = c_2$. Since L is a horizontal asymptote of $y(x)$, $\lim_{x \rightarrow \infty} y'(x) = 0$. Using the fact that $f(y)$ is continuous we have

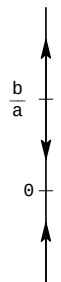
$$f(L) = f(\lim_{x \rightarrow \infty} y(x)) = \lim_{x \rightarrow \infty} f(y(x)) = \lim_{x \rightarrow \infty} y'(x) = 0.$$

But then L is a critical point of f . Since $c_1 < L \leq c_2$, and f has no critical points between c_1 and c_2 , $L = c_2$.

35. Assuming the existence of the second derivative, points of inflection of $y(x)$ occur where $y''(x) = 0$. From $dy/dx = f(y)$ we have $d^2y/dx^2 = f'(y) dy/dx$. Thus, the y -coordinate of a point of inflection can be located by solving $f'(y) = 0$. (Points where $dy/dx = 0$ correspond to constant solutions of the differential equation.)
36. Solving $y^2 - y - 6 = (y - 3)(y + 2) = 0$ we see that 3 and -2 are critical points. Now $d^2y/dx^2 = (2y - 1) dy/dx = (2y - 1)(y - 3)(y + 2)$, so the only possible point of inflection is at $y = \frac{1}{2}$, although the concavity of solutions can be different on either side of $y = -2$ and $y = 3$. Since $y''(x) < 0$ for $y < -2$ and $\frac{1}{2} < y < 3$, and $y''(x) > 0$ for $-2 < y < \frac{1}{2}$ and $y > 3$, we see that solution curves are concave down for $y < -2$ and $\frac{1}{2} < y < 3$ and concave up for $-2 < y < \frac{1}{2}$ and $y > 3$. Points of inflection of solutions of autonomous differential equations will have the same y -coordinates because between critical points they are horizontal translates of each other.



37. If (1) in the text has no critical points it has no constant solutions. The solutions have neither an upper nor lower bound. Since solutions are monotonic, every solution assumes all real values.
38. The critical points are 0 and b/a . From the phase portrait we see that 0 is an attractor and b/a is a repeller. Thus, if an initial population satisfies $P_0 > b/a$, the population becomes unbounded as t increases, most probably in finite time, i.e. $P(t) \rightarrow \infty$ as $t \rightarrow T$. If $0 < P_0 < b/a$, then the population eventually dies out, that is, $P(t) \rightarrow 0$ as $t \rightarrow \infty$. Since population $P > 0$ we do not consider the case $P_0 < 0$.

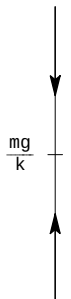


39. (a) Writing the differential equation in the form

$$\frac{dv}{dt} = \frac{k}{m} \left(\frac{mg}{k} - v \right)$$

we see that a critical point is mg/k .

From the phase portrait we see that mg/k is an asymptotically stable critical point. Thus, $\lim_{t \rightarrow \infty} v = mg/k$.



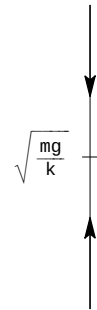
2.1 Solution Curves Without the Solution

- (b) Writing the differential equation in the form

$$\frac{dv}{dt} = \frac{k}{m} \left(\frac{mg}{k} - v^2 \right) = \frac{k}{m} \left(\sqrt{\frac{mg}{k}} - v \right) \left(\sqrt{\frac{mg}{k}} + v \right)$$

we see that the only physically meaningful critical point is $\sqrt{mg/k}$.

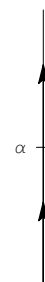
From the phase portrait we see that $\sqrt{mg/k}$ is an asymptotically stable critical point. Thus, $\lim_{t \rightarrow \infty} v = \sqrt{mg/k}$.



40. (a) From the phase portrait we see that critical points are α and β . Let $X(0) = X_0$. If $X_0 < \alpha$, we see that $X \rightarrow \alpha$ as $t \rightarrow \infty$. If $\alpha < X_0 < \beta$, we see that $X \rightarrow \alpha$ as $t \rightarrow \infty$. If $X_0 > \beta$, we see that $X(t)$ increases in an unbounded manner, but more specific behavior of $X(t)$ as $t \rightarrow \infty$ is not known.



- (b) When $\alpha = \beta$ the phase portrait is as shown. If $X_0 < \alpha$, then $X(t) \rightarrow \alpha$ as $t \rightarrow \infty$. If $X_0 > \alpha$, then $X(t)$ increases in an unbounded manner. This could happen in a finite amount of time. That is, the phase portrait does not indicate that X becomes unbounded as $t \rightarrow \infty$.



- (c) When $k = 1$ and $\alpha = \beta$ the differential equation is $dX/dt = (\alpha - X)^2$. For $X(t) = \alpha - 1/(t + c)$ we have $dX/dt = 1/(t + c)^2$ and

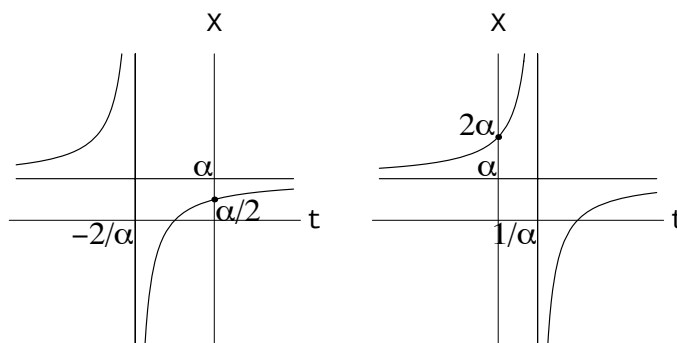
$$(\alpha - X)^2 = \left[\alpha - \left(\alpha - \frac{1}{t + c} \right) \right]^2 = \frac{1}{(t + c)^2} = \frac{dX}{dt}.$$

For $X(0) = \alpha/2$ we obtain

$$X(t) = \alpha - \frac{1}{t + 2/\alpha}.$$

For $X(0) = 2\alpha$ we obtain

$$X(t) = \alpha - \frac{1}{t - 1/\alpha}.$$



For $X_0 > \alpha$, $X(t)$ increases without bound up to $t = 1/\alpha$. For $t > 1/\alpha$, $X(t)$ increases but $X \rightarrow \alpha$ as $t \rightarrow \infty$.

EXERCISES 2.2

Separable Variables

In many of the following problems we will encounter an expression of the form $\ln |g(y)| = f(x) + c$. To solve for $g(y)$ we exponentiate both sides of the equation. This yields $|g(y)| = e^{f(x)+c} = e^c e^{f(x)}$ which implies $g(y) = \pm e^c e^{f(x)}$. Letting $c_1 = \pm e^c$ we obtain $g(y) = c_1 e^{f(x)}$.

1. From $dy = \sin 5x \, dx$ we obtain $y = -\frac{1}{5} \cos 5x + c$.
2. From $dy = (x+1)^2 \, dx$ we obtain $y = \frac{1}{3}(x+1)^3 + c$.
3. From $dy = -e^{-3x} \, dx$ we obtain $y = \frac{1}{3}e^{-3x} + c$.
4. From $\frac{1}{(y-1)^2} \, dy = dx$ we obtain $-\frac{1}{y-1} = x + c$ or $y = 1 - \frac{1}{x+c}$.
5. From $\frac{1}{y} \, dy = \frac{4}{x} \, dx$ we obtain $\ln |y| = 4 \ln |x| + c$ or $y = c_1 x^4$.
6. From $\frac{1}{y^2} \, dy = -2x \, dx$ we obtain $-\frac{1}{y} = -x^2 + c$ or $y = \frac{1}{x^2 + c_1}$.
7. From $e^{-2y} \, dy = e^{3x} \, dx$ we obtain $3e^{-2y} + 2e^{3x} = c$.
8. From $ye^y \, dy = (e^{-x} + e^{-3x}) \, dx$ we obtain $ye^y - e^y + e^{-x} + \frac{1}{3}e^{-3x} = c$.
9. From $\left(y + 2 + \frac{1}{y}\right) \, dy = x^2 \ln x \, dx$ we obtain $\frac{y^2}{2} + 2y + \ln |y| = \frac{x^3}{3} \ln |x| - \frac{1}{9}x^3 + c$.
10. From $\frac{1}{(2y+3)^2} \, dy = \frac{1}{(4x+5)^2} \, dx$ we obtain $\frac{2}{2y+3} = \frac{1}{4x+5} + c$.
11. From $\frac{1}{\csc y} \, dy = -\frac{1}{\sec^2 x} \, dx$ or $\sin y \, dy = -\cos^2 x \, dx = -\frac{1}{2}(1 + \cos 2x) \, dx$ we obtain $-\cos y = -\frac{1}{2}x - \frac{1}{4}\sin 2x + c$ or $4 \cos y = 2x + \sin 2x + c_1$.
12. From $2y \, dy = -\frac{\sin 3x}{\cos^3 3x} \, dx$ or $2y \, dy = -\tan 3x \sec^2 3x \, dx$ we obtain $y^2 = -\frac{1}{6}\sec^2 3x + c$.

2.2 Separable Variables

13. From $\frac{e^y}{(e^y + 1)^2} dy = \frac{-e^x}{(e^x + 1)^3} dx$ we obtain $-(e^y + 1)^{-1} = \frac{1}{2}(e^x + 1)^{-2} + c$.
14. From $\frac{y}{(1 + y^2)^{1/2}} dy = \frac{x}{(1 + x^2)^{1/2}} dx$ we obtain $(1 + y^2)^{1/2} = (1 + x^2)^{1/2} + c$.
15. From $\frac{1}{S} dS = k dr$ we obtain $S = ce^{kr}$.
16. From $\frac{1}{Q - 70} dQ = k dt$ we obtain $\ln |Q - 70| = kt + c$ or $Q - 70 = c_1 e^{kt}$.
17. From $\frac{1}{P - P^2} dP = \left(\frac{1}{P} + \frac{1}{1 - P} \right) dP = dt$ we obtain $\ln |P| - \ln |1 - P| = t + c$ so that $\ln \left| \frac{P}{1 - P} \right| = t + c$ or $\frac{P}{1 - P} = c_1 e^t$. Solving for P we have $P = \frac{c_1 e^t}{1 + c_1 e^t}$.
18. From $\frac{1}{N} dN = (te^{t+2} - 1) dt$ we obtain $\ln |N| = te^{t+2} - e^{t+2} - t + c$ or $N = c_1 e^{te^{t+2} - e^{t+2} - t}$.
19. From $\frac{y-2}{y+3} dy = \frac{x-1}{x+4} dx$ or $\left(1 - \frac{5}{y+3}\right) dy = \left(1 - \frac{5}{x+4}\right) dx$ we obtain $y - 5 \ln |y+3| = x - 5 \ln |x+4| + c$ or $\left(\frac{x+4}{y+3}\right)^5 = c_1 e^{x-y}$.
20. From $\frac{y+1}{y-1} dy = \frac{x+2}{x-3} dx$ or $\left(1 + \frac{2}{y-1}\right) dy = \left(1 + \frac{5}{x-3}\right) dx$ we obtain $y + 2 \ln |y-1| = x + 5 \ln |x-3| + c$ or $\frac{(y-1)^2}{(x-3)^5} = c_1 e^{x-y}$.
21. From $x dx = \frac{1}{\sqrt{1-y^2}} dy$ we obtain $\frac{1}{2}x^2 = \sin^{-1} y + c$ or $y = \sin \left(\frac{x^2}{2} + c_1 \right)$.
22. From $\frac{1}{y^2} dy = \frac{1}{e^x + e^{-x}} dx = \frac{e^x}{(e^x)^2 + 1} dx$ we obtain $-\frac{1}{y} = \tan^{-1} e^x + c$ or $y = -\frac{1}{\tan^{-1} e^x + c}$.
23. From $\frac{1}{x^2 + 1} dx = 4 dt$ we obtain $\tan^{-1} x = 4t + c$. Using $x(\pi/4) = 1$ we find $c = -3\pi/4$. The solution of the initial-value problem is $\tan^{-1} x = 4t - \frac{3\pi}{4}$ or $x = \tan \left(4t - \frac{3\pi}{4} \right)$.
24. From $\frac{1}{y^2 - 1} dy = \frac{1}{x^2 - 1} dx$ or $\frac{1}{2} \left(\frac{1}{y-1} - \frac{1}{y+1} \right) dy = \frac{1}{2} \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx$ we obtain $\ln |y-1| - \ln |y+1| = \ln |x-1| - \ln |x+1| + \ln c$ or $\frac{y-1}{y+1} = \frac{c(x-1)}{x+1}$. Using $y(2) = 2$ we find $c = 1$. A solution of the initial-value problem is $\frac{y-1}{y+1} = \frac{x-1}{x+1}$ or $y = x$.
25. From $\frac{1}{y} dy = \frac{1-x}{x^2} dx = \left(\frac{1}{x^2} - \frac{1}{x} \right) dx$ we obtain $\ln |y| = -\frac{1}{x} - \ln |x| + c$ or $xy = c_1 e^{-1/x}$. Using $y(-1) = -1$ we find $c_1 = e^{-1}$. The solution of the initial-value problem is $xy = e^{-1-1/x}$ or $y = e^{-(1+1/x)/x}$.
26. From $\frac{1}{1-2y} dy = dt$ we obtain $-\frac{1}{2} \ln |1-2y| = t + c$ or $1-2y = c_1 e^{-2t}$. Using $y(0) = 5/2$ we find $c_1 = -4$. The solution of the initial-value problem is $1-2y = -4e^{-2t}$ or $y = 2e^{-2t} + \frac{1}{2}$.
27. Separating variables and integrating we obtain

$$\frac{dx}{\sqrt{1-x^2}} - \frac{dy}{\sqrt{1-y^2}} = 0 \quad \text{and} \quad \sin^{-1} x - \sin^{-1} y = c.$$

Setting $x = 0$ and $y = \sqrt{3}/2$ we obtain $c = -\pi/3$. Thus, an implicit solution of the initial-value problem is $\sin^{-1} x - \sin^{-1} y = \pi/3$. Solving for y and using an addition formula from trigonometry, we get

$$y = \sin\left(\sin^{-1} x + \frac{\pi}{3}\right) = x \cos \frac{\pi}{3} + \sqrt{1-x^2} \sin \frac{\pi}{3} = \frac{x}{2} + \frac{\sqrt{3}\sqrt{1-x^2}}{2}.$$

28. From $\frac{1}{1+(2y)^2} dy = \frac{-x}{1+(x^2)^2} dx$ we obtain

$$\frac{1}{2} \tan^{-1} 2y = -\frac{1}{2} \tan^{-1} x^2 + c \quad \text{or} \quad \tan^{-1} 2y + \tan^{-1} x^2 = c_1.$$

Using $y(1) = 0$ we find $c_1 = \pi/4$. Thus, an implicit solution of the initial-value problem is $\tan^{-1} 2y + \tan^{-1} x^2 = \pi/4$. Solving for y and using a trigonometric identity we get

$$\begin{aligned} 2y &= \tan\left(\frac{\pi}{4} - \tan^{-1} x^2\right) \\ y &= \frac{1}{2} \tan\left(\frac{\pi}{4} - \tan^{-1} x^2\right) \\ &= \frac{1}{2} \frac{\tan \frac{\pi}{4} - \tan(\tan^{-1} x^2)}{1 + \tan \frac{\pi}{4} \tan(\tan^{-1} x^2)} \\ &= \frac{1}{2} \frac{1 - x^2}{1 + x^2}. \end{aligned}$$

29. (a) The equilibrium solutions $y(x) = 2$ and $y(x) = -2$ satisfy the initial conditions $y(0) = 2$ and $y(0) = -2$, respectively. Setting $x = \frac{1}{4}$ and $y = 1$ in $y = 2(1 + ce^{4x})/(1 - ce^{4x})$ we obtain

$$1 = 2 \frac{1 + ce}{1 - ce}, \quad 1 - ce = 2 + 2ce, \quad -1 = 3ce, \quad \text{and} \quad c = -\frac{1}{3e}.$$

The solution of the corresponding initial-value problem is

$$y = 2 \frac{1 - \frac{1}{3}e^{4x-1}}{1 + \frac{1}{3}e^{4x-1}} = 2 \frac{3 - e^{4x-1}}{3 + e^{4x-1}}.$$

- (b) Separating variables and integrating yields

$$\begin{aligned} \frac{1}{4} \ln |y - 2| - \frac{1}{4} \ln |y + 2| + \ln c_1 &= x \\ \ln |y - 2| - \ln |y + 2| + \ln c &= 4x \\ \ln \left| \frac{c(y - 2)}{y + 2} \right| &= 4x \\ c \frac{y - 2}{y + 2} &= e^{4x}. \end{aligned}$$

Solving for y we get $y = 2(c + e^{4x})/(c - e^{4x})$. The initial condition $y(0) = -2$ implies $2(c + 1)/(c - 1) = -2$ which yields $c = 0$ and $y(x) = -2$. The initial condition $y(0) = 2$ does not correspond to a value of c , and it must simply be recognized that $y(x) = 2$ is a solution of the initial-value problem. Setting $x = \frac{1}{4}$ and $y = 1$ in $y = 2(c + e^{4x})/(c - e^{4x})$ leads to $c = -3e$. Thus, a solution of the initial-value problem is

$$y = 2 \frac{-3e + e^{4x}}{-3e - e^{4x}} = 2 \frac{3 - e^{4x-1}}{3 + e^{4x-1}}.$$

30. Separating variables, we have

$$\frac{dy}{y^2 - y} = \frac{dx}{x} \quad \text{or} \quad \int \frac{dy}{y(y-1)} = \ln |x| + c.$$

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Using partial fractions, we obtain

$$\begin{aligned}\int \left(\frac{1}{y-1} - \frac{1}{y} \right) dy &= \ln |x| + c \\ \ln |y-1| - \ln |y| &= \ln |x| + c \\ \ln \left| \frac{y-1}{xy} \right| &= c \\ \frac{y-1}{xy} &= e^c = c_1.\end{aligned}$$

Solving for y we get $y = 1/(1 - c_1x)$. We note by inspection that $y = 0$ is a singular solution of the differential equation.

- (a) Setting $x = 0$ and $y = 1$ we have $1 = 1/(1 - 0)$, which is true for all values of c_1 . Thus, solutions passing through $(0, 1)$ are $y = 1/(1 - c_1x)$.
- (b) Setting $x = 0$ and $y = 0$ in $y = 1/(1 - c_1x)$ we get $0 = 1$. Thus, the only solution passing through $(0, 0)$ is $y = 0$.
- (c) Setting $x = \frac{1}{2}$ and $y = \frac{1}{2}$ we have $\frac{1}{2} = 1/(1 - \frac{1}{2}c_1)$, so $c_1 = -2$ and $y = 1/(1 + 2x)$.
- (d) Setting $x = 2$ and $y = \frac{1}{4}$ we have $\frac{1}{4} = 1/(1 - 2c_1)$, so $c_1 = -\frac{3}{2}$ and $y = 1/(1 + \frac{3}{2}x) = 2/(2 + 3x)$.
31. Singular solutions of $dy/dx = x\sqrt{1-y^2}$ are $y = -1$ and $y = 1$. A singular solution of $(e^x + e^{-x})dy/dx = y^2$ is $y = 0$.
32. Differentiating $\ln(x^2 + 10) + \csc y = c$ we get

$$\begin{aligned}\frac{2x}{x^2 + 10} - \csc y \cot y \frac{dy}{dx} &= 0, \\ \frac{2x}{x^2 + 10} - \frac{1}{\sin y} \cdot \frac{\cos y}{\sin y} \frac{dy}{dx} &= 0,\end{aligned}$$

or

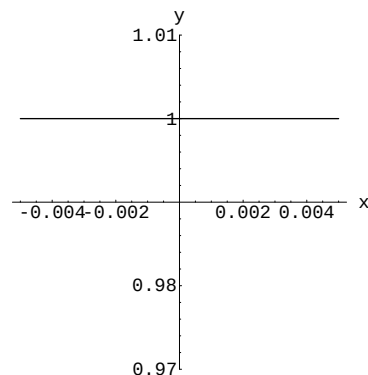
$$2x \sin^2 y \, dx - (x^2 + 10) \cos y \, dy = 0.$$

Writing the differential equation in the form

$$\frac{dy}{dx} = \frac{2x \sin^2 y}{(x^2 + 10) \cos y}$$

we see that singular solutions occur when $\sin^2 y = 0$, or $y = k\pi$, where k is an integer.

33. The singular solution $y = 1$ satisfies the initial-value problem.

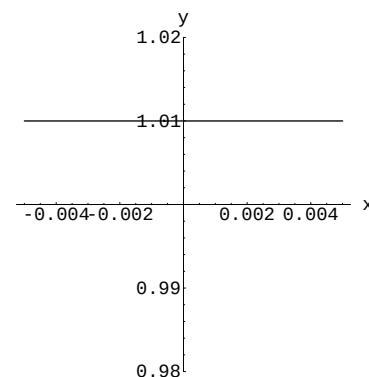


34. Separating variables we obtain $\frac{dy}{(y-1)^2} = dx$. Then

$$-\frac{1}{y-1} = x + c \quad \text{and} \quad y = \frac{x + c - 1}{x + c}.$$

Setting $x = 0$ and $y = 1.01$ we obtain $c = -100$. The solution is

$$y = \frac{x - 101}{x - 100}.$$

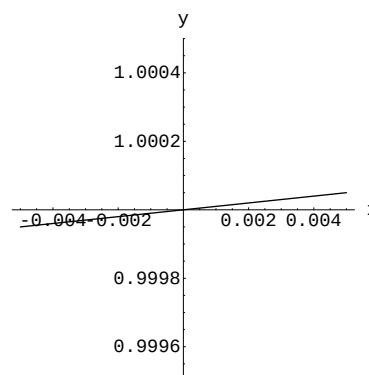


35. Separating variables we obtain $\frac{dy}{(y-1)^2 + 0.01} = dx$. Then

$$10 \tan^{-1} 10(y-1) = x + c \quad \text{and} \quad y = 1 + \frac{1}{10} \tan \frac{x+c}{10}.$$

Setting $x = 0$ and $y = 1$ we obtain $c = 0$. The solution is

$$y = 1 + \frac{1}{10} \tan \frac{x}{10}.$$



36. Separating variables we obtain $\frac{dy}{(y-1)^2 - 0.01} = dx$. Then, from (11) in this section of the manual with $u = y - 1$ and $a = \frac{1}{10}$, we get

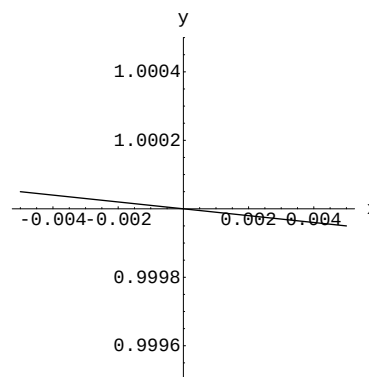
$$5 \ln \left| \frac{10y - 11}{10y - 9} \right| = x + c.$$

Setting $x = 0$ and $y = 1$ we obtain $c = 5 \ln 1 = 0$. The solution is

$$5 \ln \left| \frac{10y - 11}{10y - 9} \right| = x.$$

Solving for y we obtain

$$y = \frac{11 + 9e^{x/5}}{10 + 10e^{x/5}}.$$



Alternatively, we can use the fact that

$$\int \frac{dy}{(y-1)^2 - 0.01} = -\frac{1}{0.1} \tanh^{-1} \frac{y-1}{0.1} = -10 \tanh^{-1} 10(y-1).$$

(We use the inverse hyperbolic tangent because $|y-1| < 0.1$ or $0.9 < y < 1.1$. This follows from the initial condition $y(0) = 1$.) Solving the above equation for y we get $y = 1 + 0.1 \tanh(x/10)$.

37. Separating variables, we have

$$\frac{dy}{y-y^3} = \frac{dy}{y(1-y)(1+y)} = \left(\frac{1}{y} + \frac{1/2}{1-y} - \frac{1/2}{1+y} \right) dy = dx.$$

Integrating, we get

$$\ln |y| - \frac{1}{2} \ln |1-y| - \frac{1}{2} \ln |1+y| = x + c.$$

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When $y > 1$, this becomes

$$\ln y - \frac{1}{2} \ln(y-1) - \frac{1}{2} \ln(y+1) = \ln \frac{y}{\sqrt{y^2-1}} = x + c.$$

Letting $x = 0$ and $y = 2$ we find $c = \ln(2/\sqrt{3})$. Solving for y we get $y_1(x) = 2e^x/\sqrt{4e^{2x}-3}$, where $x > \ln(\sqrt{3}/2)$.

When $0 < y < 1$ we have

$$\ln y - \frac{1}{2} \ln(1-y) - \frac{1}{2} \ln(1+y) = \ln \frac{y}{\sqrt{1-y^2}} = x + c.$$

Letting $x = 0$ and $y = \frac{1}{2}$ we find $c = \ln(1/\sqrt{3})$. Solving for y we get $y_2(x) = e^x/\sqrt{e^{2x}+3}$, where $-\infty < x < \infty$.

When $-1 < y < 0$ we have

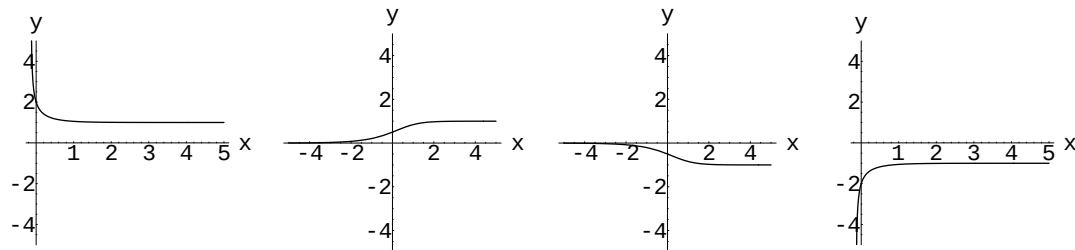
$$\ln(-y) - \frac{1}{2} \ln(1-y) - \frac{1}{2} \ln(1+y) = \ln \frac{-y}{\sqrt{1-y^2}} = x + c.$$

Letting $x = 0$ and $y = -\frac{1}{2}$ we find $c = \ln(1/\sqrt{3})$. Solving for y we get $y_3(x) = -e^x/\sqrt{e^{2x}+3}$, where $-\infty < x < \infty$.

When $y < -1$ we have

$$\ln(-y) - \frac{1}{2} \ln(1-y) - \frac{1}{2} \ln(-1-y) = \ln \frac{-y}{\sqrt{y^2-1}} = x + c.$$

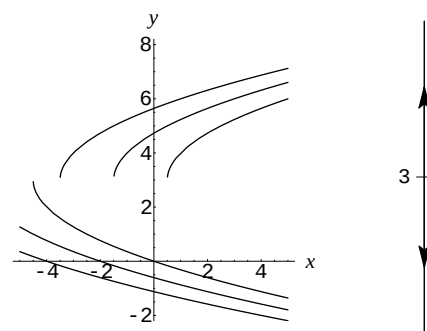
Letting $x = 0$ and $y = -2$ we find $c = \ln(2/\sqrt{3})$. Solving for y we get $y_4(x) = -2e^x/\sqrt{4e^{2x}-3}$, where $x > \ln(\sqrt{3}/2)$.



38. (a) The second derivative of y is

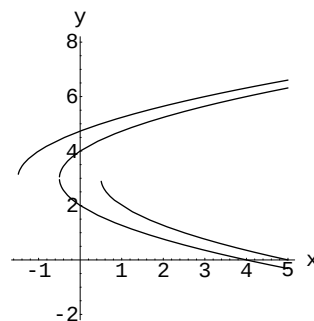
$$\frac{d^2y}{dx^2} = -\frac{dy/dx}{(y-1)^2} = -\frac{1/(y-3)}{(y-3)^2} = -\frac{1}{(y-3)^3}.$$

The solution curve is concave down when $d^2y/dx^2 < 0$ or $y > 3$, and concave up when $d^2y/dx^2 > 0$ or $y < 3$. From the phase portrait we see that the solution curve is decreasing when $y < 3$ and increasing when $y > 3$.



- (b) Separating variables and integrating we obtain

$$\begin{aligned} (y-3) dy &= dx \\ \frac{1}{2}y^2 - 3y &= x + c \\ y^2 - 6y + 9 &= 2x + c_1 \\ (y-3)^2 &= 2x + c_1 \\ y &= 3 \pm \sqrt{2x + c_1}. \end{aligned}$$



The initial condition dictates whether to use the plus or minus sign.

When $y_1(0) = 4$ we have $c_1 = 1$ and $y_1(x) = 3 + \sqrt{2x+1}$.

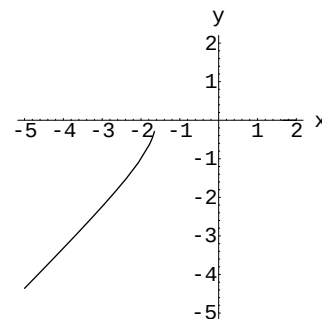
When $y_2(0) = 2$ we have $c_1 = 1$ and $y_2(x) = 3 - \sqrt{2x+1}$.

When $y_3(1) = 2$ we have $c_1 = -1$ and $y_3(x) = 3 - \sqrt{2x-1}$.

When $y_4(-1) = 4$ we have $c_1 = 3$ and $y_4(x) = 3 + \sqrt{2x+3}$.

39. (a) Separating variables we have $2y dy = (2x+1)dx$. Integrating gives $y^2 = x^2 + x + c$. When $y(-2) = -1$ we find $c = -1$, so $y^2 = x^2 + x - 1$ and $y = -\sqrt{x^2 + x - 1}$. The negative square root is chosen because of the initial condition.

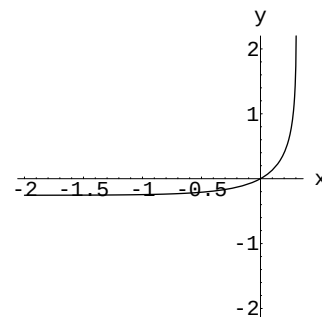
- (b) From the figure, the largest interval of definition appears to be approximately $(-\infty, -1.65)$.



- (c) Solving $x^2 + x - 1 = 0$ we get $x = -\frac{1}{2} \pm \frac{1}{2}\sqrt{5}$, so the largest interval of definition is $(-\infty, -\frac{1}{2} - \frac{1}{2}\sqrt{5})$. The right-hand endpoint of the interval is excluded because $y = -\sqrt{x^2 + x - 1}$ is not differentiable at this point.

40. (a) From Problem 7 the general solution is $3e^{-2y} + 2e^{3x} = c$. When $y(0) = 0$ we find $c = 5$, so $3e^{-2y} + 2e^{3x} = 5$. Solving for y we get $y = -\frac{1}{2} \ln \frac{1}{3}(5 - 2e^{3x})$.

- (b) The interval of definition appears to be approximately $(-\infty, 0.3)$.



- (c) Solving $\frac{1}{3}(5 - 2e^{3x}) = 0$ we get $x = \frac{1}{3} \ln(\frac{5}{2})$, so the exact interval of definition is $(-\infty, \frac{1}{3} \ln \frac{5}{2})$.

41. (a) While $y_2(x) = -\sqrt{25 - x^2}$ is defined at $x = -5$ and $x = 5$, $y'_2(x)$ is not defined at these values, and so the interval of definition is the open interval $(-5, 5)$.

- (b) At any point on the x -axis the derivative of $y(x)$ is undefined, so no solution curve can cross the x -axis. Since $-x/y$ is not defined when $y = 0$, the initial-value problem has no solution.

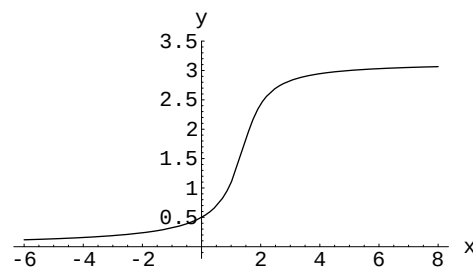
42. (a) Separating variables and integrating we obtain $x^2 - y^2 = c$. For $c \neq 0$ the graph is a hyperbola centered at the origin. All four initial conditions imply $c = 0$ and $y = \pm x$. Since the differential equation is not defined for $y = 0$, solutions are $y = \pm x$, $x < 0$ and $y = \pm x$, $x > 0$. The solution for $y(a) = a$ is $y = x$, $x > 0$; for $y(a) = -a$ is $y = -x$; for $y(-a) = a$ is $y = -x$, $x < 0$; and for $y(-a) = -a$ is $y = x$, $x < 0$.

- (b) Since x/y is not defined when $y = 0$, the initial-value problem has no solution.

- (c) Setting $x = 1$ and $y = 2$ in $x^2 - y^2 = c$ we get $c = -3$, so $y^2 = x^2 + 3$ and $y(x) = \sqrt{x^2 + 3}$, where the positive square root is chosen because of the initial condition. The domain is all real numbers since $x^2 + 3 > 0$ for all x .

2.2 Separable Variables

43. Separating variables we have $dy/(\sqrt{1+y^2} \sin^2 y) = dx$ which is not readily integrated (even by a CAS). We note that $dy/dx \geq 0$ for all values of x and y and that $dy/dx = 0$ when $y = 0$ and $y = \pi$, which are equilibrium solutions.



44. Separating variables we have $dy/(\sqrt{y} + y) = dx/(\sqrt{x} + x)$. To integrate $\int dx/(\sqrt{x} + x)$ we substitute $u^2 = x$ and get

$$\int \frac{2u}{u+u^2} du = \int \frac{2}{1+u} du = 2 \ln|1+u| + c = 2 \ln(1+\sqrt{x}) + c.$$

Integrating the separated differential equation we have

$$2 \ln(1+\sqrt{y}) = 2 \ln(1+\sqrt{x}) + c \quad \text{or} \quad \ln(1+\sqrt{y}) = \ln(1+\sqrt{x}) + \ln c_1.$$

Solving for y we get $y = [c_1(1+\sqrt{x}) - 1]^2$.

45. We are looking for a function $y(x)$ such that

$$y^2 + \left(\frac{dy}{dx}\right)^2 = 1.$$

Using the positive square root gives

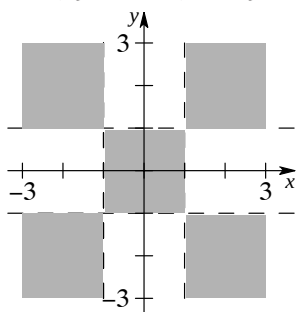
$$\frac{dy}{dx} = \sqrt{1-y^2} \implies \frac{dy}{\sqrt{1-y^2}} = dx \implies \sin^{-1} y = x + c.$$

Thus a solution is $y = \sin(x+c)$. If we use the negative square root we obtain

$$y = \sin(c-x) = -\sin(x-c) = -\sin(x+c_1).$$

Note that when $c = c_1 = 0$ and when $c = c_1 = \pi/2$ we obtain the well known particular solutions $y = \sin x$, $y = -\sin x$, $y = \cos x$, and $y = -\cos x$. Note also that $y = 1$ and $y = -1$ are singular solutions.

46. (a)



- (b) For $|x| > 1$ and $|y| > 1$ the differential equation is $dy/dx = \sqrt{y^2-1}/\sqrt{x^2-1}$. Separating variables and integrating, we obtain

$$\frac{dy}{\sqrt{y^2-1}} = \frac{dx}{\sqrt{x^2-1}} \quad \text{and} \quad \cosh^{-1} y = \cosh^{-1} x + c.$$

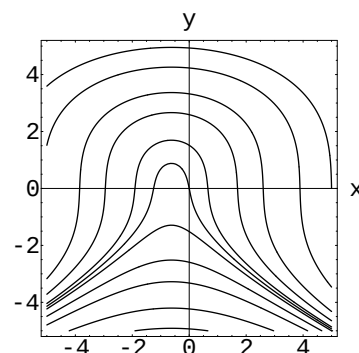
Setting $x = 2$ and $y = 2$ we find $c = \cosh^{-1} 2 - \cosh^{-1} 2 = 0$ and $\cosh^{-1} y = \cosh^{-1} x$. An explicit solution is $y = x$.

47. Since the tension T_1 (or magnitude T_1) acts at the lowest point of the cable, we use symmetry to solve the problem on the interval $[0, L/2]$. The assumption that the roadbed is uniform (that is, weighs a constant ρ

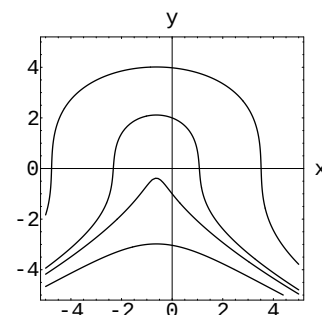
2.2 Separable Variables

pounds per horizontal foot) implies $W = \rho x$, where x is measured in feet and $0 \leq x \leq L/2$. Therefore (10) becomes $dy/dx = (\rho/T_1)x$. This last equation is a separable equation of the form given in (1) of Section 2.2 in the text. Integrating and using the initial condition $y(0) = a$ shows that the shape of the cable is a parabola: $y(x) = (\rho/2T_1)x^2 + a$. In terms of the sag h of the cable and the span L , we see from Figure 2.22 in the text that $y(L/2) = h + a$. By applying this last condition to $y(x) = (\rho/2T_1)x^2 + a$ enables us to express $\rho/2T_1$ in terms of h and L : $y(x) = (4h/L^2)x^2 + a$. Since $y(x)$ is an even function of x , the solution is valid on $-L/2 \leq x \leq L/2$.

48. (a) Separating variables and integrating, we have $(3y^2 + 1)dy = -(8x + 5)dx$ and $y^3 + y = -4x^2 - 5x + c$. Using a CAS we show various contours of $f(x, y) = y^3 + y + 4x^2 + 5x$. The plots shown on $[-5, 5] \times [-5, 5]$ correspond to c -values of 0, ± 5 , ± 20 , ± 40 , ± 80 , and ± 125 .



- (b) The value of c corresponding to $y(0) = -1$ is $f(0, -1) = -2$; to $y(0) = 2$ is $f(0, 2) = 10$; to $y(-1) = 4$ is $f(-1, 4) = 67$; and to $y(-1) = -3$ is -31 .



49. (a) An implicit solution of the differential equation $(2y + 2)dy - (4x^3 + 6x)dx = 0$ is

$$y^2 + 2y - x^4 - 3x^2 + c = 0.$$

The condition $y(0) = -3$ implies that $c = -3$. Therefore $y^2 + 2y - x^4 - 3x^2 - 3 = 0$.

- (b) Using the quadratic formula we can solve for y in terms of x :

$$y = \frac{-2 \pm \sqrt{4 + 4(x^4 + 3x^2 + 3)}}{2}.$$

The explicit solution that satisfies the initial condition is then

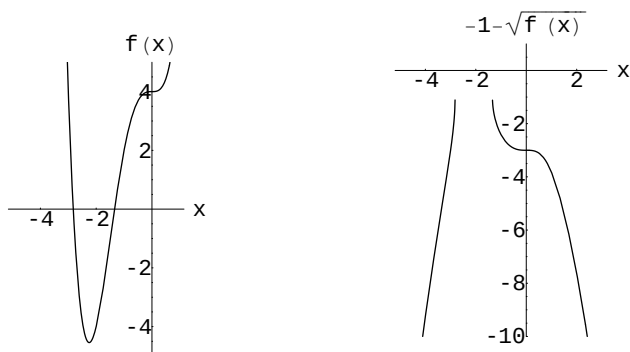
$$y = -1 - \sqrt{x^4 + 3x^2 + 4}.$$

- (c) From the graph of the function $f(x) = x^4 + 3x^2 + 4$ below we see that $f(x) \leq 0$ on the approximate interval $-2.8 \leq x \leq -1.3$. Thus the approximate domain of the function

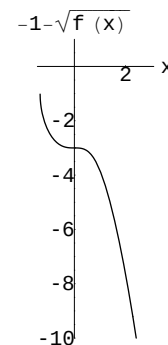
$$y = -1 - \sqrt{x^4 + 3x^2 + 4} = -1 - \sqrt{f(x)}$$

is $x \leq -2.8$ or $x \geq -1.3$. The graph of this function is shown below.

2.2 Separable Variables



- (d) Using the root finding capabilities of a CAS, the zeros of f are found to be -2.82202 and -1.3409 . The domain of definition of the solution $y(x)$ is then $x > -1.3409$. The equality has been removed since the derivative dy/dx does not exist at the points where $f(x) = 0$. The graph of the solution $y = \phi(x)$ is given on the right.



50. (a) Separating variables and integrating, we have

$$(-2y + y^2)dy = (x - x^2)dx$$

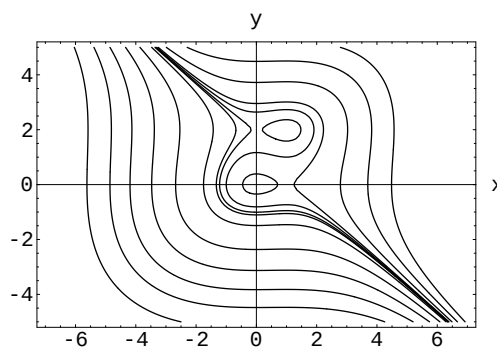
and

$$-y^2 + \frac{1}{3}y^3 = \frac{1}{2}x^2 - \frac{1}{3}x^3 + c.$$

Using a CAS we show some contours of

$$f(x, y) = 2y^3 - 6y^2 + 2x^3 - 3x^2.$$

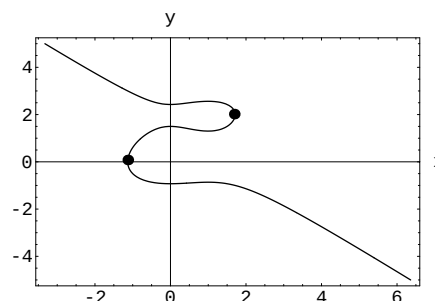
The plots shown on $[-7, 7] \times [-5, 5]$ correspond to c -values of $-450, -300, -200, -120, -60, -20, -10, -8.1, -5, -0.8, 20, 60$, and 120 .



- (b) The value of c corresponding to $y(0) = \frac{3}{2}$ is $f(0, \frac{3}{2}) = -\frac{27}{4}$. The portion of the graph between the dots corresponds to the solution curve satisfying the initial condition. To determine the interval of definition we find dy/dx for

$$2y^3 - 6y^2 + 2x^3 - 3x^2 = -\frac{27}{4}.$$

Using implicit differentiation we get $y' = (x - x^2)/(y^2 - 2y)$, which is infinite when $y = 0$ and $y = 2$. Letting $y = 0$ in $2y^3 - 6y^2 + 2x^3 - 3x^2 = -\frac{27}{4}$ and using a CAS to solve for x we get $x = -1.13232$. Similarly, letting $y = 2$, we find $x = 1.71299$. The largest interval of definition is approximately $(-1.13232, 1.71299)$.

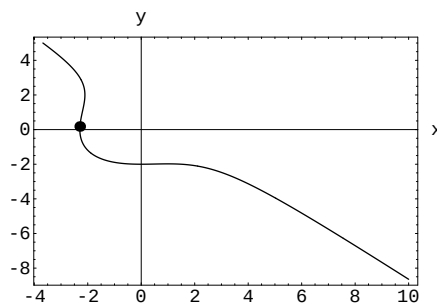


- (c) The value of c corresponding to $y(0) = -2$ is $f(0, -2) = -40$.

The portion of the graph to the right of the dot corresponds to the solution curve satisfying the initial condition. To determine the interval of definition we find dy/dx for

$$2y^3 - 6y^2 + 2x^3 - 3x^2 = -40.$$

Using implicit differentiation we get $y' = (x - x^2)/(y^2 - 2y)$, which is infinite when $y = 0$ and $y = 2$. Letting $y = 0$ in $2y^3 - 6y^2 + 2x^3 - 3x^2 = -40$ and using a CAS to solve for x we get $x = -2.29551$. The largest interval of definition is approximately $(-2.29551, \infty)$.



EXERCISES 2.3

Linear Equations

- For $y' - 5y = 0$ an integrating factor is $e^{-\int 5 dx} = e^{-5x}$ so that $\frac{d}{dx} [e^{-5x}y] = 0$ and $y = ce^{5x}$ for $-\infty < x < \infty$.
- For $y' + 2y = 0$ an integrating factor is $e^{\int 2 dx} = e^{2x}$ so that $\frac{d}{dx} [e^{2x}y] = 0$ and $y = ce^{-2x}$ for $-\infty < x < \infty$.
The transient term is ce^{-2x} .
- For $y' + y = e^{3x}$ an integrating factor is $e^{\int 1 dx} = e^x$ so that $\frac{d}{dx} [e^x y] = e^{4x}$ and $y = \frac{1}{4}e^{3x} + ce^{-x}$ for $-\infty < x < \infty$.
The transient term is ce^{-x} .
- For $y' + 4y = \frac{4}{3}$ an integrating factor is $e^{\int 4 dx} = e^{4x}$ so that $\frac{d}{dx} [e^{4x}y] = \frac{4}{3}e^{4x}$ and $y = \frac{1}{3} + ce^{-4x}$ for $-\infty < x < \infty$.
The transient term is ce^{-4x} .
- For $y' + 3x^2y = x^2$ an integrating factor is $e^{\int 3x^2 dx} = e^{x^3}$ so that $\frac{d}{dx} [e^{x^3}y] = x^2e^{x^3}$ and $y = \frac{1}{3} + ce^{-x^3}$ for $-\infty < x < \infty$.
The transient term is ce^{-x^3} .
- For $y' + 2xy = x^3$ an integrating factor is $e^{\int 2x dx} = e^{x^2}$ so that $\frac{d}{dx} [e^{x^2}y] = x^3e^{x^2}$ and $y = \frac{1}{2}x^2 - \frac{1}{2} + ce^{-x^2}$ for $-\infty < x < \infty$.
The transient term is ce^{-x^2} .
- For $y' + \frac{1}{x}y = \frac{1}{x^2}$ an integrating factor is $e^{\int (1/x) dx} = x$ so that $\frac{d}{dx} [xy] = \frac{1}{x}$ and $y = \frac{1}{x} \ln x + \frac{c}{x}$ for $0 < x < \infty$.
- For $y' - 2y = x^2 + 5$ an integrating factor is $e^{-\int 2 dx} = e^{-2x}$ so that $\frac{d}{dx} [e^{-2x}y] = x^2e^{-2x} + 5e^{-2x}$ and $y = -\frac{1}{2}x^2 - \frac{1}{2}x - \frac{11}{4} + ce^{2x}$ for $-\infty < x < \infty$.
- For $y' - \frac{1}{x}y = x \sin x$ an integrating factor is $e^{-\int (1/x) dx} = \frac{1}{x}$ so that $\frac{d}{dx} \left[\frac{1}{x}y \right] = \sin x$ and $y = cx - x \cos x$ for $0 < x < \infty$.
- For $y' + \frac{2}{x}y = \frac{3}{x}$ an integrating factor is $e^{\int (2/x) dx} = x^2$ so that $\frac{d}{dx} [x^2y] = 3x$ and $y = \frac{3}{2} + cx^{-2}$ for $0 < x < \infty$.
- For $y' + \frac{4}{x}y = x^2 - 1$ an integrating factor is $e^{\int (4/x) dx} = x^4$ so that $\frac{d}{dx} [x^4y] = x^6 - x^4$ and $y = \frac{1}{7}x^3 - \frac{1}{5}x + cx^{-4}$ for $0 < x < \infty$.

2.3 Linear Equations

12. For $y' - \frac{x}{(1+x)}y = x$ an integrating factor is $e^{-\int [x/(1+x)]dx} = (x+1)e^{-x}$ so that $\frac{d}{dx}[(x+1)e^{-x}y] = x(x+1)e^{-x}$ and $y = -x - \frac{2x+3}{x+1} + \frac{ce^x}{x+1}$ for $-1 < x < \infty$.
13. For $y' + \left(1 + \frac{2}{x}\right)y = \frac{e^x}{x^2}$ an integrating factor is $e^{\int [1+(2/x)]dx} = x^2e^x$ so that $\frac{d}{dx}[x^2e^xy] = e^{2x}$ and $y = \frac{1}{2}\frac{e^x}{x^2} + \frac{ce^{-x}}{x^2}$ for $0 < x < \infty$. The transient term is $\frac{ce^{-x}}{x^2}$.
14. For $y' + \left(1 + \frac{1}{x}\right)y = \frac{1}{x}e^{-x}\sin 2x$ an integrating factor is $e^{\int [1+(1/x)]dx} = xe^x$ so that $\frac{d}{dx}[xe^xy] = \sin 2x$ and $y = -\frac{1}{2x}e^{-x}\cos 2x + \frac{ce^{-x}}{x}$ for $0 < x < \infty$. The entire solution is transient.
15. For $\frac{dx}{dy} - \frac{4}{y}x = 4y^5$ an integrating factor is $e^{-\int (4/y)dy} = e^{\ln y^{-4}} = y^{-4}$ so that $\frac{d}{dy}[y^{-4}x] = 4y$ and $x = 2y^6 + cy^4$ for $0 < y < \infty$.
16. For $\frac{dx}{dy} + \frac{2}{y}x = e^y$ an integrating factor is $e^{\int (2/y)dy} = y^2$ so that $\frac{d}{dy}[y^2x] = y^2e^y$ and $x = e^y - \frac{2}{y}e^y + \frac{2}{y^2}e^y + \frac{c}{y^2}$ for $0 < y < \infty$. The transient term is $\frac{c}{y^2}$.
17. For $y' + (\tan x)y = \sec x$ an integrating factor is $e^{\int \tan x dx} = \sec x$ so that $\frac{d}{dx}[(\sec x)y] = \sec^2 x$ and $y = \sin x + c \cos x$ for $-\pi/2 < x < \pi/2$.
18. For $y' + (\cot x)y = \sec^2 x \csc x$ an integrating factor is $e^{\int \cot x dx} = e^{\ln |\sin x|} = \sin x$ so that $\frac{d}{dx}[(\sin x)y] = \sec^2 x$ and $y = \sec x + c \csc x$ for $0 < x < \pi/2$.
19. For $y' + \frac{x+2}{x+1}y = \frac{2xe^{-x}}{x+1}$ an integrating factor is $e^{\int [(x+2)/(x+1)]dx} = (x+1)e^x$, so $\frac{d}{dx}[(x+1)e^xy] = 2x$ and $y = \frac{x^2}{x+1}e^{-x} + \frac{c}{x+1}e^{-x}$ for $-1 < x < \infty$. The entire solution is transient.
20. For $y' + \frac{4}{x+2}y = \frac{5}{(x+2)^2}$ an integrating factor is $e^{\int [4/(x+2)]dx} = (x+2)^4$ so that $\frac{d}{dx}[(x+2)^4y] = 5(x+2)^2$ and $y = \frac{5}{3}(x+2)^{-1} + c(x+2)^{-4}$ for $-2 < x < \infty$. The entire solution is transient.
21. For $\frac{dr}{d\theta} + r \sec \theta = \cos \theta$ an integrating factor is $e^{\int \sec \theta d\theta} = e^{\ln |\sec x + \tan x|} = \sec \theta + \tan \theta$ so that $\frac{d}{d\theta}[(\sec \theta + \tan \theta)r] = 1 + \sin \theta$ and $(\sec \theta + \tan \theta)r = \theta - \cos \theta + c$ for $-\pi/2 < \theta < \pi/2$.
22. For $\frac{dP}{dt} + (2t-1)P = 4t-2$ an integrating factor is $e^{\int (2t-1)dt} = e^{t^2-t}$ so that $\frac{d}{dt}[e^{t^2-t}P] = (4t-2)e^{t^2-t}$ and $P = 2 + ce^{t-t^2}$ for $-\infty < t < \infty$. The transient term is ce^{t-t^2} .
23. For $y' + \left(3 + \frac{1}{x}\right)y = \frac{e^{-3x}}{x}$ an integrating factor is $e^{\int [3+(1/x)]dx} = xe^{3x}$ so that $\frac{d}{dx}[xe^{3x}y] = 1$ and $y = e^{-3x} + \frac{ce^{-3x}}{x}$ for $0 < x < \infty$. The transient term is ce^{-3x}/x .
24. For $y' + \frac{2}{x^2-1}y = \frac{x+1}{x-1}$ an integrating factor is $e^{\int [2/(x^2-1)]dx} = \frac{x-1}{x+1}$ so that $\frac{d}{dx}\left[\frac{x-1}{x+1}y\right] = 1$ and $(x-1)y = x(x+1) + c(x+1)$ for $-1 < x < 1$.

25. For $y' + \frac{1}{x}y = \frac{1}{x}e^x$ an integrating factor is $e^{\int(1/x)dx} = x$ so that $\frac{d}{dx}[xy] = e^x$ and $y = \frac{1}{x}e^x + \frac{c}{x}$ for $0 < x < \infty$.

If $y(1) = 2$ then $c = 2 - e$ and $y = \frac{1}{x}e^x + \frac{2-e}{x}$.

26. For $\frac{dx}{dy} - \frac{1}{y}x = 2y$ an integrating factor is $e^{-\int(1/y)dy} = \frac{1}{y}$ so that $\frac{d}{dy}\left[\frac{1}{y}x\right] = 2$ and $x = 2y^2 + cy$ for $0 < y < \infty$.

If $y(1) = 5$ then $c = -49/5$ and $x = 2y^2 - \frac{49}{5}y$.

27. For $\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$ an integrating factor is $e^{\int(R/L)dt} = e^{Rt/L}$ so that $\frac{d}{dt}[e^{Rt/L}i] = \frac{E}{L}e^{Rt/L}$ and

$i = \frac{E}{R} + ce^{-Rt/L}$ for $-\infty < t < \infty$. If $i(0) = i_0$ then $c = i_0 - E/R$ and $i = \frac{E}{R} + \left(i_0 - \frac{E}{R}\right)e^{-Rt/L}$.

28. For $\frac{dT}{dt} - kT = -T_m k$ an integrating factor is $e^{\int(-k)dt} = e^{-kt}$ so that $\frac{d}{dt}[e^{-kt}T] = -T_m k e^{-kt}$ and $T = T_m + ce^{kt}$ for $-\infty < t < \infty$. If $T(0) = T_0$ then $c = T_0 - T_m$ and $T = T_m + (T_0 - T_m)e^{kt}$.

29. For $y' + \frac{1}{x+1}y = \frac{\ln x}{x+1}$ an integrating factor is $e^{\int[1/(x+1)]dx} = x+1$ so that $\frac{d}{dx}[(x+1)y] = \ln x$ and $y = \frac{x}{x+1} \ln x - \frac{x}{x+1} + \frac{c}{x+1}$ for $0 < x < \infty$. If $y(1) = 10$ then $c = 21$ and $y = \frac{x}{x+1} \ln x - \frac{x}{x+1} + \frac{21}{x+1}$.

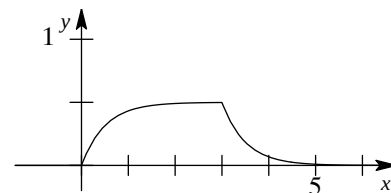
30. For $y' + (\tan x)y = \cos^2 x$ an integrating factor is $e^{\int \tan x dx} = e^{\ln|\sec x|} = \sec x$ so that $\frac{d}{dx}[(\sec x)y] = \cos x$ and $y = \sin x \cos x + c \cos x$ for $-\pi/2 < x < \pi/2$. If $y(0) = -1$ then $c = -1$ and $y = \sin x \cos x - \cos x$.

31. For $y' + 2y = f(x)$ an integrating factor is e^{2x} so that

$$ye^{2x} = \begin{cases} \frac{1}{2}e^{2x} + c_1, & 0 \leq x \leq 3 \\ c_2, & x > 3. \end{cases}$$

If $y(0) = 0$ then $c_1 = -1/2$ and for continuity we must have $c_2 = \frac{1}{2}e^6 - \frac{1}{2}$ so that

$$y = \begin{cases} \frac{1}{2}(1 - e^{-2x}), & 0 \leq x \leq 3 \\ \frac{1}{2}(e^6 - 1)e^{-2x}, & x > 3. \end{cases}$$

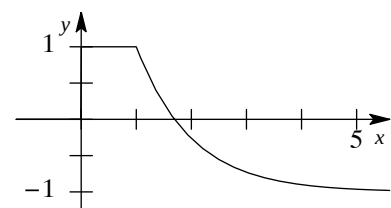


32. For $y' + y = f(x)$ an integrating factor is e^x so that

$$ye^x = \begin{cases} e^x + c_1, & 0 \leq x \leq 1 \\ -e^x + c_2, & x > 1. \end{cases}$$

If $y(0) = 1$ then $c_1 = 0$ and for continuity we must have $c_2 = 2e$ so that

$$y = \begin{cases} 1, & 0 \leq x \leq 1 \\ 2e^{1-x} - 1, & x > 1. \end{cases}$$

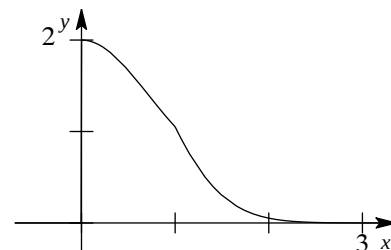


33. For $y' + 2xy = f(x)$ an integrating factor is e^{x^2} so that

$$ye^{x^2} = \begin{cases} \frac{1}{2}e^{x^2} + c_1, & 0 \leq x \leq 1 \\ c_2, & x > 1. \end{cases}$$

If $y(0) = 2$ then $c_1 = 3/2$ and for continuity we must have $c_2 = \frac{1}{2}e + \frac{3}{2}$ so that

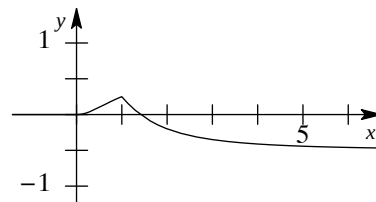
$$y = \begin{cases} \frac{1}{2} + \frac{3}{2}e^{-x^2}, & 0 \leq x \leq 1 \\ \left(\frac{1}{2}e + \frac{3}{2}\right)e^{-x^2}, & x > 1. \end{cases}$$



2.3 Linear Equations

34. For

$$y' + \frac{2x}{1+x^2}y = \begin{cases} \frac{x}{1+x^2}, & 0 \leq x \leq 1 \\ \frac{-x}{1+x^2}, & x > 1, \end{cases}$$



an integrating factor is $1+x^2$ so that

$$(1+x^2)y = \begin{cases} \frac{1}{2}x^2 + c_1, & 0 \leq x \leq 1 \\ -\frac{1}{2}x^2 + c_2, & x > 1. \end{cases}$$

If $y(0) = 0$ then $c_1 = 0$ and for continuity we must have $c_2 = 1$ so that

$$y = \begin{cases} \frac{1}{2} - \frac{1}{2(1+x^2)}, & 0 \leq x \leq 1 \\ \frac{3}{2(1+x^2)} - \frac{1}{2}, & x > 1. \end{cases}$$

35. We first solve the initial-value problem $y' + 2y = 4x$, $y(0) = 3$ on the interval $[0, 1]$.

The integrating factor is $e^{\int 2 dx} = e^{2x}$, so

$$\begin{aligned} \frac{d}{dx}[e^{2x}y] &= 4xe^{2x} \\ e^{2x}y &= \int 4xe^{2x}dx = 2xe^{2x} - e^{2x} + c_1 \\ y &= 2x - 1 + c_1e^{-2x}. \end{aligned}$$

Using the initial condition, we find $y(0) = -1 + c_1 = 3$, so $c_1 = 4$ and $y = 2x - 1 + 4e^{-2x}$, $0 \leq x \leq 1$. Now, since $y(1) = 2 - 1 + 4e^{-2} = 1 + 4e^{-2}$, we solve the initial-value problem $y' - (2/x)y = 4x$, $y(1) = 1 + 4e^{-2}$ on the interval $(1, \infty)$. The integrating factor is $e^{\int (-2/x)dx} = e^{-2 \ln x} = x^{-2}$, so

$$\begin{aligned} \frac{d}{dx}[x^{-2}y] &= 4xx^{-2} = \frac{4}{x} \\ x^{-2}y &= \int \frac{4}{x}dx = 4 \ln x + c_2 \\ y &= 4x^2 \ln x + c_2x^2. \end{aligned}$$

(We use $\ln x$ instead of $\ln |x|$ because $x > 1$.) Using the initial condition we find $y(1) = c_2 = 1 + 4e^{-2}$, so $y = 4x^2 \ln x + (1 + 4e^{-2})x^2$, $x > 1$. Thus, the solution of the original initial-value problem is

$$y = \begin{cases} 2x - 1 + 4e^{-2x}, & 0 \leq x \leq 1 \\ 4x^2 \ln x + (1 + 4e^{-2})x^2, & x > 1. \end{cases}$$

See Problem 42 in this section.

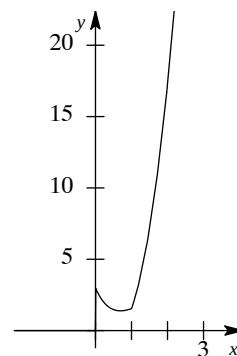
36. For $y' + e^xy = 1$ an integrating factor is e^{e^x} . Thus

$$\frac{d}{dx}[e^{e^x}y] = e^{e^x} \quad \text{and} \quad e^{e^x}y = \int_0^x e^{e^t}dt + c.$$

From $y(0) = 1$ we get $c = e$, so $y = e^{-e^x} \int_0^x e^{e^t}dt + e^{1-e^x}$.

When $y' + e^xy = 0$ we can separate variables and integrate:

$$\frac{dy}{y} = -e^x dx \quad \text{and} \quad \ln |y| = -e^x + c.$$



Thus $y = c_1 e^{-e^x}$. From $y(0) = 1$ we get $c_1 = e$, so $y = e^{1-e^x}$.

When $y' + e^x y = e^x$ we can see by inspection that $y = 1$ is a solution.

37. An integrating factor for $y' - 2xy = 1$ is e^{-x^2} . Thus

$$\begin{aligned}\frac{d}{dx}[e^{-x^2} y] &= e^{-x^2} \\ e^{-x^2} y &= \int_0^x e^{-t^2} dt = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + c \\ y &= \frac{\sqrt{\pi}}{2} e^{x^2} \operatorname{erf}(x) + c e^{x^2}.\end{aligned}$$

From $y(1) = (\sqrt{\pi}/2)e \operatorname{erf}(1) + ce = 1$ we get $c = e^{-1} - \frac{\sqrt{\pi}}{2} \operatorname{erf}(1)$. The solution of the initial-value problem is

$$\begin{aligned}y &= \frac{\sqrt{\pi}}{2} e^{x^2} \operatorname{erf}(x) + \left(e^{-1} - \frac{\sqrt{\pi}}{2} \operatorname{erf}(1)\right) e^{x^2} \\ &= e^{x^2-1} + \frac{\sqrt{\pi}}{2} e^{x^2} (\operatorname{erf}(x) - \operatorname{erf}(1)).\end{aligned}$$

38. We want 4 to be a critical point, so we use $y' = 4 - y$.
39. (a) All solutions of the form $y = x^5 e^x - x^4 e^x + cx^4$ satisfy the initial condition. In this case, since $4/x$ is discontinuous at $x = 0$, the hypotheses of Theorem 1.1 are not satisfied and the initial-value problem does not have a unique solution.
- (b) The differential equation has no solution satisfying $y(0) = y_0$, $y_0 > 0$.
- (c) In this case, since $x_0 > 0$, Theorem 1.1 applies and the initial-value problem has a unique solution given by $y = x^5 e^x - x^4 e^x + cx^4$ where $c = y_0/x_0^4 - x_0 e^{x_0} + e^{x_0}$.
40. On the interval $(-3, 3)$ the integrating factor is

$$e^{\int x dx/(x^2-9)} = e^{-\int x dx/(9-x^2)} = e^{\frac{1}{2} \ln(9-x^2)} = \sqrt{9-x^2}$$

and so

$$\frac{d}{dx} [\sqrt{9-x^2} y] = 0 \quad \text{and} \quad y = \frac{c}{\sqrt{9-x^2}}.$$

41. We want the general solution to be $y = 3x - 5 + ce^{-x}$. (Rather than e^{-x} , any function that approaches 0 as $x \rightarrow \infty$ could be used.) Differentiating we get

$$y' = 3 - ce^{-x} = 3 - (y - 3x + 5) = -y + 3x - 2,$$

so the differential equation $y' + y = 3x - 2$ has solutions asymptotic to the line $y = 3x - 5$.

42. The left-hand derivative of the function at $x = 1$ is $1/e$ and the right-hand derivative at $x = 1$ is $1 - 1/e$. Thus, y is not differentiable at $x = 1$.
43. (a) Differentiating $y_c = c/x^3$ we get

$$y'_c = -\frac{3c}{x^4} = -\frac{3}{x} \frac{c}{x^3} = -\frac{3}{x} y_c$$

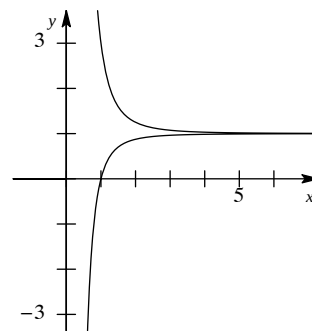
so a differential equation with general solution $y_c = c/x^3$ is $xy' + 3y = 0$. Now

$$xy'_p + 3y_p = x(3x^2) + 3(x^3) = 6x^3$$

so a differential equation with general solution $y = c/x^3 + x^3$ is $xy' + 3y = 6x^3$. This will be a general solution on $(0, \infty)$.

2.3 Linear Equations

- (b) Since $y(1) = 1^3 - 1/1^3 = 0$, an initial condition is $y(1) = 0$. Since $y(1) = 1^3 + 2/1^3 = 3$, an initial condition is $y(1) = 3$. In each case the interval of definition is $(0, \infty)$. The initial-value problem $xy' + 3y = 6x^3$, $y(0) = 0$ has solution $y = x^3$ for $-\infty < x < \infty$. In the figure the lower curve is the graph of $y(x) = x^3 - 1/x^3$, while the upper curve is the graph of $y = x^3 - 2/x^3$.



- (c) The first two initial-value problems in part (b) are not unique. For example, setting $y(2) = 2^3 - 1/2^3 = 63/8$, we see that $y(2) = 63/8$ is also an initial condition leading to the solution $y = x^3 - 1/x^3$.

44. Since $e^{\int P(x)dx+c} = e^c e^{\int P(x)dx} = c_1 e^{\int P(x)dx}$, we would have

$$c_1 e^{\int P(x)dx} y = c_2 + \int c_1 e^{\int P(x)dx} f(x) dx \quad \text{and} \quad e^{\int P(x)dx} y = c_3 + \int e^{\int P(x)dx} f(x) dx,$$

which is the same as (6) in the text.

45. We see by inspection that $y = 0$ is a solution.
46. The solution of the first equation is $x = c_1 e^{-\lambda_1 t}$. From $x(0) = x_0$ we obtain $c_1 = x_0$ and so $x = x_0 e^{-\lambda_1 t}$. The second equation then becomes

$$\frac{dy}{dt} = x_0 \lambda_1 e^{-\lambda_1 t} - \lambda_2 y \quad \text{or} \quad \frac{dy}{dt} + \lambda_2 y = x_0 \lambda_1 e^{-\lambda_1 t}$$

which is linear. An integrating factor is $e^{\lambda_2 t}$. Thus

$$\begin{aligned} \frac{d}{dt} [e^{\lambda_2 t} y] &= x_0 \lambda_1 e^{-\lambda_1 t} e^{\lambda_2 t} = x_0 \lambda_1 e^{(\lambda_2 - \lambda_1)t} \\ e^{\lambda_2 t} y &= \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} + c_2 \\ y &= \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + c_2 e^{-\lambda_2 t}. \end{aligned}$$

From $y(0) = y_0$ we obtain $c_2 = (y_0 \lambda_2 - y_0 \lambda_1 - x_0 \lambda_1)/(\lambda_2 - \lambda_1)$. The solution is

$$y = \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{y_0 \lambda_2 - y_0 \lambda_1 - x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t}.$$

47. Writing the differential equation as $\frac{dE}{dt} + \frac{1}{RC} E = 0$ we see that an integrating factor is $e^{t/RC}$. Then

$$\begin{aligned} \frac{d}{dt} [e^{t/RC} E] &= 0 \\ e^{t/RC} E &= c \\ E &= c e^{-t/RC}. \end{aligned}$$

From $E(4) = c e^{-4/RC} = E_0$ we find $c = E_0 e^{4/RC}$. Thus, the solution of the initial-value problem is

$$E = E_0 e^{4/RC} e^{-t/RC} = E_0 e^{-(t-4)/RC}.$$

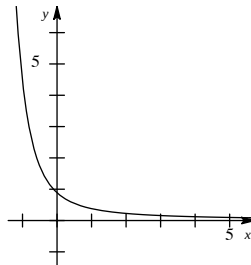
48. (a) An integrating factor for $y' - 2xy = -1$ is e^{-x^2} . Thus

$$\begin{aligned}\frac{d}{dx}[e^{-x^2}y] &= -e^{-x^2} \\ e^{-x^2}y &= -\int_0^x e^{-t^2} dt = -\frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + c.\end{aligned}$$

From $y(0) = \sqrt{\pi}/2$, and noting that $\operatorname{erf}(0) = 0$, we get $c = \sqrt{\pi}/2$. Thus

$$y = e^{x^2} \left(-\frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + \frac{\sqrt{\pi}}{2} \right) = \frac{\sqrt{\pi}}{2} e^{x^2} (1 - \operatorname{erf}(x)) = \frac{\sqrt{\pi}}{2} e^{x^2} \operatorname{erfc}(x).$$

- (b) Using a CAS we find $y(2) \approx 0.226339$.



49. (a) An integrating factor for

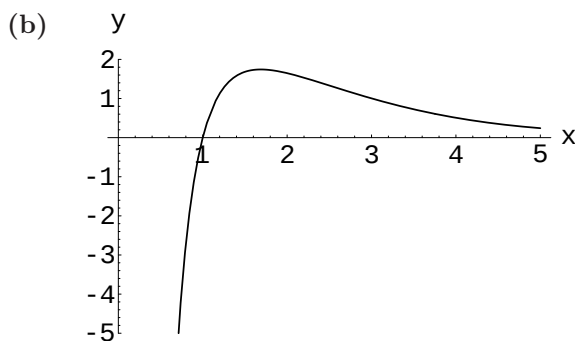
$$y' + \frac{2}{x}y = \frac{10 \sin x}{x^3}$$

is x^2 . Thus

$$\begin{aligned}\frac{d}{dx}[x^2y] &= 10 \frac{\sin x}{x} \\ x^2y &= 10 \int_0^x \frac{\sin t}{t} dt + c \\ y &= 10x^{-2} \operatorname{Si}(x) + cx^{-2}.\end{aligned}$$

From $y(1) = 0$ we get $c = -10\operatorname{Si}(1)$. Thus

$$y = 10x^{-2} \operatorname{Si}(x) - 10x^{-2} \operatorname{Si}(1) = 10x^{-2} (\operatorname{Si}(x) - \operatorname{Si}(1)).$$



- (c) From the graph in part (b) we see that the absolute maximum occurs around $x = 1.7$. Using the root-finding capability of a CAS and solving $y'(x) = 0$ for x we see that the absolute maximum is $(1.688, 1.742)$.

50. (a) The integrating factor for $y' - (\sin x^2)y = 0$ is $e^{-\int_0^x \sin t^2 dt}$. Then

$$\begin{aligned}\frac{d}{dx}[e^{-\int_0^x \sin t^2 dt}y] &= 0 \\ e^{-\int_0^x \sin t^2 dt}y &= c_1 \\ y &= c_1 e^{\int_0^x \sin t^2 dt}.\end{aligned}$$

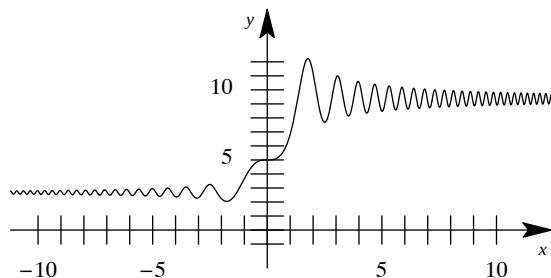
2.3 Linear Equations

Letting $t = \sqrt{\pi/2} u$ we have $dt = \sqrt{\pi/2} du$ and

$$\int_0^x \sin t^2 dt = \sqrt{\frac{\pi}{2}} \int_0^{\sqrt{2/\pi} x} \sin\left(\frac{\pi}{2} u^2\right) du = \sqrt{\frac{\pi}{2}} S\left(\sqrt{\frac{2}{\pi}} x\right)$$

so $y = c_1 e^{\sqrt{\pi/2} S(\sqrt{2/\pi} x)}$. Using $S(0) = 0$ and $y(0) = c_1 = 5$ we have $y = 5e^{\sqrt{\pi/2} S(\sqrt{2/\pi} x)}$.

(b)



- (c) From the graph we see that as $x \rightarrow \infty$, $y(x)$ oscillates with decreasing amplitudes approaching 9.35672. Since $\lim_{x \rightarrow \infty} 5S(x) = \frac{1}{2}$, we have $\lim_{x \rightarrow \infty} y(x) = 5e^{\sqrt{\pi/8}} \approx 9.357$, and since $\lim_{x \rightarrow -\infty} S(x) = -\frac{1}{2}$, we have $\lim_{x \rightarrow -\infty} y(x) = 5e^{-\sqrt{\pi/8}} \approx 2.672$.
- (d) From the graph in part (b) we see that the absolute maximum occurs around $x = 1.7$ and the absolute minimum occurs around $x = -1.8$. Using the root-finding capability of a CAS and solving $y'(x) = 0$ for x , we see that the absolute maximum is $(1.772, 12.235)$ and the absolute minimum is $(-1.772, 2.044)$.

EXERCISES 2.4

Exact Equations

- Let $M = 2x - 1$ and $N = 3y + 7$ so that $M_y = 0 = N_x$. From $f_x = 2x - 1$ we obtain $f = x^2 - x + h(y)$, $h'(y) = 3y + 7$, and $h(y) = \frac{3}{2}y^2 + 7y$. A solution is $x^2 - x + \frac{3}{2}y^2 + 7y = c$.
- Let $M = 2x + y$ and $N = -x - 6y$. Then $M_y = 1$ and $N_x = -1$, so the equation is not exact.
- Let $M = 5x + 4y$ and $N = 4x - 8y^3$ so that $M_y = 4 = N_x$. From $f_x = 5x + 4y$ we obtain $f = \frac{5}{2}x^2 + 4xy + h(y)$, $h'(y) = -8y^3$, and $h(y) = -2y^4$. A solution is $\frac{5}{2}x^2 + 4xy - 2y^4 = c$.
- Let $M = \sin y - y \sin x$ and $N = \cos x + x \cos y - y$ so that $M_y = \cos y - \sin x = N_x$. From $f_x = \sin y - y \sin x$ we obtain $f = x \sin y + y \cos x + h(y)$, $h'(y) = -y$, and $h(y) = -\frac{1}{2}y^2$. A solution is $x \sin y + y \cos x - \frac{1}{2}y^2 = c$.
- Let $M = 2y^2x - 3$ and $N = 2yx^2 + 4$ so that $M_y = 4xy = N_x$. From $f_x = 2y^2x - 3$ we obtain $f = x^2y^2 - 3x + h(y)$, $h'(y) = 4$, and $h(y) = 4y$. A solution is $x^2y^2 - 3x + 4y = c$.
- Let $M = 4x^3 - 3y \sin 3x - y/x^2$ and $N = 2y - 1/x + \cos 3x$ so that $M_y = -3 \sin 3x - 1/x^2$ and $N_x = 1/x^2 - 3 \sin 3x$. The equation is not exact.
- Let $M = x^2 - y^2$ and $N = x^2 - 2xy$ so that $M_y = -2y$ and $N_x = 2x - 2y$. The equation is not exact.
- Let $M = 1 + \ln x + y/x$ and $N = -1 + \ln x$ so that $M_y = 1/x = N_x$. From $f_y = -1 + \ln x$ we obtain $f = -y + y \ln x + h(y)$, $h'(x) = 1 + \ln x$, and $h(y) = x \ln x$. A solution is $-y + y \ln x + x \ln x = c$.

2.4 Exact Equations

9. Let $M = y^3 - y^2 \sin x - x$ and $N = 3xy^2 + 2y \cos x$ so that $M_y = 3y^2 - 2y \sin x = N_x$. From $f_x = y^3 - y^2 \sin x - x$ we obtain $f = xy^3 + y^2 \cos x - \frac{1}{2}x^2 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution is $xy^3 + y^2 \cos x - \frac{1}{2}x^2 = c$.
10. Let $M = x^3 + y^3$ and $N = 3xy^2$ so that $M_y = 3y^2 = N_x$. From $f_x = x^3 + y^3$ we obtain $f = \frac{1}{4}x^4 + xy^3 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution is $\frac{1}{4}x^4 + xy^3 = c$.
11. Let $M = y \ln y - e^{-xy}$ and $N = 1/y + x \ln y$ so that $M_y = 1 + \ln y + xe^{-xy}$ and $N_x = \ln y$. The equation is not exact.
12. Let $M = 3x^2y + e^y$ and $N = x^3 + xe^y - 2y$ so that $M_y = 3x^2 + e^y = N_x$. From $f_x = 3x^2y + e^y$ we obtain $f = x^3y + xe^y + h(y)$, $h'(y) = -2y$, and $h(y) = -y^2$. A solution is $x^3y + xe^y - y^2 = c$.
13. Let $M = y - 6x^2 - 2xe^x$ and $N = x$ so that $M_y = 1 = N_x$. From $f_x = y - 6x^2 - 2xe^x$ we obtain $f = xy - 2x^3 - 2xe^x + 2e^x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution is $xy - 2x^3 - 2xe^x + 2e^x = c$.
14. Let $M = 1 - 3/x + y$ and $N = 1 - 3/y + x$ so that $M_y = 1 = N_x$. From $f_x = 1 - 3/x + y$ we obtain $f = x - 3 \ln |x| + xy + h(y)$, $h'(y) = 1 - \frac{3}{y}$, and $h(y) = y - 3 \ln |y|$. A solution is $x + y + xy - 3 \ln |xy| = c$.
15. Let $M = x^2y^3 - 1/(1 + 9x^2)$ and $N = x^3y^2$ so that $M_y = 3x^2y^2 = N_x$. From $f_x = x^2y^3 - 1/(1 + 9x^2)$ we obtain $f = \frac{1}{3}x^3y^3 - \frac{1}{3} \arctan(3x) + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution is $x^3y^3 - \arctan(3x) = c$.
16. Let $M = -2y$ and $N = 5y - 2x$ so that $M_y = -2 = N_x$. From $f_x = -2y$ we obtain $f = -2xy + h(y)$, $h'(y) = 5y$, and $h(y) = \frac{5}{2}y^2$. A solution is $-2xy + \frac{5}{2}y^2 = c$.
17. Let $M = \tan x - \sin x \sin y$ and $N = \cos x \cos y$ so that $M_y = -\sin x \cos y = N_x$. From $f_x = \tan x - \sin x \sin y$ we obtain $f = \ln |\sec x| + \cos x \sin y + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution is $\ln |\sec x| + \cos x \sin y = c$.
18. Let $M = 2y \sin x \cos x - y + 2y^2e^{xy^2}$ and $N = -x + \sin^2 x + 4xye^{xy^2}$ so that

$$M_y = 2 \sin x \cos x - 1 + 4xy^3e^{xy^2} + 4ye^{xy^2} = N_x.$$

From $f_x = 2y \sin x \cos x - y + 2y^2e^{xy^2}$ we obtain $f = y \sin^2 x - xy + 2e^{xy^2} + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution is $y \sin^2 x - xy + 2e^{xy^2} = c$.

19. Let $M = 4t^3y - 15t^2 - y$ and $N = t^4 + 3y^2 - t$ so that $M_y = 4t^3 - 1 = N_t$. From $f_t = 4t^3y - 15t^2 - y$ we obtain $f = t^4y - 5t^3 - ty + h(y)$, $h'(y) = 3y^2$, and $h(y) = y^3$. A solution is $t^4y - 5t^3 - ty + y^3 = c$.
20. Let $M = 1/t + 1/t^2 - y/(t^2 + y^2)$ and $N = ye^y + t/(t^2 + y^2)$ so that $M_y = (y^2 - t^2)/(t^2 + y^2)^2 = N_t$. From $f_t = 1/t + 1/t^2 - y/(t^2 + y^2)$ we obtain $f = \ln |t| - \frac{1}{t} - \arctan\left(\frac{t}{y}\right) + h(y)$, $h'(y) = ye^y$, and $h(y) = ye^y - e^y$. A solution is

$$\ln |t| - \frac{1}{t} - \arctan\left(\frac{t}{y}\right) + ye^y - e^y = c.$$

21. Let $M = x^2 + 2xy + y^2$ and $N = 2xy + x^2 - 1$ so that $M_y = 2(x + y) = N_x$. From $f_x = x^2 + 2xy + y^2$ we obtain $f = \frac{1}{3}x^3 + x^2y + xy^2 + h(y)$, $h'(y) = -1$, and $h(y) = -y$. The solution is $\frac{1}{3}x^3 + x^2y + xy^2 - y = c$. If $y(1) = 1$ then $c = 4/3$ and a solution of the initial-value problem is $\frac{1}{3}x^3 + x^2y + xy^2 - y = \frac{4}{3}$.
22. Let $M = e^x + y$ and $N = 2 + x + ye^y$ so that $M_y = 1 = N_x$. From $f_x = e^x + y$ we obtain $f = e^x + xy + h(y)$, $h'(y) = 2 + ye^y$, and $h(y) = 2y + ye^y - y$. The solution is $e^x + xy + 2y + ye^y - e^y = c$. If $y(0) = 1$ then $c = 3$ and a solution of the initial-value problem is $e^x + xy + 2y + ye^y - e^y = 3$.
23. Let $M = 4y + 2t - 5$ and $N = 6y + 4t - 1$ so that $M_y = 4 = N_t$. From $f_t = 4y + 2t - 5$ we obtain $f = 4ty + t^2 - 5t + h(y)$, $h'(y) = 6y - 1$, and $h(y) = 3y^2 - y$. The solution is $4ty + t^2 - 5t + 3y^2 - y = c$. If $y(-1) = 2$ then $c = 8$ and a solution of the initial-value problem is $4ty + t^2 - 5t + 3y^2 - y = 8$.

2.4 Exact Equations

24. Let $M = t/2y^4$ and $N = (3y^2 - t^2)/y^5$ so that $M_y = -2t/y^5 = N_t$. From $f_t = t/2y^4$ we obtain $f = \frac{t^2}{4y^4} + h(y)$, $h'(y) = \frac{3}{y^3}$, and $h(y) = -\frac{3}{2y^2}$. The solution is $\frac{t^2}{4y^4} - \frac{3}{2y^2} = c$. If $y(1) = 1$ then $c = -5/4$ and a solution of the initial-value problem is $\frac{t^2}{4y^4} - \frac{3}{2y^2} = -\frac{5}{4}$.
25. Let $M = y^2 \cos x - 3x^2y - 2x$ and $N = 2y \sin x - x^3 + \ln y$ so that $M_y = 2y \cos x - 3x^2 = N_x$. From $f_x = y^2 \cos x - 3x^2y - 2x$ we obtain $f = y^2 \sin x - x^3y - x^2 + h(y)$, $h'(y) = \ln y$, and $h(y) = y \ln y - y$. The solution is $y^2 \sin x - x^3y - x^2 + y \ln y - y = c$. If $y(0) = e$ then $c = 0$ and a solution of the initial-value problem is $y^2 \sin x - x^3y - x^2 + y \ln y - y = 0$.
26. Let $M = y^2 + y \sin x$ and $N = 2xy - \cos x - 1/(1 + y^2)$ so that $M_y = 2y + \sin x = N_x$. From $f_x = y^2 + y \sin x$ we obtain $f = xy^2 - y \cos x + h(y)$, $h'(y) = \frac{-1}{1 + y^2}$, and $h(y) = -\tan^{-1} y$. The solution is $xy^2 - y \cos x - \tan^{-1} y = c$. If $y(0) = 1$ then $c = -1 - \pi/4$ and a solution of the initial-value problem is $xy^2 - y \cos x - \tan^{-1} y = -1 - \frac{\pi}{4}$.
27. Equating $M_y = 3y^2 + 4kxy^3$ and $N_x = 3y^2 + 40xy^3$ we obtain $k = 10$.
28. Equating $M_y = 18xy^2 - \sin y$ and $N_x = 4kxy^2 - \sin y$ we obtain $k = 9/2$.
29. Let $M = -x^2y^2 \sin x + 2xy^2 \cos x$ and $N = 2x^2y \cos x$ so that $M_y = -2x^2y \sin x + 4xy \cos x = N_x$. From $f_y = 2x^2y \cos x$ we obtain $f = x^2y^2 \cos x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution of the differential equation is $x^2y^2 \cos x = c$.
30. Let $M = (x^2 + 2xy - y^2)/(x^2 + 2xy + y^2)$ and $N = (y^2 + 2xy - x^2)/(y^2 + 2xy + x^2)$ so that $M_y = -4xy/(x + y)^3 = N_x$. From $f_x = (x^2 + 2xy + y^2 - 2y^2)/(x + y)^2$ we obtain $f = x + \frac{2y^2}{x + y} + h(y)$, $h'(y) = -1$, and $h(y) = -y$. A solution of the differential equation is $x^2 + y^2 = c(x + y)$.
31. We note that $(M_y - N_x)/N = 1/x$, so an integrating factor is $e^{\int dx/x} = x$. Let $M = 2xy^2 + 3x^2$ and $N = 2x^2y$ so that $M_y = 4xy = N_x$. From $f_x = 2xy^2 + 3x^2$ we obtain $f = x^2y^2 + x^3 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution of the differential equation is $x^2y^2 + x^3 = c$.
32. We note that $(M_y - N_x)/N = 1$, so an integrating factor is $e^{\int dx} = e^x$. Let $M = xye^x + y^2e^x + ye^x$ and $N = xe^x + 2ye^x$ so that $M_y = xe^x + 2ye^x + e^x = N_x$. From $f_y = xe^x + 2ye^x$ we obtain $f = xye^x + y^2e^x + h(x)$, $h'(x) = 0$, and $h(x) = 0$. A solution of the differential equation is $xye^x + y^2e^x = c$.
33. We note that $(N_x - M_y)/M = 2/y$, so an integrating factor is $e^{\int 2dy/y} = y^2$. Let $M = 6xy^3$ and $N = 4y^3 + 9x^2y^2$ so that $M_y = 18xy^2 = N_x$. From $f_x = 6xy^3$ we obtain $f = 3x^2y^3 + h(y)$, $h'(y) = 4y^3$, and $h(y) = y^4$. A solution of the differential equation is $3x^2y^3 + y^4 = c$.
34. We note that $(M_y - N_x)/N = -\cot x$, so an integrating factor is $e^{-\int \cot x dx} = \csc x$. Let $M = \cos x \csc x = \cot x$ and $N = (1 + 2/y) \sin x \csc x = 1 + 2/y$, so that $M_y = 0 = N_x$. From $f_x = \cot x$ we obtain $f = \ln(\sin x) + h(y)$, $h'(y) = 1 + 2/y$, and $h(y) = y + \ln y^2$. A solution of the differential equation is $\ln(\sin x) + y + \ln y^2 = c$.
35. We note that $(M_y - N_x)/N = 3$, so an integrating factor is $e^{\int 3dx} = e^{3x}$. Let $M = (10 - 6y + e^{-3x})e^{3x} = 10e^{3x} - 6ye^{3x} + 1$ and $N = -2e^{3x}$, so that $M_y = -6e^{3x} = N_x$. From $f_x = 10e^{3x} - 6ye^{3x} + 1$ we obtain $f = \frac{10}{3}e^{3x} - 2ye^{3x} + x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. A solution of the differential equation is $\frac{10}{3}e^{3x} - 2ye^{3x} + x = c$.
36. We note that $(N_x - M_y)/M = -3/y$, so an integrating factor is $e^{-3 \int dy/y} = 1/y^3$. Let $M = (y^2 + xy^3)/y^3 = 1/y + x$ and $N = (5y^2 - xy + y^3 \sin y)/y^3 = 5/y - x/y^2 + \sin y$, so that $M_y = -1/y^2 = N_x$. From $f_x = 1/y + x$ we obtain $f = x/y + \frac{1}{2}x^2 + h(y)$, $h'(y) = 5/y + \sin y$, and $h(y) = 5 \ln |y| - \cos y$. A solution of the differential equation is $x/y + \frac{1}{2}x^2 + 5 \ln |y| - \cos y = c$.

37. We note that $(M_y - N_x)/N = 2x/(4 + x^2)$, so an integrating factor is $e^{-2 \int x dx/(4+x^2)} = 1/(4 + x^2)$. Let $M = x/(4 + x^2)$ and $N = (x^2y + 4y)/(4 + x^2) = y$, so that $M_y = 0 = N_x$. From $f_x = x/(4 + x^2)$ we obtain $f = \frac{1}{2} \ln(4 + x^2) + h(y)$, $h'(y) = y$, and $h(y) = \frac{1}{2}y^2$. A solution of the differential equation is $\frac{1}{2} \ln(4 + x^2) + \frac{1}{2}y^2 = c$.
38. We note that $(M_y - N_x)/N = -3/(1 + x)$, so an integrating factor is $e^{-3 \int dx/(1+x)} = 1/(1 + x)^3$. Let $M = (x^2 + y^2 - 5)/(1 + x)^3$ and $N = -(y + xy)/(1 + x)^3 = -y/(1 + x)^2$, so that $M_y = 2y/(1 + x)^3 = N_x$. From $f_y = -y/(1 + x)^2$ we obtain $f = -\frac{1}{2}y^2/(1 + x)^2 + h(x)$, $h'(x) = (x^2 - 5)/(1 + x)^3$, and $h(x) = 2/(1 + x)^2 + 2/(1 + x) + \ln|1 + x|$. A solution of the differential equation is

$$-\frac{y^2}{2(1+x)^2} + \frac{2}{(1+x)^2} + \frac{2}{(1+x)} + \ln|1+x| = c.$$

39. (a) Implicitly differentiating $x^3 + 2x^2y + y^2 = c$ and solving for dy/dx we obtain

$$3x^2 + 2x^2 \frac{dy}{dx} + 4xy + 2y \frac{dy}{dx} = 0 \quad \text{and} \quad \frac{dy}{dx} = -\frac{3x^2 + 4xy}{2x^2 + 2y}.$$

By writing the last equation in differential form we get $(4xy + 3x^2)dx + (2y + 2x^2)dy = 0$.

- (b) Setting $x = 0$ and $y = -2$ in $x^3 + 2x^2y + y^2 = c$ we find $c = 4$, and setting $x = y = 1$ we also find $c = 4$. Thus, both initial conditions determine the same implicit solution.

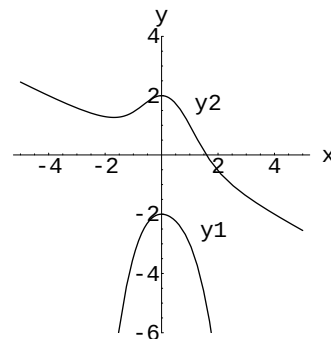
- (c) Solving $x^3 + 2x^2y + y^2 = 4$ for y we get

$$y_1(x) = -x^2 - \sqrt{4 - x^3 + x^4}$$

and

$$y_2(x) = -x^2 + \sqrt{4 - x^3 + x^4}.$$

Observe in the figure that $y_1(0) = -2$ and $y_2(1) = 1$.



40. To see that the equations are not equivalent consider $dx = -(x/y)dy$. An integrating factor is $\mu(x, y) = y$ resulting in $y dx + x dy = 0$. A solution of the latter equation is $y = 0$, but this is not a solution of the original equation.
41. The explicit solution is $y = \sqrt{(3 + \cos^2 x)/(1 - x^2)}$. Since $3 + \cos^2 x > 0$ for all x we must have $1 - x^2 > 0$ or $-1 < x < 1$. Thus, the interval of definition is $(-1, 1)$.
42. (a) Since $f_y = N(x, y) = xe^{xy} + 2xy + 1/x$ we obtain $f = e^{xy} + xy^2 + \frac{y}{x} + h(x)$ so that $f_x = ye^{xy} + y^2 - \frac{y}{x^2} + h'(x)$. Let $M(x, y) = ye^{xy} + y^2 - \frac{y}{x^2}$.
- (b) Since $f_x = M(x, y) = y^{1/2}x^{-1/2} + x(x^2 + y)^{-1}$ we obtain $f = 2y^{1/2}x^{1/2} + \frac{1}{2} \ln|x^2 + y| + g(y)$ so that $f_y = y^{-1/2}x^{1/2} + \frac{1}{2}(x^2 + y)^{-1} + g'(y)$. Let $N(x, y) = y^{-1/2}x^{1/2} + \frac{1}{2}(x^2 + y)^{-1}$.
43. First note that

$$d\left(\sqrt{x^2 + y^2}\right) = \frac{x}{\sqrt{x^2 + y^2}} dx + \frac{y}{\sqrt{x^2 + y^2}} dy.$$

Then $x dx + y dy = \sqrt{x^2 + y^2} dx$ becomes

$$\frac{x}{\sqrt{x^2 + y^2}} dx + \frac{y}{\sqrt{x^2 + y^2}} dy = d\left(\sqrt{x^2 + y^2}\right) = dx.$$

2.4 Exact Equations

The left side is the total differential of $\sqrt{x^2 + y^2}$ and the right side is the total differential of $x + c$. Thus $\sqrt{x^2 + y^2} = x + c$ is a solution of the differential equation.

44. To see that the statement is true, write the separable equation as $-g(x)dx + dy/h(y) = 0$. Identifying $M = -g(x)$ and $N = 1/h(y)$, we see that $M_y = 0 = N_x$, so the differential equation is exact.
45. (a) In differential form we have $(v^2 - 32x)dx + xv dv = 0$. This is not an exact form, but $\mu(x) = x$ is an integrating factor. Multiplying by x we get $(xv^2 - 32x^2)dx + x^2v dv = 0$. This form is the total differential of $u = \frac{1}{2}x^2v^2 - \frac{32}{3}x^3$, so an implicit solution is $\frac{1}{2}x^2v^2 - \frac{32}{3}x^3 = c$. Letting $x = 3$ and $v = 0$ we find $c = -288$. Solving for v we get

$$v = 8\sqrt{\frac{x}{3} - \frac{9}{x^2}}.$$

- (b) The chain leaves the platform when $x = 8$, so the velocity at this time is

$$v(8) = 8\sqrt{\frac{8}{3} - \frac{9}{64}} \approx 12.7 \text{ ft/s.}$$

46. (a) Letting

$$M(x, y) = \frac{2xy}{(x^2 + y^2)^2} \quad \text{and} \quad N(x, y) = 1 + \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

we compute

$$M_y = \frac{2x^3 - 8xy^2}{(x^2 + y^2)^3} = N_x,$$

so the differential equation is exact. Then we have

$$\begin{aligned} \frac{\partial f}{\partial x} &= M(x, y) = \frac{2xy}{(x^2 + y^2)^2} = 2xy(x^2 + y^2)^{-2} \\ f(x, y) &= -y(x^2 + y^2)^{-1} + g(y) = -\frac{y}{x^2 + y^2} + g(y) \\ \frac{\partial f}{\partial y} &= \frac{y^2 - x^2}{(x^2 + y^2)^2} + g'(y) = N(x, y) = 1 + \frac{y^2 - x^2}{(x^2 + y^2)^2}. \end{aligned}$$

Thus, $g'(y) = 1$ and $g(y) = y$. The solution is $y - \frac{y}{x^2 + y^2} = c$. When $c = 0$ the solution is $x^2 + y^2 = 1$.

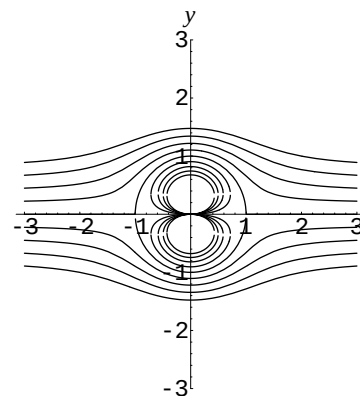
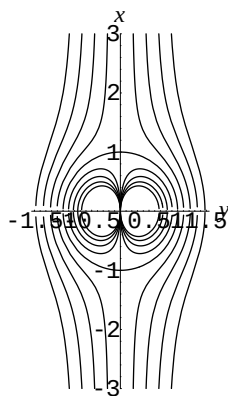
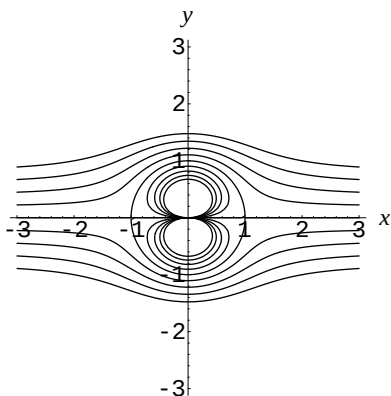
- (b) The first graph below is obtained in *Mathematica* using $f(x, y) = y - y/(x^2 + y^2)$ and

```
ContourPlot[f[x, y], {x, -3, 3}, {y, -3, 3},
  Axes->True, AxesOrigin->{0, 0}, AxesLabel->{x, y},
  Frame->False, PlotPoints->100, ContourShading->False,
  Contours->{0, -0.2, 0.2, -0.4, 0.4, -0.6, 0.6, -0.8, 0.8}]
```

The second graph uses

$$x = -\sqrt{\frac{y^3 - cy^2 - y}{c - y}} \quad \text{and} \quad x = \sqrt{\frac{y^3 - cy^2 - y}{c - y}}.$$

In this case the x -axis is vertical and the y -axis is horizontal. To obtain the third graph, we solve $y - y/(x^2 + y^2) = c$ for y in a CAS. This appears to give one real and two complex solutions. When graphed in *Mathematica* however, all three solutions contribute to the graph. This is because the solutions involve the square root of expressions containing c . For some values of c the expression is negative, causing an apparent complex solution to actually be real.



EXERCISES 2.5

Solutions by Substitutions

1. Letting $y = ux$ we have

$$\begin{aligned}(x - ux) dx + x(u dx + x du) &= 0 \\ dx + x du &= 0 \\ \frac{dx}{x} + du &= 0 \\ \ln |x| + u &= c \\ x \ln |x| + y &= cx.\end{aligned}$$

2. Letting $y = ux$ we have

$$\begin{aligned}(x + ux) dx + x(u dx + x du) &= 0 \\ (1 + 2u) dx + x du &= 0 \\ \frac{dx}{x} + \frac{du}{1 + 2u} &= 0 \\ \ln |x| + \frac{1}{2} \ln |1 + 2u| &= c \\ x^2 \left(1 + 2\frac{y}{x}\right) &= c_1 \\ x^2 + 2xy &= c_1.\end{aligned}$$

2.5 Solutions by Substitutions

3. Letting $x = vy$ we have

$$\begin{aligned}
 vy(v\,dy + y\,dv) + (y - 2vy)\,dy &= 0 \\
 vy^2\,dv + y(v^2 - 2v + 1)\,dy &= 0 \\
 \frac{v\,dv}{(v-1)^2} + \frac{dy}{y} &= 0 \\
 \ln|v-1| - \frac{1}{v-1} + \ln|y| &= c \\
 \ln\left|\frac{x}{y} - 1\right| - \frac{1}{x/y - 1} + \ln y &= c \\
 (x-y)\ln|x-y| - y &= c(x-y).
 \end{aligned}$$

4. Letting $x = vy$ we have

$$\begin{aligned}
 y(v\,dy + y\,dv) - 2(vy + y)\,dy &= 0 \\
 y\,dv - (v+2)\,dy &= 0 \\
 \frac{dv}{v+2} - \frac{dy}{y} &= 0 \\
 \ln|v+2| - \ln|y| &= c \\
 \ln\left|\frac{x}{y} + 2\right| - \ln|y| &= c \\
 x + 2y &= c_1 y^2.
 \end{aligned}$$

5. Letting $y = ux$ we have

$$\begin{aligned}
 (u^2 x^2 + ux^2)\,dx - x^2(u\,dx + x\,du) &= 0 \\
 u^2\,dx - x\,du &= 0 \\
 \frac{dx}{x} - \frac{du}{u^2} &= 0 \\
 \ln|x| + \frac{1}{u} &= c \\
 \ln|x| + \frac{x}{y} &= c \\
 y\ln|x| + x &= cy.
 \end{aligned}$$

6. Letting $y = ux$ and using partial fractions, we have

$$\begin{aligned}
 (u^2 x^2 + ux^2)\,dx + x^2(u\,dx + x\,du) &= 0 \\
 x^2(u^2 + 2u)\,dx + x^3\,du &= 0 \\
 \frac{dx}{x} + \frac{du}{u(u+2)} &= 0 \\
 \ln|x| + \frac{1}{2}\ln|u| - \frac{1}{2}\ln|u+2| &= c \\
 \frac{x^2 u}{u+2} &= c_1 \\
 x^2 \frac{y}{x} &= c_1 \left(\frac{y}{x} + 2\right) \\
 x^2 y &= c_1(y + 2x).
 \end{aligned}$$

7. Letting $y = ux$ we have

$$\begin{aligned}
 (ux - x) dx - (ux + x)(u dx + x du) &= 0 \\
 (u^2 + 1) dx + x(u + 1) du &= 0 \\
 \frac{dx}{x} + \frac{u + 1}{u^2 + 1} du &= 0 \\
 \ln |x| + \frac{1}{2} \ln(u^2 + 1) + \tan^{-1} u &= c \\
 \ln x^2 \left(\frac{y^2}{x^2} + 1 \right) + 2 \tan^{-1} \frac{y}{x} &= c_1 \\
 \ln(x^2 + y^2) + 2 \tan^{-1} \frac{y}{x} &= c_1.
 \end{aligned}$$

8. Letting $y = ux$ we have

$$\begin{aligned}
 (x + 3ux) dx - (3x + ux)(u dx + x du) &= 0 \\
 (u^2 - 1) dx + x(u + 3) du &= 0 \\
 \frac{dx}{x} + \frac{u + 3}{(u - 1)(u + 1)} du &= 0 \\
 \ln |x| + 2 \ln |u - 1| - \ln |u + 1| &= c \\
 \frac{x(u - 1)^2}{u + 1} &= c_1 \\
 x \left(\frac{y}{x} - 1 \right)^2 &= c_1 \left(\frac{y}{x} + 1 \right) \\
 (y - x)^2 &= c_1(y + x).
 \end{aligned}$$

9. Letting $y = ux$ we have

$$\begin{aligned}
 -ux dx + (x + \sqrt{u}x)(u dx + x du) &= 0 \\
 (x^2 + x^2\sqrt{u}) du + xu^{3/2} dx &= 0 \\
 \left(u^{-3/2} + \frac{1}{u} \right) du + \frac{dx}{x} &= 0 \\
 -2u^{-1/2} + \ln |u| + \ln |x| &= c \\
 \ln |y/x| + \ln |x| &= 2\sqrt{x/y} + c \\
 y(\ln |y| - c)^2 &= 4x.
 \end{aligned}$$

10. Letting $y = ux$ we have

$$\begin{aligned}
 (ux + \sqrt{x^2 - (ux)^2}) dx - x(udx + xdu) du &= 0 \\
 \sqrt{x^2 - u^2x^2} dx - x^2 du &= 0 \\
 x\sqrt{1 - u^2} dx - x^2 du &= 0, \quad (x > 0) \\
 \frac{dx}{x} - \frac{du}{\sqrt{1 - u^2}} &= 0 \\
 \ln x - \sin^{-1} u &= c \\
 \sin^{-1} u &= \ln x + c_1
 \end{aligned}$$

2.5 Solutions by Substitutions

$$\begin{aligned}\sin^{-1} \frac{y}{x} &= \ln x + c_2 \\ \frac{y}{x} &= \sin(\ln x + c_2) \\ y &= x \sin(\ln x + c_2).\end{aligned}$$

See Problem 33 in this section for an analysis of the solution.

11. Letting $y = ux$ we have

$$\begin{aligned}(x^3 - u^3 x^3) dx + u^2 x^3 (u dx + x du) &= 0 \\ dx + u^2 x du &= 0 \\ \frac{dx}{x} + u^2 du &= 0 \\ \ln |x| + \frac{1}{3} u^3 &= c \\ 3x^3 \ln |x| + y^3 &= c_1 x^3.\end{aligned}$$

Using $y(1) = 2$ we find $c_1 = 8$. The solution of the initial-value problem is $3x^3 \ln |x| + y^3 = 8x^3$.

12. Letting $y = ux$ we have

$$\begin{aligned}(x^2 + 2u^2 x^2) dx - ux^2 (u dx + x du) &= 0 \\ x^2 (1 + u^2) dx - ux^3 du &= 0 \\ \frac{dx}{x} - \frac{u du}{1 + u^2} &= 0 \\ \ln |x| - \frac{1}{2} \ln(1 + u^2) &= c \\ \frac{x^2}{1 + u^2} &= c_1 \\ x^4 &= c_1 (x^2 + y^2).\end{aligned}$$

Using $y(-1) = 1$ we find $c_1 = 1/2$. The solution of the initial-value problem is $2x^4 = y^2 + x^2$.

13. Letting $y = ux$ we have

$$\begin{aligned}(x + uxe^u) dx - xe^u (u dx + x du) &= 0 \\ dx - xe^u du &= 0 \\ \frac{dx}{x} - e^u du &= 0 \\ \ln |x| - e^u &= c \\ \ln |x| - e^{y/x} &= c.\end{aligned}$$

Using $y(1) = 0$ we find $c = -1$. The solution of the initial-value problem is $\ln |x| = e^{y/x} - 1$.

14. Letting $x = vy$ we have

$$\begin{aligned}y(v dy + y dv) + vy(\ln vy - \ln y - 1) dy &= 0 \\ y dv + v \ln v dy &= 0 \\ \frac{dv}{v \ln v} + \frac{dy}{y} &= 0 \\ \ln |\ln |v|| + \ln |y| &= c \\ y \ln \left| \frac{x}{y} \right| &= c_1.\end{aligned}$$

2.5 Solutions by Substitutions

Using $y(1) = e$ we find $c_1 = -e$. The solution of the initial-value problem is $y \ln \left| \frac{x}{y} \right| = -e$.

15. From $y' + \frac{1}{x}y = \frac{1}{x}y^{-2}$ and $w = y^3$ we obtain $\frac{dw}{dx} + \frac{3}{x}w = \frac{3}{x}$. An integrating factor is x^3 so that $x^3w = x^3 + c$ or $y^3 = 1 + cx^{-3}$.
16. From $y' - y = e^xy^2$ and $w = y^{-1}$ we obtain $\frac{dw}{dx} + w = -e^x$. An integrating factor is e^x so that $e^xw = -\frac{1}{2}e^{2x} + c$ or $y^{-1} = -\frac{1}{2}e^x + ce^{-x}$.
17. From $y' + y = xy^4$ and $w = y^{-3}$ we obtain $\frac{dw}{dx} - 3w = -3x$. An integrating factor is e^{-3x} so that $e^{-3x}w = xe^{-3x} + \frac{1}{3}e^{-3x} + c$ or $y^{-3} = x + \frac{1}{3} + ce^{3x}$.
18. From $y' - \left(1 + \frac{1}{x}\right)y = y^2$ and $w = y^{-1}$ we obtain $\frac{dw}{dx} + \left(1 + \frac{1}{x}\right)w = -1$. An integrating factor is xe^x so that $xe^xw = -xe^x + e^x + c$ or $y^{-1} = -1 + \frac{1}{x} + \frac{c}{x}e^{-x}$.
19. From $y' - \frac{1}{t}y = -\frac{1}{t^2}y^2$ and $w = y^{-1}$ we obtain $\frac{dw}{dt} + \frac{1}{t}w = \frac{1}{t^2}$. An integrating factor is t so that $tw = \ln t + c$ or $y^{-1} = \frac{1}{t} \ln t + \frac{c}{t}$. Writing this in the form $\frac{t}{y} = \ln t + c$, we see that the solution can also be expressed in the form $e^{t/y} = c_1t$.
20. From $y' + \frac{2}{3(1+t^2)}y = \frac{2t}{3(1+t^2)}y^4$ and $w = y^{-3}$ we obtain $\frac{dw}{dt} - \frac{2t}{1+t^2}w = \frac{-2t}{1+t^2}$. An integrating factor is $\frac{1}{1+t^2}$ so that $\frac{w}{1+t^2} = \frac{1}{1+t^2} + c$ or $y^{-3} = 1 + c(1+t^2)$.
21. From $y' - \frac{2}{x}y = \frac{3}{x^2}y^4$ and $w = y^{-3}$ we obtain $\frac{dw}{dx} + \frac{6}{x}w = -\frac{9}{x^2}$. An integrating factor is x^6 so that $x^6w = -\frac{9}{5}x^5 + c$ or $y^{-3} = -\frac{9}{5}x^{-1} + cx^{-6}$. If $y(1) = \frac{1}{2}$ then $c = \frac{49}{5}$ and $y^{-3} = -\frac{9}{5}x^{-1} + \frac{49}{5}x^{-6}$.
22. From $y' + y = y^{-1/2}$ and $w = y^{3/2}$ we obtain $\frac{dw}{dx} + \frac{3}{2}w = \frac{3}{2}$. An integrating factor is $e^{3x/2}$ so that $e^{3x/2}w = e^{3x/2} + c$ or $y^{3/2} = 1 + ce^{-3x/2}$. If $y(0) = 4$ then $c = 7$ and $y^{3/2} = 1 + 7e^{-3x/2}$.
23. Let $u = x + y + 1$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = u^2$ or $\frac{1}{1+u^2} du = dx$. Thus $\tan^{-1} u = x + c$ or $u = \tan(x + c)$, and $x + y + 1 = \tan(x + c)$ or $y = \tan(x + c) - x - 1$.
24. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \frac{1-u}{u}$ or $u du = dx$. Thus $\frac{1}{2}u^2 = x + c$ or $u^2 = 2x + c_1$, and $(x + y)^2 = 2x + c_1$.
25. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \tan^2 u$ or $\cos^2 u du = dx$. Thus $\frac{1}{2}u + \frac{1}{4}\sin 2u = x + c$ or $2u + \sin 2u = 4x + c_1$, and $2(x + y) + \sin 2(x + y) = 4x + c_1$ or $2y + \sin 2(x + y) = 2x + c_1$.
26. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \sin u$ or $\frac{1}{1 + \sin u} du = dx$. Multiplying by $(1 - \sin u)/(1 - \sin u)$ we have $\frac{1 - \sin u}{\cos^2 u} du = dx$ or $(\sec^2 u - \sec u \tan u) du = dx$. Thus $\tan u - \sec u = x + c$ or $\tan(x + y) - \sec(x + y) = x + c$.

2.5 Solutions by Substitutions

27. Let $u = y - 2x + 3$ so that $du/dx = dy/dx - 2$. Then $\frac{du}{dx} + 2 = 2 + \sqrt{u}$ or $\frac{1}{\sqrt{u}} du = dx$. Thus $2\sqrt{u} = x + c$ and $2\sqrt{y - 2x + 3} = x + c$.

28. Let $u = y - x + 5$ so that $du/dx = dy/dx - 1$. Then $\frac{du}{dx} + 1 = 1 + e^u$ or $e^{-u} du = dx$. Thus $-e^{-u} = x + c$ and $-e^{y-x+5} = x + c$.

29. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \cos u$ and $\frac{1}{1 + \cos u} du = dx$. Now

$$\frac{1}{1 + \cos u} = \frac{1 - \cos u}{1 - \cos^2 u} = \frac{1 - \cos u}{\sin^2 u} = \csc^2 u - \csc u \cot u$$

so we have $\int (\csc^2 u - \csc u \cot u) du = \int dx$ and $-\cot u + \csc u = x + c$. Thus $-\cot(x + y) + \csc(x + y) = x + c$. Setting $x = 0$ and $y = \pi/4$ we obtain $c = \sqrt{2} - 1$. The solution is

$$\csc(x + y) - \cot(x + y) = x + \sqrt{2} - 1.$$

30. Let $u = 3x + 2y$ so that $du/dx = 3 + 2 dy/dx$. Then $\frac{du}{dx} = 3 + \frac{2u}{u+2} = \frac{5u+6}{u+2}$ and $\frac{u+2}{5u+6} du = dx$. Now by long division

$$\frac{u+2}{5u+6} = \frac{1}{5} + \frac{4}{25u+30}$$

so we have

$$\int \left(\frac{1}{5} + \frac{4}{25u+30} \right) du = dx$$

and $\frac{1}{5}u + \frac{4}{25} \ln |25u + 30| = x + c$. Thus

$$\frac{1}{5}(3x + 2y) + \frac{4}{25} \ln |75x + 50y + 30| = x + c.$$

Setting $x = -1$ and $y = -1$ we obtain $c = \frac{4}{25} \ln 95$. The solution is

$$\frac{1}{5}(3x + 2y) + \frac{4}{25} \ln |75x + 50y + 30| = x + \frac{4}{25} \ln 95$$

or

$$5y - 5x + 2 \ln |75x + 50y + 30| = 2 \ln 95.$$

31. We write the differential equation $M(x, y)dx + N(x, y)dy = 0$ as $dy/dx = f(x, y)$ where

$$f(x, y) = -\frac{M(x, y)}{N(x, y)}.$$

The function $f(x, y)$ must necessarily be homogeneous of degree 0 when M and N are homogeneous of degree α . Since M is homogeneous of degree α , $M(tx, ty) = t^\alpha M(x, y)$, and letting $t = 1/x$ we have

$$M(1, y/x) = \frac{1}{x^\alpha} M(x, y) \quad \text{or} \quad M(x, y) = x^\alpha M(1, y/x).$$

Thus

$$\frac{dy}{dx} = f(x, y) = -\frac{x^\alpha M(1, y/x)}{x^\alpha N(1, y/x)} = -\frac{M(1, y/x)}{N(1, y/x)} = F\left(\frac{y}{x}\right).$$

32. Rewrite $(5x^2 - 2y^2)dx - xy dy = 0$ as

$$xy \frac{dy}{dx} = 5x^2 - 2y^2$$

and divide by xy , so that

$$\frac{dy}{dx} = 5 \frac{x}{y} - 2 \frac{y}{x}.$$

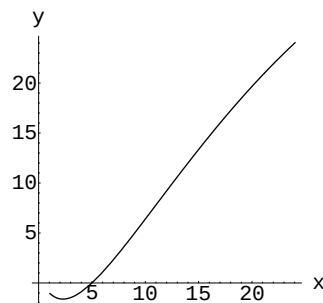
We then identify

$$F\left(\frac{y}{x}\right) = 5\left(\frac{y}{x}\right)^{-1} - 2\left(\frac{y}{x}\right).$$

- 33. (a)** By inspection $y = x$ and $y = -x$ are solutions of the differential equation and not members of the family $y = x \sin(\ln x + c_2)$.

- (b)** Letting $x = 5$ and $y = 0$ in $\sin^{-1}(y/x) = \ln x + c_2$ we get $\sin^{-1} 0 = \ln 5 + c$ or $c = -\ln 5$. Then $\sin^{-1}(y/x) = \ln x - \ln 5 = \ln(x/5)$. Because the range of the arcsine function is $[-\pi/2, \pi/2]$ we must have

$$\begin{aligned} -\frac{\pi}{2} &\leq \ln \frac{x}{5} \leq \frac{\pi}{2} \\ e^{-\pi/2} &\leq \frac{x}{5} \leq e^{\pi/2} \\ 5e^{-\pi/2} &\leq x \leq 5e^{\pi/2}. \end{aligned}$$



The interval of definition of the solution is approximately $[1.04, 24.05]$.

- 34.** As $x \rightarrow -\infty$, $e^{6x} \rightarrow 0$ and $y \rightarrow 2x + 3$. Now write $(1 + ce^{6x})/(1 - ce^{6x})$ as $(e^{-6x} + c)/(e^{-6x} - c)$. Then, as $x \rightarrow \infty$, $e^{-6x} \rightarrow 0$ and $y \rightarrow 2x - 3$.

- 35. (a)** The substitutions $y = y_1 + u$ and

$$\frac{dy}{dx} = \frac{dy_1}{dx} + \frac{du}{dx}$$

lead to

$$\begin{aligned} \frac{dy_1}{dx} + \frac{du}{dx} &= P + Q(y_1 + u) + R(y_1 + u)^2 \\ &= P + Qy_1 + Ry_1^2 + Qu + 2y_1Ru + Ru^2 \end{aligned}$$

or

$$\frac{du}{dx} - (Q + 2y_1R)u = Ru^2.$$

This is a Bernoulli equation with $n = 2$ which can be reduced to the linear equation

$$\frac{dw}{dx} + (Q + 2y_1R)w = -R$$

by the substitution $w = u^{-1}$.

- (b)** Identify $P(x) = -4/x^2$, $Q(x) = -1/x$, and $R(x) = 1$. Then $\frac{dw}{dx} + \left(-\frac{1}{x} + \frac{4}{x}\right)w = -1$. An integrating factor is x^3 so that $x^3w = -\frac{1}{4}x^4 + c$ or $u = \left[-\frac{1}{4}x + cx^{-3}\right]^{-1}$. Thus, $y = \frac{2}{x} + u$.

- 36.** Write the differential equation in the form $x(y'/y) = \ln x + \ln y$ and let $u = \ln y$. Then $du/dx = y'/y$ and the differential equation becomes $x(du/dx) = \ln x + u$ or $du/dx - u/x = (\ln x)/x$, which is first-order and linear. An integrating factor is $e^{-\int dx/x} = 1/x$, so that (using integration by parts)

$$\frac{d}{dx} \left[\frac{1}{x} u \right] = \frac{\ln x}{x^2} \quad \text{and} \quad \frac{u}{x} = -\frac{1}{x} - \frac{\ln x}{x} + c.$$

The solution is

$$\ln y = -1 - \ln x + cx \quad \text{or} \quad y = \frac{e^{cx-1}}{x}.$$

- 37.** Write the differential equation as

$$\frac{dv}{dx} + \frac{1}{x}v = 32v^{-1},$$

2.5 Solutions by Substitutions

and let $u = v^2$ or $v = u^{1/2}$. Then

$$\frac{dv}{dx} = \frac{1}{2}u^{-1/2} \frac{du}{dx},$$

and substituting into the differential equation, we have

$$\frac{1}{2}u^{-1/2} \frac{du}{dx} + \frac{1}{x}u^{1/2} = 32u^{-1/2} \quad \text{or} \quad \frac{du}{dx} + \frac{2}{x}u = 64.$$

The latter differential equation is linear with integrating factor $e^{\int (2/x)dx} = x^2$, so

$$\frac{d}{dx}[x^2u] = 64x^2$$

and

$$x^2u = \frac{64}{3}x^3 + c \quad \text{or} \quad v^2 = \frac{64}{3}x + \frac{c}{x^2}.$$

38. Write the differential equation as $dP/dt - aP = -bP^2$ and let $u = P^{-1}$ or $P = u^{-1}$. Then

$$\frac{dp}{dt} = -u^{-2} \frac{du}{dt},$$

and substituting into the differential equation, we have

$$-u^{-2} \frac{du}{dt} - au^{-1} = -bu^{-2} \quad \text{or} \quad \frac{du}{dt} + au = b.$$

The latter differential equation is linear with integrating factor $e^{\int a dt} = e^{at}$, so

$$\frac{d}{dt}[e^{at}u] = be^{at}$$

and

$$\begin{aligned} e^{at}u &= \frac{b}{a}e^{at} + c \\ e^{at}P^{-1} &= \frac{b}{a}e^{at} + c \\ P^{-1} &= \frac{b}{a} + ce^{-at} \\ P &= \frac{1}{b/a + ce^{-at}} = \frac{a}{b + c_1e^{-at}}. \end{aligned}$$

EXERCISES 2.6

A Numerical Method

1. We identify $f(x, y) = 2x - 3y + 1$. Then, for $h = 0.1$,

$$y_{n+1} = y_n + 0.1(2x_n - 3y_n + 1) = 0.2x_n + 0.7y_n + 0.1,$$

and

$$y(1.1) \approx y_1 = 0.2(1) + 0.7(5) + 0.1 = 3.8$$

$$y(1.2) \approx y_2 = 0.2(1.1) + 0.7(3.8) + 0.1 = 2.98.$$

For $h = 0.05$,

$$y_{n+1} = y_n + 0.05(2x_n - 3y_n + 1) = 0.1x_n + 0.85y_n + 0.1,$$

and

$$\begin{aligned}y(1.05) &\approx y_1 = 0.1(1) + 0.85(5) + 0.1 = 4.4 \\y(1.1) &\approx y_2 = 0.1(1.05) + 0.85(4.4) + 0.1 = 3.895 \\y(1.15) &\approx y_3 = 0.1(1.1) + 0.85(3.895) + 0.1 = 3.47075 \\y(1.2) &\approx y_4 = 0.1(1.15) + 0.85(3.47075) + 0.1 = 3.11514.\end{aligned}$$

2. We identify $f(x, y) = x + y^2$. Then, for $h = 0.1$,

$$y_{n+1} = y_n + 0.1(x_n + y_n^2) = 0.1x_n + y_n + 0.1y_n^2,$$

and

$$\begin{aligned}y(0.1) &\approx y_1 = 0.1(0) + 0 + 0.1(0)^2 = 0 \\y(0.2) &\approx y_2 = 0.1(0.1) + 0 + 0.1(0)^2 = 0.01.\end{aligned}$$

For $h = 0.05$,

$$y_{n+1} = y_n + 0.05(x_n + y_n^2) = 0.05x_n + y_n + 0.05y_n^2,$$

and

$$\begin{aligned}y(0.05) &\approx y_1 = 0.05(0) + 0 + 0.05(0)^2 = 0 \\y(0.1) &\approx y_2 = 0.05(0.05) + 0 + 0.05(0)^2 = 0.0025 \\y(0.15) &\approx y_3 = 0.05(0.1) + 0.0025 + 0.05(0.0025)^2 = 0.0075 \\y(0.2) &\approx y_4 = 0.05(0.15) + 0.0075 + 0.05(0.0075)^2 = 0.0150.\end{aligned}$$

3. Separating variables and integrating, we have

$$\frac{dy}{y} = dx \quad \text{and} \quad \ln |y| = x + c.$$

Thus $y = c_1 e^x$ and, using $y(0) = 1$, we find $c = 1$, so $y = e^x$ is the solution of the initial-value problem.

$h=0.1$

x_n	y_n	Actual Value	Abs. Error	% Rel. Error
0.00	1.0000	1.0000	0.0000	0.00
0.10	1.1000	1.1052	0.0052	0.47
0.20	1.2100	1.2214	0.0114	0.93
0.30	1.3310	1.3499	0.0189	1.40
0.40	1.4641	1.4918	0.0277	1.86
0.50	1.6105	1.6487	0.0382	2.32
0.60	1.7716	1.8221	0.0506	2.77
0.70	1.9487	2.0138	0.0650	3.23
0.80	2.1436	2.2255	0.0820	3.68
0.90	2.3579	2.4596	0.1017	4.13
1.00	2.5937	2.7183	0.1245	4.58

$h=0.05$

x_n	y_n	Actual Value	Abs. Error	% Rel. Error
0.00	1.0000	1.0000	0.0000	0.00
0.05	1.0500	1.0513	0.0013	0.12
0.10	1.1025	1.1052	0.0027	0.24
0.15	1.1576	1.1618	0.0042	0.36
0.20	1.2155	1.2214	0.0059	0.48
0.25	1.2763	1.2840	0.0077	0.60
0.30	1.3401	1.3499	0.0098	0.72
0.35	1.4071	1.4191	0.0120	0.84
0.40	1.4775	1.4918	0.0144	0.96
0.45	1.5513	1.5683	0.0170	1.08
0.50	1.6289	1.6487	0.0198	1.20
0.55	1.7103	1.7333	0.0229	1.32
0.60	1.7959	1.8221	0.0263	1.44
0.65	1.8856	1.9155	0.0299	1.56
0.70	1.9799	2.0138	0.0338	1.68
0.75	2.0789	2.1170	0.0381	1.80
0.80	2.1829	2.2255	0.0427	1.92
0.85	2.2920	2.3396	0.0476	2.04
0.90	2.4066	2.4596	0.0530	2.15
0.95	2.5270	2.5857	0.0588	2.27
1.00	2.6533	2.7183	0.0650	2.39

2.6 A Numerical Method

4. Separating variables and integrating, we have

$$\frac{dy}{y} = 2x dx \quad \text{and} \quad \ln|y| = x^2 + c.$$

Thus $y = c_1 e^{x^2}$ and, using $y(1) = 1$, we find $c = e^{-1}$, so $y = e^{x^2-1}$ is the solution of the initial-value problem.

$h=0.1$

x_n	y_n	Actual Value	Abs. Error	% Rel. Error
1.00	1.0000	1.0000	0.0000	0.00
1.10	1.2000	1.2337	0.0337	2.73
1.20	1.4640	1.5527	0.0887	5.71
1.30	1.8154	1.9937	0.1784	8.95
1.40	2.2874	2.6117	0.3243	12.42
1.50	2.9278	3.4903	0.5625	16.12

$h=0.05$

x_n	y_n	Actual Value	Abs. Error	% Rel. Error
1.00	1.0000	1.0000	0.0000	0.00
1.05	1.1000	1.1079	0.0079	0.72
1.10	1.2155	1.2337	0.0182	1.47
1.15	1.3492	1.3806	0.0314	2.27
1.20	1.5044	1.5527	0.0483	3.11
1.25	1.6849	1.7551	0.0702	4.00
1.30	1.8955	1.9937	0.0982	4.93
1.35	2.1419	2.2762	0.1343	5.90
1.40	2.4311	2.6117	0.1806	6.92
1.45	2.7714	3.0117	0.2403	7.98
1.50	3.1733	3.4903	0.3171	9.08

5. $h=0.1$

x_n	y_n
0.00	0.0000
0.10	0.1000
0.20	0.1905
0.30	0.2731
0.40	0.3492
0.50	0.4198

$h=0.05$

x_n	y_n
0.00	0.0000
0.05	0.0500
0.10	0.0976
0.15	0.1429
0.20	0.1863
0.25	0.2278
0.30	0.2676
0.35	0.3058
0.40	0.3427
0.45	0.3782
0.50	0.4124

6. $h=0.1$

x_n	y_n
0.00	1.0000
0.10	1.1000
0.20	1.2220
0.30	1.3753
0.40	1.5735
0.50	1.8371

$h=0.05$

x_n	y_n
0.00	1.0000
0.05	1.0500
0.10	1.1053
0.15	1.1668
0.20	1.2360
0.25	1.3144
0.30	1.4039
0.35	1.5070
0.40	1.6267
0.45	1.7670
0.50	1.9332

7. $h=0.1$

x_n	y_n
0.00	0.5000
0.10	0.5250
0.20	0.5431
0.30	0.5548
0.40	0.5613
0.50	0.5639

$h=0.05$

x_n	y_n
0.00	0.5000
0.05	0.5125
0.10	0.5232
0.15	0.5322
0.20	0.5395
0.25	0.5452
0.30	0.5496
0.35	0.5527
0.40	0.5547
0.45	0.5559
0.50	0.5565

8. $h=0.1$

x_n	y_n
0.00	1.0000
0.10	1.1000
0.20	1.2159
0.30	1.3505
0.40	1.5072
0.50	1.6902

$h=0.05$

x_n	y_n
0.00	1.0000
0.05	1.0500
0.10	1.1039
0.15	1.1619
0.20	1.2245
0.25	1.2921
0.30	1.3651
0.35	1.4440
0.40	1.5293
0.45	1.6217
0.50	1.7219

9. $h=0.1$

x_n	y_n
1.00	1.0000
1.10	1.0000
1.20	1.0191
1.30	1.0588
1.40	1.1231
1.50	1.2194

$h=0.05$

x_n	y_n
1.00	1.0000
1.05	1.0000
1.10	1.0049
1.15	1.0147
1.20	1.0298
1.25	1.0506
1.30	1.0775
1.35	1.1115
1.40	1.1538
1.45	1.2057
1.50	1.2696

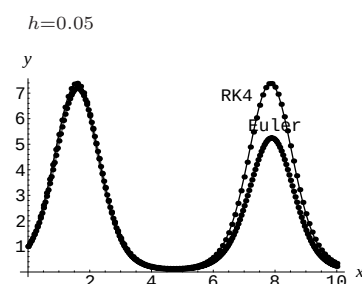
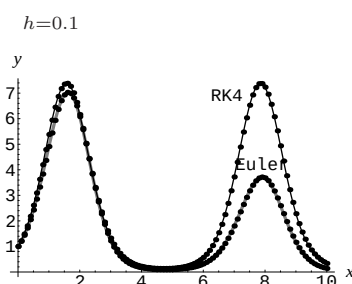
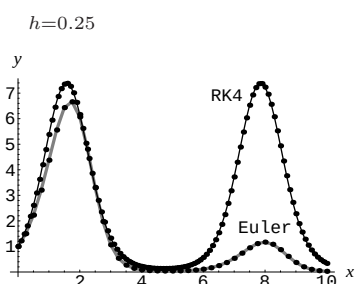
10. $h=0.1$

x_n	y_n
0.00	0.5000
0.10	0.5250
0.20	0.5499
0.30	0.5747
0.40	0.5991
0.50	0.6231

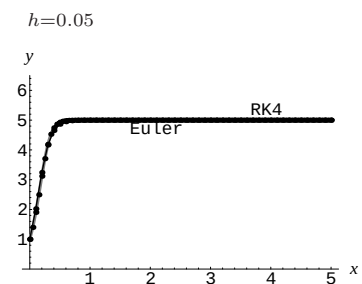
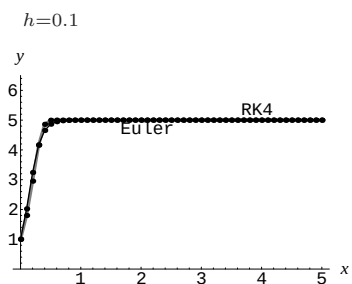
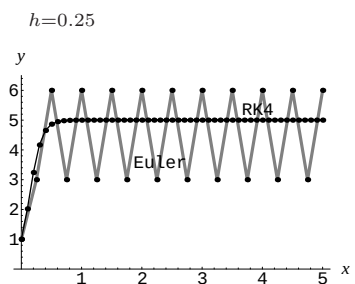
$h=0.05$

x_n	y_n
0.00	0.5000
0.05	0.5125
0.10	0.5250
0.15	0.5375
0.20	0.5499
0.25	0.5623
0.30	0.5746
0.35	0.5868
0.40	0.5989
0.45	0.6109
0.50	0.6228

11. Tables of values were computed using the Euler and RK4 methods. The resulting points were plotted and joined using **ListPlot** in *Mathematica*.

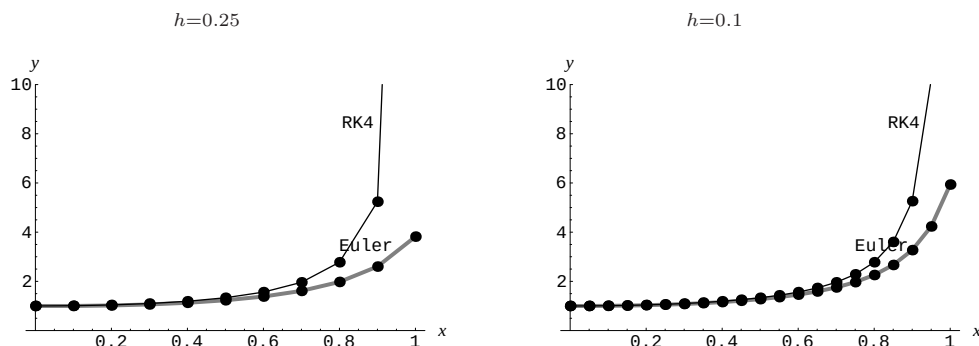


12. See the comments in Problem 11 above.



13. Using separation of variables we find that the solution of the differential equation is $y = 1/(1 - x^2)$, which is undefined at $x = 1$, where the graph has a vertical asymptote. Because the actual solution of the differential equation becomes unbounded as x approaches 1, very small changes in the inputs x will result in large changes in the corresponding outputs y . This can be expected to have a serious effect on numerical procedures. The graphs below were obtained as described above in Problem 11.

2.6 A Numerical Method



EXERCISES 2.7

Linear Models

- Let $P = P(t)$ be the population at time t , and P_0 the initial population. From $dP/dt = kP$ we obtain $P = P_0 e^{kt}$. Using $P(5) = 2P_0$ we find $k = \frac{1}{5} \ln 2$ and $P = P_0 e^{(\ln 2)t/5}$. Setting $P(t) = 3P_0$ we have $3 = e^{(\ln 2)t/5}$, so

$$\ln 3 = \frac{(\ln 2)t}{5} \quad \text{and} \quad t = \frac{5 \ln 3}{\ln 2} \approx 7.9 \text{ years.}$$

Setting $P(t) = 4P_0$ we have $4 = e^{(\ln 2)t/5}$, so

$$\ln 4 = \frac{(\ln 2)t}{5} \quad \text{and} \quad t \approx 10 \text{ years.}$$

- From Problem 1 the growth constant is $k = \frac{1}{5} \ln 2$. Then $P = P_0 e^{(1/5)(\ln 2)t}$ and $10,000 = P_0 e^{(3/5) \ln 2}$. Solving for P_0 we get $P_0 = 10,000 e^{-(3/5) \ln 2} = 6,597.5$. Now

$$P(10) = P_0 e^{(1/5)(\ln 2)(10)} = 6,597.5 e^{2 \ln 2} = 4P_0 = 26,390.$$

The rate at which the population is growing is

$$P'(10) = kP(10) = \frac{1}{5} (\ln 2) 26,390 = 3658 \text{ persons/year.}$$

- Let $P = P(t)$ be the population at time t . Then $dP/dt = kP$ and $P = ce^{kt}$. From $P(0) = c = 500$ we see that $P = 500e^{kt}$. Since 15% of 500 is 75, we have $P(10) = 500e^{10k} = 575$. Solving for k , we get $k = \frac{1}{10} \ln \frac{575}{500} = \frac{1}{10} \ln 1.15$. When $t = 30$,

$$P(30) = 500e^{(1/10)(\ln 1.15)30} = 500e^{3 \ln 1.15} = 760 \text{ years}$$

and

$$P'(30) = kP(30) = \frac{1}{10} (\ln 1.15) 760 = 10.62 \text{ persons/year.}$$

- Let $P = P(t)$ be bacteria population at time t and P_0 the initial number. From $dP/dt = kP$ we obtain $P = P_0 e^{kt}$. Using $P(3) = 400$ and $P(10) = 2000$ we find $400 = P_0 e^{3k}$ or $e^k = (400/P_0)^{1/3}$. From $P(10) = 2000$ we then have $2000 = P_0 e^{10k} = P_0 (400/P_0)^{10/3}$, so

$$\frac{2000}{400^{10/3}} = P_0^{-7/3} \quad \text{and} \quad P_0 = \left(\frac{2000}{400^{10/3}} \right)^{-3/7} \approx 201.$$

5. Let $A = A(t)$ be the amount of lead present at time t . From $dA/dt = kA$ and $A(0) = 1$ we obtain $A = e^{kt}$. Using $A(3.3) = 1/2$ we find $k = \frac{1}{3.3} \ln(1/2)$. When 90% of the lead has decayed, 0.1 grams will remain. Setting $A(t) = 0.1$ we have $e^{t(1/3.3) \ln(1/2)} = 0.1$, so

$$\frac{t}{3.3} \ln \frac{1}{2} = \ln 0.1 \quad \text{and} \quad t = \frac{3.3 \ln 0.1}{\ln(1/2)} \approx 10.96 \text{ hours.}$$

6. Let $A = A(t)$ be the amount present at time t . From $dA/dt = kA$ and $A(0) = 100$ we obtain $A = 100e^{kt}$. Using $A(6) = 97$ we find $k = \frac{1}{6} \ln 0.97$. Then $A(24) = 100e^{(1/6)(\ln 0.97)24} = 100(0.97)^4 \approx 88.5$ mg.
7. Setting $A(t) = 50$ in Problem 6 we obtain $50 = 100e^{kt}$, so

$$kt = \ln \frac{1}{2} \quad \text{and} \quad t = \frac{\ln(1/2)}{(1/6) \ln 0.97} \approx 136.5 \text{ hours.}$$

8. (a) The solution of $dA/dt = kA$ is $A(t) = A_0 e^{kt}$. Letting $A = \frac{1}{2}A_0$ and solving for t we obtain the half-life $T = -(\ln 2)/k$.

- (b) Since $k = -(\ln 2)/T$ we have

$$A(t) = A_0 e^{-(\ln 2)t/T} = A_0 2^{-t/T}.$$

- (c) Writing $\frac{1}{8}A_0 = A_0 2^{-t/T}$ as $2^{-3} = 2^{-t/T}$ and solving for t we get $t = 3T$. Thus, an initial amount A_0 will decay to $\frac{1}{8}A_0$ in three half-lives.

9. Let $I = I(t)$ be the intensity, t the thickness, and $I(0) = I_0$. If $dI/dt = kI$ and $I(3) = 0.25I_0$, then $I = I_0 e^{kt}$, $k = \frac{1}{3} \ln 0.25$, and $I(15) = 0.00098I_0$.

10. From $dS/dt = rS$ we obtain $S = S_0 e^{rt}$ where $S(0) = S_0$.

- (a) If $S_0 = \$5000$ and $r = 5.75\%$ then $S(5) = \$6665.45$.

- (b) If $S(t) = \$10,000$ then $t = 12$ years.

- (c) $S \approx \$6651.82$

11. Assume that $A = A_0 e^{kt}$ and $k = -0.00012378$. If $A(t) = 0.145A_0$ then $t \approx 15,600$ years.

12. From Example 3 in the text, the amount of carbon present at time t is $A(t) = A_0 e^{-0.00012378t}$. Letting $t = 660$ and solving for A_0 we have $A(660) = A_0 e^{-0.0001237(660)} = 0.921553A_0$. Thus, approximately 92% of the original amount of C-14 remained in the cloth as of 1988.

13. Assume that $dT/dt = k(T - 10)$ so that $T = 10 + ce^{kt}$. If $T(0) = 70^\circ$ and $T(1/2) = 50^\circ$ then $c = 60$ and $k = 2 \ln(2/3)$ so that $T(1) = 36.67^\circ$. If $T(t) = 15^\circ$ then $t = 3.06$ minutes.

14. Assume that $dT/dt = k(T - 5)$ so that $T = 5 + ce^{kt}$. If $T(1) = 55^\circ$ and $T(5) = 30^\circ$ then $k = -\frac{1}{4} \ln 2$ and $c = 59.4611$ so that $T(0) = 64.4611^\circ$.

15. Assume that $dT/dt = k(T - 100)$ so that $T = 100 + ce^{kt}$. If $T(0) = 20^\circ$ and $T(1) = 22^\circ$, then $c = -80$ and $k = \ln(39/40)$ so that $T(t) = 90^\circ$, which implies $t = 82.1$ seconds. If $T(t) = 98^\circ$ then $t = 145.7$ seconds.

16. The differential equation for the first container is $dT_1/dt = k_1(T_1 - 0) = k_1 T_1$, whose solution is $T_1(t) = c_1 e^{k_1 t}$. Since $T_1(0) = 100$ (the initial temperature of the metal bar), we have $100 = c_1$ and $T_1(t) = 100e^{k_1 t}$. After 1 minute, $T_1(1) = 100e^{k_1} = 90^\circ\text{C}$, so $k_1 = \ln 0.9$ and $T_1(t) = 100e^{t \ln 0.9}$. After 2 minutes, $T_1(2) = 100e^{2 \ln 0.9} = 100(0.9)^2 = 81^\circ\text{C}$.

The differential equation for the second container is $dT_2/dt = k_2(T_2 - 100)$, whose solution is $T_2(t) = 100 + c_2 e^{k_2 t}$. When the metal bar is immersed in the second container, its initial temperature is $T_2(0) = 81$, so

$$T_2(0) = 100 + c_2 e^{k_2(0)} = 100 + c_2 = 81$$

2.7 Linear Models

and $c_2 = -19$. Thus, $T_2(t) = 100 - 19e^{k_2 t}$. After 1 minute in the second tank, the temperature of the metal bar is 91°C , so

$$T_2(1) = 100 - 19e^{k_2} = 91$$

$$e^{k_2} = \frac{9}{19}$$

$$k_2 = \ln \frac{9}{19}$$

and $T_2(t) = 100 - 19e^{t \ln(9/19)}$. Setting $T_2(t) = 99.9$ we have

$$100 - 19e^{t \ln(9/19)} = 99.9$$

$$e^{t \ln(9/19)} = \frac{0.1}{19}$$

$$t = \frac{\ln(0.1/19)}{\ln(9/19)} \approx 7.02.$$

Thus, from the start of the “double dipping” process, the total time until the bar reaches 99.9°C in the second container is approximately 9.02 minutes.

17. Using separation of variables to solve $dT/dt = k(T - T_m)$ we get $T(t) = T_m + ce^{kt}$. Using $T(0) = 70$ we find $c = 70 - T_m$, so $T(t) = T_m + (70 - T_m)e^{kt}$. Using the given observations, we obtain

$$T\left(\frac{1}{2}\right) = T_m + (70 - T_m)e^{k/2} = 110$$

$$T(1) = T_m + (70 - T_m)e^k = 145.$$

Then, from the first equation, $e^{k/2} = (110 - T_m)/(70 - T_m)$ and

$$e^k = (e^{k/2})^2 = \left(\frac{110 - T_m}{70 - T_m}\right)^2 = \frac{145 - T_m}{70 - T_m}$$

$$\frac{(110 - T_m)^2}{70 - T_m} = 145 - T_m$$

$$12100 - 220T_m + T_m^2 = 10150 - 250T_m + T_m^2$$

$$T_m = 390.$$

The temperature in the oven is 390° .

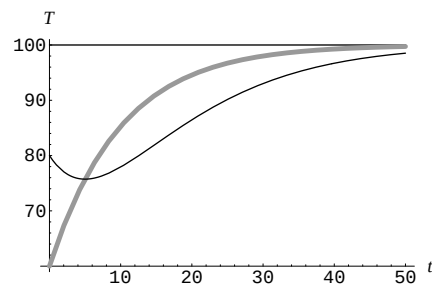
18. (a) The initial temperature of the bath is $T_m(0) = 60^\circ$, so in the short term the temperature of the chemical, which starts at 80° , should decrease or cool. Over time, the temperature of the bath will increase toward 100° since $e^{-0.1t}$ decreases from 1 toward 0 as t increases from 0. Thus, in the long term, the temperature of the chemical should increase or warm toward 100° .
- (b) Adapting the model for Newton’s law of cooling, we have

$$\frac{dT}{dt} = -0.1(T - 100 + 40e^{-0.1t}), \quad T(0) = 80.$$

Writing the differential equation in the form

$$\frac{dT}{dt} + 0.1T = 10 - 4e^{-0.1t}$$

we see that it is linear with integrating factor $e^{\int 0.1 dt} = e^{0.1t}$.



Thus

$$\begin{aligned}\frac{d}{dt}[e^{0.1t}T] &= 10e^{0.1t} - 4 \\ e^{0.1t}T &= 100e^{0.1t} - 4t + c\end{aligned}$$

and

$$T(t) = 100 - 4te^{-0.1t} + ce^{-0.1t}.$$

Now $T(0) = 80$ so $100 + c = 80$, $c = -20$ and

$$T(t) = 100 - 4te^{-0.1t} - 20e^{-0.1t} = 100 - (4t + 20)e^{-0.1t}.$$

The thinner curve verifies the prediction of cooling followed by warming toward 100° . The wider curve shows the temperature T_m of the liquid bath.

19. From $dA/dt = 4 - A/50$ we obtain $A = 200 + ce^{-t/50}$. If $A(0) = 30$ then $c = -170$ and $A = 200 - 170e^{-t/50}$.
20. From $dA/dt = 0 - A/50$ we obtain $A = ce^{-t/50}$. If $A(0) = 30$ then $c = 30$ and $A = 30e^{-t/50}$.
21. From $dA/dt = 10 - A/100$ we obtain $A = 1000 + ce^{-t/100}$. If $A(0) = 0$ then $c = -1000$ and $A(t) = 1000 - 1000e^{-t/100}$.
22. From Problem 21 the number of pounds of salt in the tank at time t is $A(t) = 1000 - 1000e^{-t/100}$. The concentration at time t is $c(t) = A(t)/500 = 2 - 2e^{-t/100}$. Therefore $c(5) = 2 - 2e^{-1/20} = 0.0975$ lb/gal and $\lim_{t \rightarrow \infty} c(t) = 2$. Solving $c(t) = 1 = 2 - 2e^{-t/100}$ for t we obtain $t = 100 \ln 2 \approx 69.3$ min.
23. From

$$\frac{dA}{dt} = 10 - \frac{10A}{500 - (10 - 5)t} = 10 - \frac{2A}{100 - t}$$

we obtain $A = 1000 - 10t + c(100 - t)^2$. If $A(0) = 0$ then $c = -\frac{1}{10}$. The tank is empty in 100 minutes.

24. With $c_{in}(t) = 2 + \sin(t/4)$ lb/gal, the initial-value problem is

$$\frac{dA}{dt} + \frac{1}{100}A = 6 + 3 \sin \frac{t}{4}, \quad A(0) = 50.$$

The differential equation is linear with integrating factor $e^{\int dt/100} = e^{t/100}$, so

$$\begin{aligned}\frac{d}{dt}[e^{t/100}A(t)] &= \left(6 + 3 \sin \frac{t}{4}\right)e^{t/100} \\ e^{t/100}A(t) &= 600e^{t/100} + \frac{150}{313}e^{t/100} \sin \frac{t}{4} - \frac{3750}{313}e^{t/100} \cos \frac{t}{4} + c,\end{aligned}$$

and

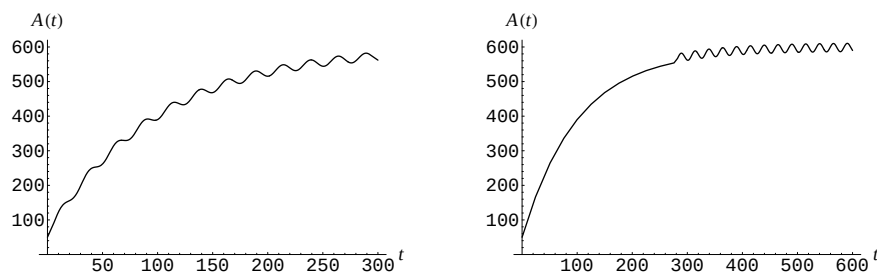
$$A(t) = 600 + \frac{150}{313} \sin \frac{t}{4} - \frac{3750}{313} \cos \frac{t}{4} + ce^{-t/100}.$$

Letting $t = 0$ and $A = 50$ we have $600 - 3750/313 + c = 50$ and $c = -168400/313$. Then

$$A(t) = 600 + \frac{150}{313} \sin \frac{t}{4} - \frac{3750}{313} \cos \frac{t}{4} - \frac{168400}{313} e^{-t/100}.$$

The graphs on $[0, 300]$ and $[0, 600]$ below show the effect of the sine function in the input when compared with the graph in Figure 2.38(a) in the text.

2.7 Linear Models



25. From

$$\frac{dA}{dt} = 3 - \frac{4A}{100 + (6-4)t} = 3 - \frac{2A}{50+t}$$

we obtain $A = 50 + t + c(50 + t)^{-2}$. If $A(0) = 10$ then $c = -100,000$ and $A(30) = 64.38$ pounds.

26. (a) Initially the tank contains 300 gallons of solution. Since brine is pumped in at a rate of 3 gal/min and the mixture is pumped out at a rate of 2 gal/min, the net change is an increase of 1 gal/min. Thus, in 100 minutes the tank will contain its capacity of 400 gallons.

(b) The differential equation describing the amount of salt in the tank is $A'(t) = 6 - 2A/(300 + t)$ with solution

$$A(t) = 600 + 2t - (4.95 \times 10^7)(300 + t)^{-2}, \quad 0 \leq t \leq 100,$$

as noted in the discussion following Example 5 in the text. Thus, the amount of salt in the tank when it overflows is

$$A(100) = 800 - (4.95 \times 10^7)(400)^{-2} = 490.625 \text{ lbs.}$$

(c) When the tank is overflowing the amount of salt in the tank is governed by the differential equation

$$\begin{aligned} \frac{dA}{dt} &= (3 \text{ gal/min})(2 \text{ lb/gal}) - \left(\frac{A}{400} \text{ lb/gal} \right) (3 \text{ gal/min}) \\ &= 6 - \frac{3A}{400}, \quad A(100) = 490.625. \end{aligned}$$

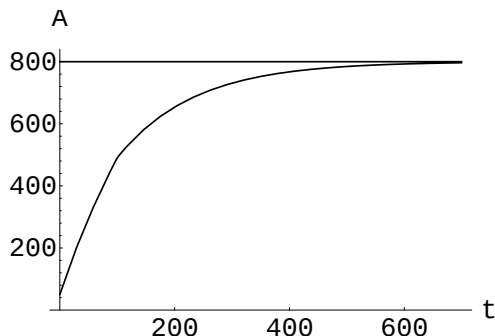
Solving the equation, we obtain $A(t) = 800 + ce^{-3t/400}$. The initial condition yields $c = -654.947$, so that

$$A(t) = 800 - 654.947e^{-3t/400}.$$

When $t = 150$, $A(150) = 587.37$ lbs.

(d) As $t \rightarrow \infty$, the amount of salt is 800 lbs, which is to be expected since $(400 \text{ gal})(2 \text{ lb/gal}) = 800 \text{ lbs}$.

(e)



27. Assume $L di/dt + Ri = E(t)$, $L = 0.1$, $R = 50$, and $E(t) = 50$ so that $i = \frac{3}{5} + ce^{-500t}$. If $i(0) = 0$ then $c = -3/5$ and $\lim_{t \rightarrow \infty} i(t) = 3/5$.

28. Assume $L di/dt + Ri = E(t)$, $E(t) = E_0 \sin \omega t$, and $i(0) = i_0$ so that

$$i = \frac{E_0 R}{L^2 \omega^2 + R^2} \sin \omega t - \frac{E_0 L \omega}{L^2 \omega^2 + R^2} \cos \omega t + c e^{-Rt/L}.$$

Since $i(0) = i_0$ we obtain $c = i_0 + \frac{E_0 L \omega}{L^2 \omega^2 + R^2}$.

29. Assume $R dq/dt + (1/C)q = E(t)$, $R = 200$, $C = 10^{-4}$, and $E(t) = 100$ so that $q = 1/100 + c e^{-50t}$. If $q(0) = 0$ then $c = -1/100$ and $i = \frac{1}{2} e^{-50t}$.
30. Assume $R dq/dt + (1/C)q = E(t)$, $R = 1000$, $C = 5 \times 10^{-6}$, and $E(t) = 200$. Then $q = \frac{1}{1000} + c e^{-200t}$ and $i = -200 c e^{-200t}$. If $i(0) = 0.4$ then $c = -\frac{1}{500}$, $q(0.005) = 0.003$ coulombs, and $i(0.005) = 0.1472$ amps. We have $q \rightarrow \frac{1}{1000}$ as $t \rightarrow \infty$.
31. For $0 \leq t \leq 20$ the differential equation is $20 di/dt + 2i = 120$. An integrating factor is $e^{t/10}$, so $(d/dt)[e^{t/10}i] = 6e^{t/10}$ and $i = 60 + c_1 e^{-t/10}$. If $i(0) = 0$ then $c_1 = -60$ and $i = 60 - 60e^{-t/10}$. For $t > 20$ the differential equation is $20 di/dt + 2i = 0$ and $i = c_2 e^{-t/10}$. At $t = 20$ we want $c_2 e^{-2} = 60 - 60e^{-2}$ so that $c_2 = 60(e^2 - 1)$. Thus

$$i(t) = \begin{cases} 60 - 60e^{-t/10}, & 0 \leq t \leq 20 \\ 60(e^2 - 1)e^{-t/10}, & t > 20. \end{cases}$$

32. Separating variables, we obtain

$$\begin{aligned} \frac{dq}{E_0 - q/C} &= \frac{dt}{k_1 + k_2 t} \\ -C \ln \left| E_0 - \frac{q}{C} \right| &= \frac{1}{k_2} \ln |k_1 + k_2 t| + c_1 \\ \frac{(E_0 - q/C)^{-C}}{(k_1 + k_2 t)^{1/k_2}} &= c_2. \end{aligned}$$

Setting $q(0) = q_0$ we find $c_2 = (E_0 - q_0/C)^{-C} / k_1^{1/k_2}$, so

$$\begin{aligned} \frac{(E_0 - q/C)^{-C}}{(k_1 + k_2 t)^{1/k_2}} &= \frac{(E_0 - q_0/C)^{-C}}{k_1^{1/k_2}} \\ \left(E_0 - \frac{q}{C} \right)^{-C} &= \left(E_0 - \frac{q_0}{C} \right)^{-C} \left(\frac{k_1}{k_1 + k_2 t} \right)^{-1/k_2} \\ E_0 - \frac{q}{C} &= \left(E_0 - \frac{q_0}{C} \right) \left(\frac{k_1}{k_1 + k_2 t} \right)^{1/Ck_2} \\ q &= E_0 C + (q_0 - E_0 C) \left(\frac{k_1}{k_1 + k_2 t} \right)^{1/Ck_2}. \end{aligned}$$

33. (a) From $m dv/dt = mg - kv$ we obtain $v = mg/k + c e^{-kt/m}$. If $v(0) = v_0$ then $c = v_0 - mg/k$ and the solution of the initial-value problem is

$$v(t) = \frac{mg}{k} + \left(v_0 - \frac{mg}{k} \right) e^{-kt/m}.$$

(b) As $t \rightarrow \infty$ the limiting velocity is mg/k .

(c) From $ds/dt = v$ and $s(0) = 0$ we obtain

$$s(t) = \frac{mg}{k} t - \frac{m}{k} \left(v_0 - \frac{mg}{k} \right) e^{-kt/m} + \frac{m}{k} \left(v_0 - \frac{mg}{k} \right).$$

34. (a) Integrating $d^2s/dt^2 = -g$ we get $v(t) = ds/dt = -gt + c$. From $v(0) = 300$ we find $c = 300$, and we are given $g = 32$, so the velocity is $v(t) = -32t + 300$.

2.7 Linear Models

(b) Integrating again and using $s(0) = 0$ we get $s(t) = -16t^2 + 300t$. The maximum height is attained when $v = 0$, that is, at $t_a = 9.375$. The maximum height will be $s(9.375) = 1406.25$ ft.

35. When air resistance is proportional to velocity, the model for the velocity is $m dv/dt = -mg - kv$ (using the fact that the positive direction is upward.) Solving the differential equation using separation of variables we obtain $v(t) = -mg/k + ce^{-kt/m}$. From $v(0) = 300$ we get

$$v(t) = -\frac{mg}{k} + \left(300 + \frac{mg}{k}\right)e^{-kt/m}.$$

Integrating and using $s(0) = 0$ we find

$$s(t) = -\frac{mg}{k}t + \frac{m}{k}\left(300 + \frac{mg}{k}\right)(1 - e^{-kt/m}).$$

Setting $k = 0.0025$, $m = 16/32 = 0.5$, and $g = 32$ we have

$$s(t) = 1,340,000 - 6,400t - 1,340,000e^{-0.005t}$$

and

$$v(t) = -6,400 + 6,700e^{-0.005t}.$$

The maximum height is attained when $v = 0$, that is, at $t_a = 9.162$. The maximum height will be $s(9.162) = 1363.79$ ft, which is less than the maximum height in Problem 34.

36. Assuming that the air resistance is proportional to velocity and the positive direction is downward with $s(0) = 0$, the model for the velocity is $m dv/dt = mg - kv$. Using separation of variables to solve this differential equation, we obtain $v(t) = mg/k + ce^{-kt/m}$. Then, using $v(0) = 0$, we get $v(t) = (mg/k)(1 - e^{-kt/m})$. Letting $k = 0.5$, $m = (125 + 35)/32 = 5$, and $g = 32$, we have $v(t) = 320(1 - e^{-0.1t})$. Integrating, we find $s(t) = 320t + 3200e^{-0.1t} + c_1$. Solving $s(0) = 0$ for c_1 we find $c_1 = -3200$, therefore $s(t) = 320t + 3200e^{-0.1t} - 3200$. At $t = 15$, when the parachute opens, $v(15) = 248.598$ and $s(15) = 2314.02$. At this time the value of k changes to $k = 10$ and the new initial velocity is $v_0 = 248.598$. With the parachute open, the skydiver's velocity is $v_p(t) = mg/k + c_2e^{-kt/m}$, where t is reset to 0 when the parachute opens. Letting $m = 5$, $g = 32$, and $k = 10$, this gives $v_p(t) = 16 + c_2e^{-2t}$. From $v(0) = 248.598$ we find $c_2 = 232.598$, so $v_p(t) = 16 + 232.598e^{-2t}$. Integrating, we get $s_p(t) = 16t - 116.299e^{-2t} + c_3$. Solving $s_p(0) = 0$ for c_3 , we find $c_3 = 116.299$, so $s_p(t) = 16t - 116.299e^{-2t} + 116.299$. Twenty seconds after leaving the plane is five seconds after the parachute opens. The skydiver's velocity at this time is $v_p(5) = 16.0106$ ft/s and she has fallen a total of $s(15) + s_p(5) = 2314.02 + 196.294 = 2510.31$ ft. Her terminal velocity is $\lim_{t \rightarrow \infty} v_p(t) = 16$, so she has very nearly reached her terminal velocity five seconds after the parachute opens. When the parachute opens, the distance to the ground is $15,000 - s(15) = 15,000 - 2,314 = 12,686$ ft. Solving $s_p(t) = 12,686$ we get $t = 785.6$ s = 13.1 min. Thus, it will take her approximately 13.1 minutes to reach the ground after her parachute has opened and a total of $(785.6 + 15)/60 = 13.34$ minutes after she exits the plane.

37. (a) The differential equation is first-order and linear. Letting $b = k/\rho$, the integrating factor is $e^{\int 3b dt/(bt+r_0)} = (r_0 + bt)^3$. Then

$$\frac{d}{dt}[(r_0 + bt)^3 v] = g(r_0 + bt)^3 \quad \text{and} \quad (r_0 + bt)^3 v = \frac{g}{4b}(r_0 + bt)^4 + c.$$

The solution of the differential equation is $v(t) = (g/4b)(r_0 + bt) + c(r_0 + bt)^{-3}$. Using $v(0) = 0$ we find $c = -gr_0^4/4b$, so that

$$v(t) = \frac{g}{4b}(r_0 + bt) - \frac{gr_0^4}{4b(r_0 + bt)^3} = \frac{g\rho}{4k}\left(r_0 + \frac{k}{\rho}t\right) - \frac{g\rho r_0^4}{4k(r_0 + kt/\rho)^3}.$$

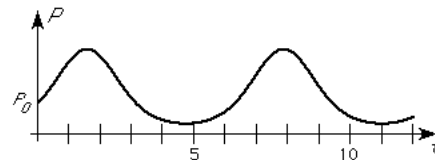
- (b) Integrating $dr/dt = k/\rho$ we get $r = kt/\rho + c$. Using $r(0) = r_0$ we have $c = r_0$, so $r(t) = kt/\rho + r_0$.

- (c) If $r = 0.007$ ft when $t = 10$ s, then solving $r(10) = 0.007$ for k/ρ , we obtain $k/\rho = -0.0003$ and $r(t) = 0.01 - 0.0003t$. Solving $r(t) = 0$ we get $t = 33.3$, so the raindrop will have evaporated completely at 33.3 seconds.

38. Separating variables, we obtain $dP/P = k \cos t \, dt$, so

$$\ln |P| = k \sin t + c \quad \text{and} \quad P = c_1 e^{k \sin t}.$$

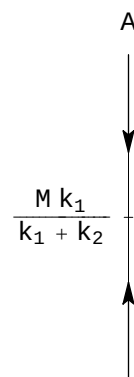
If $P(0) = P_0$, then $c_1 = P_0$ and $P = P_0 e^{k \sin t}$.



39. (a) From $dP/dt = (k_1 - k_2)P$ we obtain $P = P_0 e^{(k_1 - k_2)t}$ where $P_0 = P(0)$.

(b) If $k_1 > k_2$ then $P \rightarrow \infty$ as $t \rightarrow \infty$. If $k_1 = k_2$ then $P = P_0$ for every t . If $k_1 < k_2$ then $P \rightarrow 0$ as $t \rightarrow \infty$.

40. (a) Solving $k_1(M - A) - k_2A = 0$ for A we find the equilibrium solution $A = k_1M/(k_1 + k_2)$. From the phase portrait we see that $\lim_{t \rightarrow \infty} A(t) = k_1M/(k_1 + k_2)$. Since $k_2 > 0$, the material will never be completely memorized and the larger k_2 is, the less the amount of material will be memorized over time.



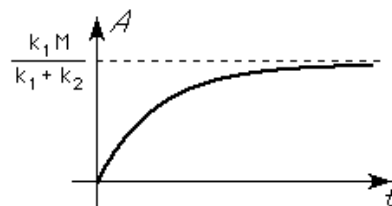
- (b) Write the differential equation in the form $dA/dt + (k_1 + k_2)A = k_1M$.

Then an integrating factor is $e^{(k_1 + k_2)t}$, and

$$\frac{d}{dt} \left[e^{(k_1 + k_2)t} A \right] = k_1 M e^{(k_1 + k_2)t}$$

$$e^{(k_1 + k_2)t} A = \frac{k_1 M}{k_1 + k_2} e^{(k_1 + k_2)t} + c$$

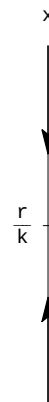
$$A = \frac{k_1 M}{k_1 + k_2} + c e^{-(k_1 + k_2)t}.$$



Using $A(0) = 0$ we find $c = -\frac{k_1 M}{k_1 + k_2}$ and $A = \frac{k_1 M}{k_1 + k_2} (1 - e^{-(k_1 + k_2)t})$. As $t \rightarrow \infty$,

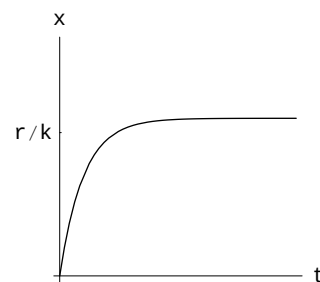
$$A \rightarrow \frac{k_1 M}{k_1 + k_2}.$$

41. (a) Solving $r - kx = 0$ for x we find the equilibrium solution $x = r/k$. When $x < r/k$, $dx/dt > 0$ and when $x > r/k$, $dx/dt < 0$. From the phase portrait we see that $\lim_{t \rightarrow \infty} x(t) = r/k$.



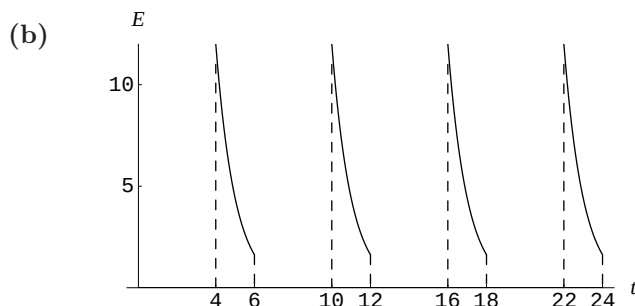
2.7 Linear Models

- (b) From $dx/dt = r - kx$ and $x(0) = 0$ we obtain $x = r/k - (r/k)e^{-kt}$ so that $x \rightarrow r/k$ as $t \rightarrow \infty$. If $x(T) = r/2k$ then $T = (\ln 2)/k$.



42. The bar removed from the oven has an initial temperature of 300°F and, after being removed from the oven, approaches a temperature of 70°F . The bar taken from the room and placed in the oven has an initial temperature of 70°F and approaches a temperature of 300°F in the oven. Since the two temperature functions are continuous they must intersect at some time, t^* .
43. (a) For $0 \leq t < 4$, $6 \leq t < 10$ and $12 \leq t < 16$, no voltage is applied to the heart and $E(t) = 0$. At the other times, the differential equation is $dE/dt = -E/RC$. Separating variables, integrating, and solving for e , we get $E = ke^{-t/RC}$, subject to $E(4) = E(10) = E(16) = 12$. These initial conditions yield, respectively, $k = 12e^{4/RC}$, $k = 12e^{10/RC}$, $k = 12e^{16/RC}$, and $k = 12e^{22/RC}$. Thus

$$E(t) = \begin{cases} 0, & 0 \leq t < 4, \quad 6 \leq t < 10, \quad 12 \leq t < 16 \\ 12e^{(4-t)/RC}, & 4 \leq t < 6 \\ 12e^{(10-t)/RC}, & 10 \leq t < 12 \\ 12e^{(16-t)/RC}, & 16 \leq t < 18 \\ 12e^{(22-t)/RC}, & 22 \leq t < 24. \end{cases}$$



44. (a) (i) Using Newton's second law of motion, $F = ma = m dv/dt$, the differential equation for the velocity v is

$$m \frac{dv}{dt} = mg \sin \theta \quad \text{or} \quad \frac{dv}{dt} = g \sin \theta,$$

where $mg \sin \theta$, $0 < \theta < \pi/2$, is the component of the weight along the plane in the direction of motion.

(ii) The model now becomes

$$m \frac{dv}{dt} = mg \sin \theta - \mu mg \cos \theta,$$

where $\mu mg \cos \theta$ is the component of the force of sliding friction (which acts perpendicular to the plane) along the plane. The negative sign indicates that this component of force is a retarding force which acts in the direction opposite to that of motion.

(iii) If air resistance is taken to be proportional to the instantaneous velocity of the body, the model becomes

$$m \frac{dv}{dt} = mg \sin \theta - \mu mg \cos \theta - kv,$$

where k is a constant of proportionality.

- (b) (i) With $m = 3$ slugs, the differential equation is

$$3 \frac{dv}{dt} = (96) \cdot \frac{1}{2} \quad \text{or} \quad \frac{dv}{dt} = 16.$$

Integrating the last equation gives $v(t) = 16t + c_1$. Since $v(0) = 0$, we have $c_1 = 0$ and so $v(t) = 16t$.

- (ii) With $m = 3$ slugs, the differential equation is

$$3 \frac{dv}{dt} = (96) \cdot \frac{1}{2} - \frac{\sqrt{3}}{4} \cdot (96) \cdot \frac{\sqrt{3}}{2} \quad \text{or} \quad \frac{dv}{dt} = 4.$$

In this case $v(t) = 4t$.

- (iii) When the retarding force due to air resistance is taken into account, the differential equation for velocity v becomes

$$3 \frac{dv}{dt} = (96) \cdot \frac{1}{2} - \frac{\sqrt{3}}{4} \cdot (96) \cdot \frac{\sqrt{3}}{2} - \frac{1}{4} v \quad \text{or} \quad 3 \frac{dv}{dt} = 12 - \frac{1}{4} v.$$

The last differential equation is linear and has solution $v(t) = 48 + c_1 e^{-t/12}$. Since $v(0) = 0$, we find $c_1 = -48$, so $v(t) = 48 - 48e^{-t/12}$.

45. (a) (i) If $s(t)$ is distance measured down the plane from the highest point, then $ds/dt = v$. Integrating $ds/dt = 16t$ gives $s(t) = 8t^2 + c_2$. Using $s(0) = 0$ then gives $c_2 = 0$. Now the length L of the plane is $L = 50/\sin 30^\circ = 100$ ft. The time it takes the box to slide completely down the plane is the solution of $s(t) = 100$ or $t^2 = 25/2$, so $t \approx 3.54$ s.

- (ii) Integrating $ds/dt = 4t$ gives $s(t) = 2t^2 + c_2$. Using $s(0) = 0$ gives $c_2 = 0$, so $s(t) = 2t^2$ and the solution of $s(t) = 100$ is now $t \approx 7.07$ s.

- (iii) Integrating $ds/dt = 48 - 48e^{-t/12}$ and using $s(0) = 0$ to determine the constant of integration, we obtain $s(t) = 48t + 576e^{-t/12} - 576$. With the aid of a CAS we find that the solution of $s(t) = 100$, or

$$100 = 48t + 576e^{-t/12} - 576 \quad \text{or} \quad 0 = 48t + 576e^{-t/12} - 676,$$

is now $t \approx 7.84$ s.

- (b) The differential equation $m dv/dt = mg \sin \theta - \mu mg \cos \theta$ can be written

$$m \frac{dv}{dt} = mg \cos \theta (\tan \theta - \mu).$$

If $\tan \theta = \mu$, $dv/dt = 0$ and $v(0) = 0$ implies that $v(t) = 0$. If $\tan \theta < \mu$ and $v(0) = 0$, then integration implies $v(t) = g \cos \theta (\tan \theta - \mu)t < 0$ for all time t .

- (c) Since $\tan 23^\circ = 0.4245$ and $\mu = \sqrt{3}/4 = 0.4330$, we see that $\tan 23^\circ < 0.4330$. The differential equation is $dv/dt = 32 \cos 23^\circ (\tan 23^\circ - \sqrt{3}/4) = -0.251493$. Integration and the use of the initial condition gives $v(t) = -0.251493t + 1$. When the box stops, $v(t) = 0$ or $0 = -0.251493t + 1$ or $t = 3.976254$ s. From $s(t) = -0.125747t^2 + t$ we find $s(3.976254) = 1.988119$ ft.

- (d) With $v_0 > 0$, $v(t) = -0.251493t + v_0$ and $s(t) = -0.125747t^2 + v_0 t$. Because two real positive solutions of the equation $s(t) = 100$, or $0 = -0.125747t^2 + v_0 t - 100$, would be physically meaningless, we use the quadratic formula and require that $b^2 - 4ac = 0$ or $v_0^2 - 50.2987 = 0$. From this last equality we find $v_0 \approx 7.092164$ ft/s. For the time it takes the box to traverse the entire inclined plane, we must have $0 = -0.125747t^2 + 7.092164t - 100$. *Mathematica* gives complex roots for the last equation: $t = 28.2001 \pm 0.0124458i$. But, for

$$0 = -0.125747t^2 + 7.092164691t - 100,$$

2.7 Linear Models

the roots are $t = 28.1999$ s and $t = 28.2004$ s. So if $v_0 > 7.092164$, we are guaranteed that the box will slide completely down the plane.

46. (a) We saw in part (b) of Problem 34 that the ascent time is $t_a = 9.375$. To find when the cannonball hits the ground we solve $s(t) = -16t^2 + 300t = 0$, getting a total time in flight of $t = 18.75$ s. Thus, the time of descent is $t_d = 18.75 - 9.375 = 9.375$. The impact velocity is $v_i = v(18.75) = -300$, which has the same magnitude as the initial velocity.
- (b) We saw in Problem 35 that the ascent time in the case of air resistance is $t_a = 9.162$. Solving $s(t) = 1,340,000 - 6,400t - 1,340,000e^{-0.005t} = 0$ we see that the total time of flight is 18.466 s. Thus, the descent time is $t_d = 18.466 - 9.162 = 9.304$. The impact velocity is $v_i = v(18.466) = -290.91$, compared to an initial velocity of $v_0 = 300$.

EXERCISES 2.8

Nonlinear Models

1. (a) Solving $N(1 - 0.0005N) = 0$ for N we find the equilibrium solutions $N = 0$ and $N = 2000$. When $0 < N < 2000$, $dN/dt > 0$. From the phase portrait we see that $\lim_{t \rightarrow \infty} N(t) = 2000$. A graph of the solution is shown in part (b).

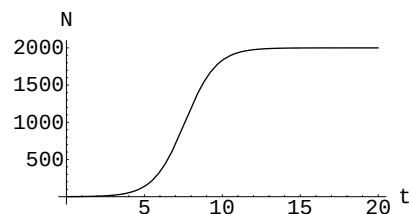


- (b) Separating variables and integrating we have

$$\frac{dN}{N(1 - 0.0005N)} = \left(\frac{1}{N} - \frac{1}{N - 2000} \right) dN = dt$$

and

$$\ln N - \ln(N - 2000) = t + c.$$



Solving for N we get $N(t) = 2000e^{c+t}/(1 + e^{c+t}) = 2000e^c e^t / (1 + e^c e^t)$. Using $N(0) = 1$ and solving for e^c we find $e^c = 1/1999$ and so $N(t) = 2000e^t / (1999 + e^t)$. Then $N(10) = 1833.59$, so 1834 companies are expected to adopt the new technology when $t = 10$.

2. From $dN/dt = N(a - bN)$ and $N(0) = 500$ we obtain

$$N = \frac{500a}{500b + (a - 500b)e^{-at}}.$$

Since $\lim_{t \rightarrow \infty} N = a/b = 50,000$ and $N(1) = 1000$ we have $a = 0.7033$, $b = 0.00014$, and $N = 50,000 / (1 + 99e^{-0.7033t})$.

3. From $dP/dt = P(10^{-1} - 10^{-7}P)$ and $P(0) = 5000$ we obtain $P = 500/(0.0005 + 0.0995e^{-0.1t})$ so that $P \rightarrow 1,000,000$ as $t \rightarrow \infty$. If $P(t) = 500,000$ then $t = 52.9$ months.

4. (a) We have $dP/dt = P(a - bP)$ with $P(0) = 3.929$ million. Using separation of variables we obtain

$$P(t) = \frac{3.929a}{3.929b + (a - 3.929b)e^{-at}} = \frac{a/b}{1 + (a/3.929b - 1)e^{-at}}$$

$$= \frac{c}{1 + (c/3.929 - 1)e^{-at}},$$

where $c = a/b$. At $t = 60(1850)$ the population is 23.192 million, so

$$23.192 = \frac{c}{1 + (c/3.929 - 1)e^{-60a}}$$

or $c = 23.192 + 23.192(c/3.929 - 1)e^{-60a}$. At $t = 120(1910)$,

$$91.972 = \frac{c}{1 + (c/3.929 - 1)e^{-120a}}$$

or $c = 91.972 + 91.972(c/3.929 - 1)(e^{-60a})^2$. Combining the two equations for c we get

$$\left(\frac{(c - 23.192)/23.192}{c/3.929 - 1} \right)^2 \left(\frac{c}{3.929} - 1 \right) = \frac{c - 91.972}{91.972}$$

or

$$91.972(3.929)(c - 23.192)^2 = (23.192)^2(c - 91.972)(c - 3.929).$$

The solution of this quadratic equation is $c = 197.274$. This in turn gives $a = 0.0313$. Therefore,

$$P(t) = \frac{197.274}{1 + 49.21e^{-0.0313t}}.$$

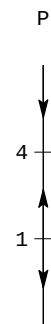
(b)

Year	Census Population	Predicted Population	Error	% Error
1790	3.929	3.929	0.000	0.00
1800	5.308	5.334	-0.026	-0.49
1810	7.240	7.222	0.018	0.24
1820	9.638	9.746	-0.108	-1.12
1830	12.866	13.090	-0.224	-1.74
1840	17.069	17.475	-0.406	-2.38
1850	23.192	23.143	0.049	0.21
1860	31.433	30.341	1.092	3.47
1870	38.558	39.272	-0.714	-1.85
1880	50.156	50.044	0.112	0.22
1890	62.948	62.600	0.348	0.55
1900	75.996	76.666	-0.670	-0.88
1910	91.972	91.739	0.233	0.25
1920	105.711	107.143	-1.432	-1.35
1930	122.775	122.140	0.635	0.52
1940	131.669	136.068	-4.399	-3.34
1950	150.697	148.445	2.252	1.49

The model predicts a population of 159.0 million for 1960 and 167.8 million for 1970. The census populations for these years were 179.3 and 203.3, respectively. The percentage errors are 12.8 and 21.2, respectively.

2.8 Nonlinear Models

5. (a) The differential equation is $dP/dt = P(5 - P) - 4$. Solving $P(5 - P) - 4 = 0$ for P we obtain equilibrium solutions $P = 1$ and $P = 4$. The phase portrait is shown on the right and solution curves are shown in part (b). We see that for $P_0 > 4$ and $1 < P_0 < 4$ the population approaches 4 as t increases. For $0 < P < 1$ the population decreases to 0 in finite time.



- (b) The differential equation is

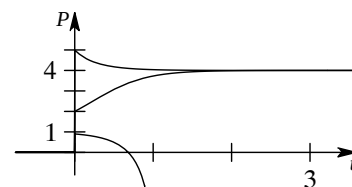
$$\frac{dP}{dt} = P(5 - P) - 4 = -(P^2 - 5P + 4) = -(P - 4)(P - 1).$$

Separating variables and integrating, we obtain

$$\begin{aligned} \frac{dP}{(P - 4)(P - 1)} &= -dt \\ \left(\frac{1/3}{P - 4} - \frac{1/3}{P - 1} \right) dP &= -dt \\ \frac{1}{3} \ln \left| \frac{P - 4}{P - 1} \right| &= -t + c \\ \frac{P - 4}{P - 1} &= c_1 e^{-3t}. \end{aligned}$$

Setting $t = 0$ and $P = P_0$ we find $c_1 = (P_0 - 4)/(P_0 - 1)$. Solving for P we obtain

$$P(t) = \frac{4(P_0 - 1) - (P_0 - 4)e^{-3t}}{(P_0 - 1) - (P_0 - 4)e^{-3t}}.$$



- (c) To find when the population becomes extinct in the case $0 < P_0 < 1$ we set $P = 0$ in

$$\frac{P - 4}{P - 1} = \frac{P_0 - 4}{P_0 - 1} e^{-3t}$$

from part (a) and solve for t . This gives the time of extinction

$$t = -\frac{1}{3} \ln \frac{4(P_0 - 1)}{P_0 - 4}.$$

6. Solving $P(5 - P) - \frac{25}{4} = 0$ for P we obtain the equilibrium solution $P = \frac{5}{2}$. For $P \neq \frac{5}{2}$, $dP/dt < 0$. Thus, if $P_0 < \frac{5}{2}$, the population becomes extinct (otherwise there would be another equilibrium solution.) Using separation of variables to solve the initial-value problem, we get

$$P(t) = [4P_0 + (10P_0 - 25)t]/[4 + (4P_0 - 10)t].$$

To find when the population becomes extinct for $P_0 < \frac{5}{2}$ we solve $P(t) = 0$ for t . We see that the time of extinction is $t = 4P_0/5(5 - 2P_0)$.

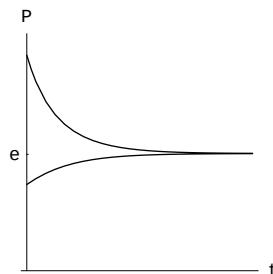
7. Solving $P(5 - P) - 7 = 0$ for P we obtain complex roots, so there are no equilibrium solutions. Since $dP/dt < 0$ for all values of P , the population becomes extinct for any initial condition. Using separation of variables to solve the initial-value problem, we get

$$P(t) = \frac{5}{2} + \frac{\sqrt{3}}{2} \tan \left[\tan^{-1} \left(\frac{2P_0 - 5}{\sqrt{3}} \right) - \frac{\sqrt{3}}{2} t \right].$$

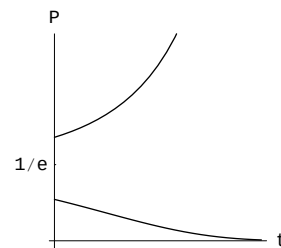
Solving $P(t) = 0$ for t we see that the time of extinction is

$$t = \frac{2}{3} \left(\sqrt{3} \tan^{-1}(5/\sqrt{3}) + \sqrt{3} \tan^{-1}[(2P_0 - 5)/\sqrt{3}] \right).$$

8. (a) The differential equation is $dP/dt = P(1 - \ln P)$, which has the equilibrium solution $P = e$. When $P_0 > e$, $dP/dt < 0$, and when $P_0 < e$, $dP/dt > 0$.



- (b) The differential equation is $dP/dt = P(1 + \ln P)$, which has the equilibrium solution $P = 1/e$. When $P_0 > 1/e$, $dP/dt > 0$, and when $P_0 < 1/e$, $dP/dt < 0$.



- (c) From $dP/dt = P(a - b \ln P)$ we obtain $-(1/b) \ln |a - b \ln P| = t + c_1$ so that $P = e^{a/b} e^{-ce^{-bt}}$. If $P(0) = P_0$ then $c = (a/b) - \ln P_0$.
9. Let $X = X(t)$ be the amount of C at time t and $dX/dt = k(120 - 2X)(150 - X)$. If $X(0) = 0$ and $X(5) = 10$, then

$$X(t) = \frac{150 - 150e^{180kt}}{1 - 2.5e^{180kt}},$$

where $k = .0001259$ and $X(20) = 29.3$ grams. Now by L'Hôpital's rule, $X \rightarrow 60$ as $t \rightarrow \infty$, so that the amount of $A \rightarrow 0$ and the amount of $B \rightarrow 30$ as $t \rightarrow \infty$.

10. From $dX/dt = k(150 - X)^2$, $X(0) = 0$, and $X(5) = 10$ we obtain $X = 150 - 150/(150kt + 1)$ where $k = .000095238$. Then $X(20) = 33.3$ grams and $X \rightarrow 150$ as $t \rightarrow \infty$ so that the amount of $A \rightarrow 0$ and the amount of $B \rightarrow 0$ as $t \rightarrow \infty$. If $X(t) = 75$ then $t = 70$ minutes.

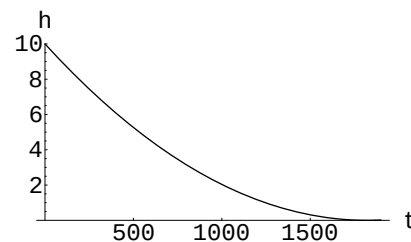
11. (a) The initial-value problem is $dh/dt = -8A_h\sqrt{h}/A_w$, $h(0) = H$.

Separating variables and integrating we have

$$\frac{dh}{\sqrt{h}} = -\frac{8A_h}{A_w} dt \quad \text{and} \quad 2\sqrt{h} = -\frac{8A_h}{A_w}t + c.$$

Using $h(0) = H$ we find $c = 2\sqrt{H}$, so the solution of the initial-value problem is $\sqrt{h(t)} = (A_w\sqrt{H} - 4A_h t)/A_w$, where $A_w\sqrt{H} - 4A_h t \geq 0$. Thus,

$$h(t) = (A_w\sqrt{H} - 4A_h t)^2/A_w^2 \quad \text{for} \quad 0 \leq t \leq A_w H/4A_h.$$



- (b) Identifying $H = 10$, $A_w = 4\pi$, and $A_h = \pi/576$ we have $h(t) = t^2/331,776 - (\sqrt{5/2}/144)t + 10$. Solving $h(t) = 0$ we see that the tank empties in $576\sqrt{10}$ seconds or 30.36 minutes.
12. To obtain the solution of this differential equation we use $h(t)$ from Problem 13 in Exercises 1.3. Then $h(t) = (A_w\sqrt{H} - 4cA_h t)^2/A_w^2$. Solving $h(t) = 0$ with $c = 0.6$ and the values from Problem 11 we see that the tank empties in 3035.79 seconds or 50.6 minutes.

2.8 Nonlinear Models

13. (a) Separating variables and integrating gives

$$6h^{3/2}dh = -5t \quad \text{and} \quad \frac{12}{5}h^{5/2} = -5t + c.$$

Using $h(0) = 20$ we find $c = 1920\sqrt{5}$, so the solution of the initial-value problem is $h(t) = (800\sqrt{5} - \frac{25}{12}t)^{2/5}$. Solving $h(t) = 0$ we see that the tank empties in $384\sqrt{5}$ seconds or 14.31 minutes.

- (b) When the height of the water is h , the radius of the top of the water is $r = h \tan 30^\circ = h/\sqrt{3}$ and $A_w = \pi h^2/3$. The differential equation is

$$\frac{dh}{dt} = -c \frac{A_h}{A_w} \sqrt{2gh} = -0.6 \frac{\pi(2/12)^2}{\pi h^2/3} \sqrt{64h} = -\frac{2}{5h^{3/2}}.$$

Separating variables and integrating gives

$$5h^{3/2}dh = -2dt \quad \text{and} \quad 2h^{5/2} = -2t + c.$$

Using $h(0) = 9$ we find $c = 486$, so the solution of the initial-value problem is $h(t) = (243 - t)^{2/5}$. Solving $h(t) = 0$ we see that the tank empties in 24.3 seconds or 4.05 minutes.

14. When the height of the water is h , the radius of the top of the water is $\frac{2}{5}(20 - h)$ and $A_w = 4\pi(20 - h)^2/25$. The differential equation is

$$\frac{dh}{dt} = -c \frac{A_h}{A_w} \sqrt{2gh} = -0.6 \frac{\pi(2/12)^2}{4\pi(20 - h)^2/25} \sqrt{64h} = -\frac{5}{6} \frac{\sqrt{h}}{(20 - h)^2}.$$

Separating variables and integrating we have

$$\frac{(20 - h)^2}{\sqrt{h}} dh = -\frac{5}{6} dt \quad \text{and} \quad 800\sqrt{h} - \frac{80}{3}h^{3/2} + \frac{2}{5}h^{5/2} = -\frac{5}{6}t + c.$$

Using $h(0) = 20$ we find $c = 2560\sqrt{5}/3$, so an implicit solution of the initial-value problem is

$$800\sqrt{h} - \frac{80}{3}h^{3/2} + \frac{2}{5}h^{5/2} = -\frac{5}{6}t + \frac{2560\sqrt{5}}{3}.$$

To find the time it takes the tank to empty we set $h = 0$ and solve for t . The tank empties in $1024\sqrt{5}$ seconds or 38.16 minutes. Thus, the tank empties more slowly when the base of the cone is on the bottom.

15. (a) After separating variables we obtain

$$\begin{aligned} \frac{m dv}{mg - kv^2} &= dt \\ \frac{1}{g} \frac{dv}{1 - (\sqrt{k}v/\sqrt{mg})^2} &= dt \\ \frac{\sqrt{mg}}{\sqrt{k}g} \frac{\sqrt{k/mg} dv}{1 - (\sqrt{k}v/\sqrt{mg})^2} &= dt \\ \sqrt{\frac{m}{kg}} \tanh^{-1} \frac{\sqrt{k}v}{\sqrt{mg}} &= t + c \\ \tanh^{-1} \frac{\sqrt{k}v}{\sqrt{mg}} &= \sqrt{\frac{kg}{m}} t + c_1. \end{aligned}$$

Thus the velocity at time t is

$$v(t) = \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} t + c_1 \right).$$

Setting $t = 0$ and $v = v_0$ we find $c_1 = \tanh^{-1}(\sqrt{k}v_0/\sqrt{mg})$.

(b) Since $\tanh t \rightarrow 1$ as $t \rightarrow \infty$, we have $v \rightarrow \sqrt{mg/k}$ as $t \rightarrow \infty$.

(c) Integrating the expression for $v(t)$ in part (a) we obtain an integral of the form $\int du/u$:

$$s(t) = \sqrt{\frac{mg}{k}} \int \tanh \left(\sqrt{\frac{kg}{m}} t + c_1 \right) dt = \frac{m}{k} \ln \left[\cosh \left(\sqrt{\frac{kg}{m}} t + c_1 \right) \right] + c_2.$$

Setting $t = 0$ and $s = 0$ we find $c_2 = -(m/k) \ln(\cosh c_1)$, where c_1 is given in part (a).

16. The differential equation is $m dv/dt = -mg - kv^2$. Separating variables and integrating, we have

$$\begin{aligned} \frac{dv}{mg + kv^2} &= -\frac{dt}{m} \\ \frac{1}{\sqrt{mgk}} \tan^{-1} \left(\frac{\sqrt{k} v}{\sqrt{mg}} \right) &= -\frac{1}{m} t + c \\ \tan^{-1} \left(\frac{\sqrt{k} v}{\sqrt{mg}} \right) &= -\sqrt{\frac{gk}{m}} t + c_1 \\ v(t) &= \sqrt{\frac{mg}{k}} \tan \left(c_1 - \sqrt{\frac{gk}{m}} t \right). \end{aligned}$$

Setting $v(0) = 300$, $m = \frac{16}{32} = \frac{1}{2}$, $g = 32$, and $k = 0.0003$, we find $v(t) = 230.94 \tan(c_1 - 0.138564t)$ and $c_1 = 0.914743$. Integrating

$$v(t) = 230.94 \tan(0.914743 - 0.138564t)$$

we get

$$s(t) = 1666.67 \ln |\cos(0.914743 - 0.138564t)| + c_2.$$

Using $s(0) = 0$ we find $c_2 = 823.843$. Solving $v(t) = 0$ we see that the maximum height is attained when $t = 6.60159$. The maximum height is $s(6.60159) = 823.843$ ft.

17. (a) Let ρ be the weight density of the water and V the volume of the object. Archimedes' principle states that the upward buoyant force has magnitude equal to the weight of the water displaced. Taking the positive direction to be down, the differential equation is

$$m \frac{dv}{dt} = mg - kv^2 - \rho V.$$

(b) Using separation of variables we have

$$\begin{aligned} \frac{m dv}{(mg - \rho V) - kv^2} &= dt \\ \frac{m}{\sqrt{k}} \frac{\sqrt{k} dv}{(\sqrt{mg - \rho V})^2 - (\sqrt{k} v)^2} &= dt \\ \frac{m}{\sqrt{k}} \frac{1}{\sqrt{mg - \rho V}} \tanh^{-1} \frac{\sqrt{k} v}{\sqrt{mg - \rho V}} &= t + c. \end{aligned}$$

Thus

$$v(t) = \sqrt{\frac{mg - \rho V}{k}} \tanh \left(\frac{\sqrt{kmg - k\rho V}}{m} t + c_1 \right).$$

(c) Since $\tanh t \rightarrow 1$ as $t \rightarrow \infty$, the terminal velocity is $\sqrt{(mg - \rho V)/k}$.

2.8 Nonlinear Models

18. (a) Writing the equation in the form $(x - \sqrt{x^2 + y^2})dx + y dy = 0$ we identify $M = x - \sqrt{x^2 + y^2}$ and $N = y$. Since M and N are both homogeneous functions of degree 1 we use the substitution $y = ux$. It follows that

$$\begin{aligned} (x - \sqrt{x^2 + u^2 x^2}) dx + ux(u dx + x du) &= 0 \\ x [1 - \sqrt{1 + u^2} + u^2] dx + x^2 u du &= 0 \\ -\frac{u du}{1 + u^2 - \sqrt{1 + u^2}} &= \frac{dx}{x} \\ \frac{u du}{\sqrt{1 + u^2} (1 - \sqrt{1 + u^2})} &= \frac{dx}{x}. \end{aligned}$$

Letting $w = 1 - \sqrt{1 + u^2}$ we have $dw = -u du / \sqrt{1 + u^2}$ so that

$$\begin{aligned} -\ln |1 - \sqrt{1 + u^2}| &= \ln |x| + c \\ \frac{1}{1 - \sqrt{1 + u^2}} &= c_1 x \\ 1 - \sqrt{1 + u^2} &= -\frac{c_2}{x} \quad (-c_2 = 1/c_1) \\ 1 + \frac{c_2}{x} &= \sqrt{1 + \frac{y^2}{x^2}} \\ 1 + \frac{2c_2}{x} + \frac{c_2^2}{x^2} &= 1 + \frac{y^2}{x^2}. \end{aligned}$$

Solving for y^2 we have

$$y^2 = 2c_2 x + c_2^2 = 4 \left(\frac{c_2}{2} \right) \left(x + \frac{c_2}{2} \right)$$

which is a family of parabolas symmetric with respect to the x -axis with vertex at $(-c_2/2, 0)$ and focus at the origin.

- (b) Let $u = x^2 + y^2$ so that

$$\frac{du}{dx} = 2x + 2y \frac{dy}{dx}.$$

Then

$$y \frac{dy}{dx} = \frac{1}{2} \frac{du}{dx} - x$$

and the differential equation can be written in the form

$$\frac{1}{2} \frac{du}{dx} - x = -x + \sqrt{u} \quad \text{or} \quad \frac{1}{2} \frac{du}{dx} = \sqrt{u}.$$

Separating variables and integrating gives

$$\begin{aligned} \frac{du}{2\sqrt{u}} &= dx \\ \sqrt{u} &= x + c \\ u &= x^2 + 2cx + c^2 \\ x^2 + y^2 &= x^2 + 2cx + c^2 \\ y^2 &= 2cx + c^2. \end{aligned}$$

19. (a) From $2W^2 - W^3 = W^2(2 - W) = 0$ we see that $W = 0$ and $W = 2$ are constant solutions.

- (b) Separating variables and using a CAS to integrate we get

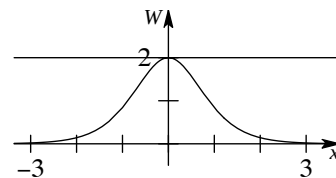
$$\frac{dW}{W\sqrt{4-2W}} = dx \quad \text{and} \quad -\tanh^{-1}\left(\frac{1}{2}\sqrt{4-2W}\right) = x + c.$$

Using the facts that the hyperbolic tangent is an odd function and $1 - \tanh^2 x = \operatorname{sech}^2 x$ we have

$$\begin{aligned}\frac{1}{2}\sqrt{4-2W} &= \tanh(-x-c) = -\tanh(x+c) \\ \frac{1}{4}(4-2W) &= \tanh^2(x+c) \\ 1 - \frac{1}{2}W &= \tanh^2(x+c) \\ \frac{1}{2}W &= 1 - \tanh^2(x+c) = \operatorname{sech}^2(x+c).\end{aligned}$$

Thus, $W(x) = 2\operatorname{sech}^2(x+c)$.

- (c) Letting $x = 0$ and $W = 2$ we find that $\operatorname{sech}^2(c) = 1$ and $c = 0$.



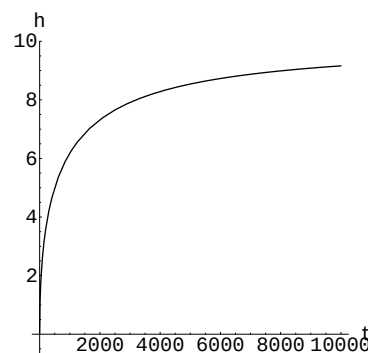
20. (a) Solving $r^2 + (10 - h)^2 = 10^2$ for r^2 we see that $r^2 = 20h - h^2$. Combining the rate of input of water, π , with the rate of output due to evaporation, $k\pi r^2 = k\pi(20h - h^2)$, we have $dV/dt = \pi - k\pi(20h - h^2)$. Using $V = 10\pi h^2 - \frac{1}{3}\pi h^3$, we see also that $dV/dt = (20\pi h - \pi h^2)dh/dt$. Thus,

$$(20\pi h - \pi h^2)\frac{dh}{dt} = \pi - k\pi(20h - h^2) \quad \text{and} \quad \frac{dh}{dt} = \frac{1 - 20kh + kh^2}{20h - h^2}.$$

- (b) Letting $k = 1/100$, separating variables and integrating (with the help of a CAS), we get

$$\frac{100h(h-20)}{(h-10)^2} dh = dt \quad \text{and} \quad \frac{100(h^2 - 10h + 100)}{10 - h} = t + c.$$

Using $h(0) = 0$ we find $c = 1000$, and solving for h we get $h(t) = 0.005(\sqrt{t^2 + 4000t} - t)$, where the positive square root is chosen because $h \geq 0$.



- (c) The volume of the tank is $V = \frac{2}{3}\pi(10)^3$ feet, so at a rate of π cubic feet per minute, the tank will fill in $\frac{2}{3}(10)^3 \approx 666.67$ minutes ≈ 11.11 hours.
- (d) At 666.67 minutes, the depth of the water is $h(666.67) = 5.486$ feet. From the graph in (b) we suspect that $\lim_{t \rightarrow \infty} h(t) = 10$, in which case the tank will never completely fill. To prove this we compute the limit of $h(t)$:

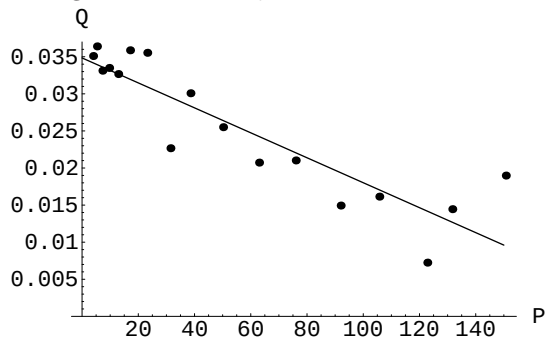
$$\begin{aligned}\lim_{t \rightarrow \infty} h(t) &= 0.005 \lim_{t \rightarrow \infty} \left(\sqrt{t^2 + 4000t} - t \right) = 0.005 \lim_{t \rightarrow \infty} \frac{t^2 + 4000t - t^2}{\sqrt{t^2 + 4000t} + t} \\ &= 0.005 \lim_{t \rightarrow \infty} \frac{4000t}{t\sqrt{1 + 4000/t} + t} = 0.005 \frac{4000}{1 + 1} = 0.005(2000) = 10.\end{aligned}$$

2.8 Nonlinear Models

21. (a)

t	P(t)	Q(t)
0	3.929	0.035
10	5.308	0.036
20	7.240	0.033
30	9.638	0.033
40	12.866	0.033
50	17.069	0.036
60	23.192	0.036
70	31.433	0.023
80	38.558	0.030
90	50.156	0.026
100	62.948	0.021
110	75.996	0.021
120	91.972	0.015
130	105.711	0.016
140	122.775	0.007
150	131.669	0.014
160	150.697	0.019
170	179.300	

(b) The regression line is $Q = 0.0348391 - 0.000168222P$.



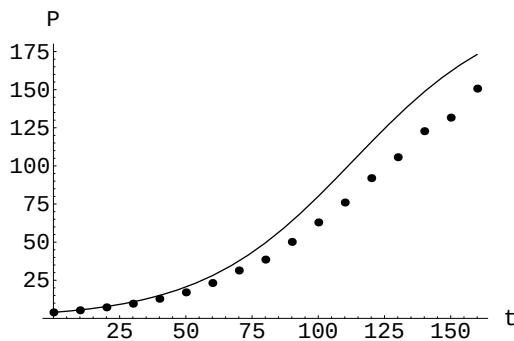
(c) The solution of the logistic equation is given in equation (5) in the text. Identifying $a = 0.0348391$ and $b = 0.000168222$ we have

$$P(t) = \frac{aP_0}{bP_0 + (a - bP_0)e^{-at}}.$$

(d) With $P_0 = 3.929$ the solution becomes

$$P(t) = \frac{0.136883}{0.000660944 + 0.0341781e^{-0.0348391t}}.$$

(e)



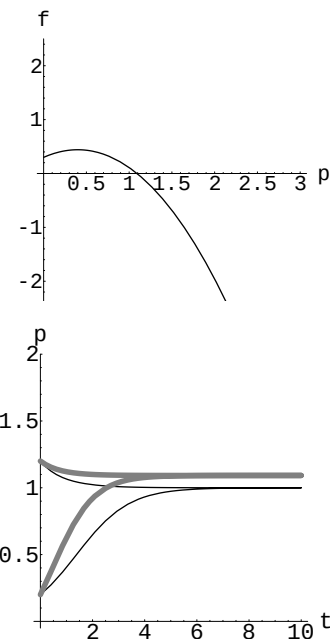
(f) We identify $t = 180$ with 1970, $t = 190$ with 1980, and $t = 200$ with 1990. The model predicts $P(180) = 188.661$, $P(190) = 193.735$, and $P(200) = 197.485$. The actual population figures for these years are 203.303, 226.542, and 248.765 millions. As $t \rightarrow \infty$, $P(t) \rightarrow a/b = 207.102$.

22. (a) Using a CAS to solve $P(1 - P) + 0.3e^{-P} = 0$ for P we see that $P = 1.09216$ is an equilibrium solution.

(b) Since $f(P) > 0$ for $0 < P < 1.09216$, the solution $P(t)$ of

$$dP/dt = P(1 - P) + 0.3e^{-P}, \quad P(0) = P_0,$$

is increasing for $P_0 < 1.09216$. Since $f(P) < 0$ for $P > 1.09216$, the solution $P(t)$ is decreasing for $P_0 > 1.09216$. Thus $P = 1.09216$ is an attractor.



(c) The curves for the second initial-value problem are thicker. The equilibrium solution for the logic model is $P = 1$. Comparing 1.09216 and 1, we see that the percentage increase is 9.216%.

23. To find t_d we solve

$$m \frac{dv}{dt} = mg - kv^2, \quad v(0) = 0$$

using separation of variables. This gives

$$v(t) = \sqrt{\frac{mg}{k}} \tanh \sqrt{\frac{kg}{m}} t.$$

Integrating and using $s(0) = 0$ gives

$$s(t) = \frac{m}{k} \ln \left(\cosh \sqrt{\frac{kg}{m}} t \right).$$

To find the time of descent we solve $s(t) = 823.84$ and find $t_d = 7.77882$. The impact velocity is $v(t_d) = 182.998$, which is positive because the positive direction is downward.

24. (a) Solving $v_t = \sqrt{mg/k}$ for k we obtain $k = mg/v_t^2$. The differential equation then becomes

$$m \frac{dv}{dt} = mg - \frac{mg}{v_t^2} v^2 \quad \text{or} \quad \frac{dv}{dt} = g \left(1 - \frac{1}{v_t^2} v^2 \right).$$

Separating variables and integrating gives

$$v_t \tanh^{-1} \frac{v}{v_t} = gt + c_1.$$

The initial condition $v(0) = 0$ implies $c_1 = 0$, so

$$v(t) = v_t \tanh \frac{gt}{v_t}.$$

We find the distance by integrating:

$$s(t) = \int v_t \tanh \frac{gt}{v_t} dt = \frac{v_t^2}{g} \ln \left(\cosh \frac{gt}{v_t} \right) + c_2.$$

2.8 Nonlinear Models

The initial condition $s(0) = 0$ implies $c_2 = 0$, so

$$s(t) = \frac{v_t^2}{g} \ln \left(\cosh \frac{gt}{v_t} \right).$$

In 25 seconds she has fallen $20,000 - 14,800 = 5,200$ feet. Using a CAS to solve

$$5200 = (v_t^2/32) \ln \left(\cosh \frac{32(25)}{v_t} \right)$$

for v_t gives $v_t \approx 271.711$ ft/s. Then

$$s(t) = \frac{v_t^2}{g} \ln \left(\cosh \frac{gt}{v_t} \right) = 2307.08 \ln(\cosh 0.117772t).$$

(b) At $t = 15$, $s(15) = 2,542.94$ ft and $v(15) = s'(15) = 256.287$ ft/sec.

25. While the object is in the air its velocity is modeled by the linear differential equation $m dv/dt = mg - kv$. Using $m = 160$, $k = \frac{1}{4}$, and $g = 32$, the differential equation becomes $dv/dt + (1/640)v = 32$. The integrating factor is $e^{\int dt/640} = e^{t/640}$ and the solution of the differential equation is $e^{t/640}v = \int 32e^{t/640}dt = 20,480e^{t/640} + c$. Using $v(0) = 0$ we see that $c = -20,480$ and $v(t) = 20,480 - 20,480e^{-t/640}$. Integrating we get $s(t) = 20,480t + 13,107,200e^{-t/640} + c$. Since $s(0) = 0$, $c = -13,107,200$ and $s(t) = -13,107,200 + 20,480t + 13,107,200e^{-t/640}$. To find when the object hits the liquid we solve $s(t) = 500 - 75 = 425$, obtaining $t_a = 5.16018$. The velocity at the time of impact with the liquid is $v_a = v(t_a) = 164.482$. When the object is in the liquid its velocity is modeled by the nonlinear differential equation $m dv/dt = mg - kv^2$. Using $m = 160$, $g = 32$, and $k = 0.1$ this becomes $dv/dt = (51,200 - v^2)/1600$. Separating variables and integrating we have

$$\frac{dv}{51,200 - v^2} = \frac{dt}{1600} \quad \text{and} \quad \frac{\sqrt{2}}{640} \ln \left| \frac{v - 160\sqrt{2}}{v + 160\sqrt{2}} \right| = \frac{1}{1600}t + c.$$

Solving $v(0) = v_a = 164.482$ we obtain $c = -0.00407537$. Then, for $v < 160\sqrt{2} = 226.274$,

$$\left| \frac{v - 160\sqrt{2}}{v + 160\sqrt{2}} \right| = e^{\sqrt{2}t/5 - 1.8443} \quad \text{or} \quad -\frac{v - 160\sqrt{2}}{v + 160\sqrt{2}} = e^{\sqrt{2}t/5 - 1.8443}.$$

Solving for v we get

$$v(t) = \frac{13964.6 - 2208.29e^{\sqrt{2}t/5}}{61.7153 + 9.75937e^{\sqrt{2}t/5}}.$$

Integrating we find

$$s(t) = 226.275t - 1600 \ln(6.3237 + e^{\sqrt{2}t/5}) + c.$$

Solving $s(0) = 0$ we see that $c = 3185.78$, so

$$s(t) = 3185.78 + 226.275t - 1600 \ln(6.3237 + e^{\sqrt{2}t/5}).$$

To find when the object hits the bottom of the tank we solve $s(t) = 75$, obtaining $t_b = 0.466273$. The time from when the object is dropped from the helicopter to when it hits the bottom of the tank is $t_a + t_b = 5.62708$ seconds.

EXERCISES 2.9

Modeling with Systems of First-Order DEs

1. The linear equation $dx/dt = -\lambda_1 x$ can be solved by either separation of variables or by an integrating factor. Integrating both sides of $dx/x = -\lambda_1 dt$ we obtain $\ln|x| = -\lambda_1 t + c$ from which we get $x = c_1 e^{-\lambda_1 t}$. Using $x(0) = x_0$ we find $c_1 = x_0$ so that $x = x_0 e^{-\lambda_1 t}$. Substituting this result into the second differential equation we have

$$\frac{dy}{dt} + \lambda_2 y = \lambda_1 x_0 e^{-\lambda_1 t}$$

which is linear. An integrating factor is $e^{\lambda_2 t}$ so that

$$\frac{d}{dt} [e^{\lambda_2 t} y] = \lambda_1 x_0 e^{(\lambda_2 - \lambda_1)t} + c_2$$

$$y = \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} e^{-\lambda_2 t} + c_2 e^{-\lambda_2 t} = \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + c_2 e^{-\lambda_2 t}.$$

Using $y(0) = 0$ we find $c_2 = -\lambda_1 x_0 / (\lambda_2 - \lambda_1)$. Thus

$$y = \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}).$$

Substituting this result into the third differential equation we have

$$\frac{dz}{dt} = \frac{\lambda_1 \lambda_2 x_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}).$$

Integrating we find

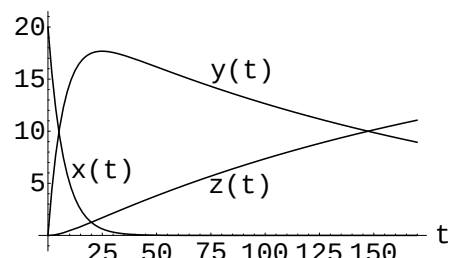
$$z = -\frac{\lambda_2 x_0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} + c_3.$$

Using $z(0) = 0$ we find $c_3 = x_0$. Thus

$$z = x_0 \left(1 - \frac{\lambda_2}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{\lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} \right).$$

2. We see from the graph that the half-life of A is approximately 4.7 days. To determine the half-life of B we use $t = 50$ as a base, since at this time the amount of substance A is so small that it contributes very little to substance B . Now we see from the graph that $y(50) \approx 16.2$ and $y(191) \approx 8.1$. Thus, the half-life of B is approximately 141 days.

x, y, z



3. The amounts x and y are the same at about $t = 5$ days. The amounts x and z are the same at about $t = 20$ days. The amounts y and z are the same at about $t = 147$ days. The time when y and z are the same makes sense because most of A and half of B are gone, so half of C should have been formed.
4. Suppose that the series is described schematically by $W \Rightarrow -\lambda_1 X \Rightarrow -\lambda_2 Y \Rightarrow -\lambda_3 Z$ where $-\lambda_1$, $-\lambda_2$, and $-\lambda_3$ are the decay constants for W , X and Y , respectively, and Z is a stable element. Let $w(t)$, $x(t)$, $y(t)$, and

2.9 Modeling with Systems of First-Order DEs

$z(t)$ denote the amounts of substances W , X , Y , and Z , respectively. A model for the radioactive series is

$$\begin{aligned}\frac{dw}{dt} &= -\lambda_1 w \\ \frac{dx}{dt} &= \lambda_1 w - \lambda_2 x \\ \frac{dy}{dt} &= \lambda_2 x - \lambda_3 y \\ \frac{dz}{dt} &= \lambda_3 y.\end{aligned}$$

5. The system is

$$\begin{aligned}x'_1 &= 2 \cdot 3 + \frac{1}{50}x_2 - \frac{1}{50}x_1 \cdot 4 = -\frac{2}{25}x_1 + \frac{1}{50}x_2 + 6 \\ x'_2 &= \frac{1}{50}x_1 \cdot 4 - \frac{1}{50}x_2 - \frac{1}{50}x_2 \cdot 3 = \frac{2}{25}x_1 - \frac{2}{25}x_2.\end{aligned}$$

6. Let x_1 , x_2 , and x_3 be the amounts of salt in tanks A , B , and C , respectively, so that

$$\begin{aligned}x'_1 &= \frac{1}{100}x_2 \cdot 2 - \frac{1}{100}x_1 \cdot 6 = \frac{1}{50}x_2 - \frac{3}{50}x_1 \\ x'_2 &= \frac{1}{100}x_1 \cdot 6 + \frac{1}{100}x_3 - \frac{1}{100}x_2 \cdot 2 - \frac{1}{100}x_2 \cdot 5 = \frac{3}{50}x_1 - \frac{7}{100}x_2 + \frac{1}{100}x_3 \\ x'_3 &= \frac{1}{100}x_2 \cdot 5 - \frac{1}{100}x_3 - \frac{1}{100}x_3 \cdot 4 = \frac{1}{20}x_2 - \frac{1}{20}x_3.\end{aligned}$$

7. (a) A model is

$$\begin{aligned}\frac{dx_1}{dt} &= 3 \cdot \frac{x_2}{100-t} - 2 \cdot \frac{x_1}{100+t}, & x_1(0) &= 100 \\ \frac{dx_2}{dt} &= 2 \cdot \frac{x_1}{100+t} - 3 \cdot \frac{x_2}{100-t}, & x_2(0) &= 50.\end{aligned}$$

(b) Since the system is closed, no salt enters or leaves the system and $x_1(t) + x_2(t) = 100 + 50 = 150$ for all time. Thus $x_1 = 150 - x_2$ and the second equation in part (a) becomes

$$\frac{dx_2}{dt} = \frac{2(150 - x_2)}{100 + t} - \frac{3x_2}{100 - t} = \frac{300}{100 + t} - \frac{2x_2}{100 + t} - \frac{3x_2}{100 - t}$$

or

$$\frac{dx_2}{dt} + \left(\frac{2}{100+t} + \frac{3}{100-t} \right) x_2 = \frac{300}{100+t},$$

which is linear in x_2 . An integrating factor is

$$e^{2 \ln(100+t) - 3 \ln(100-t)} = (100+t)^2(100-t)^{-3}$$

so

$$\frac{d}{dt}[(100+t)^2(100-t)^{-3}x_2] = 300(100+t)(100-t)^{-3}.$$

Using integration by parts, we obtain

$$(100+t)^2(100-t)^{-3}x_2 = 300 \left[\frac{1}{2}(100+t)(100-t)^{-2} - \frac{1}{2}(100-t)^{-1} + c \right].$$

Thus

$$\begin{aligned}x_2 &= \frac{300}{(100+t)^2} \left[c(100-t)^3 - \frac{1}{2}(100-t)^2 + \frac{1}{2}(100+t)(100-t) \right] \\ &= \frac{300}{(100+t)^2} [c(100-t)^3 + t(100-t)].\end{aligned}$$

2.9 Modeling with Systems of First-Order DEs

Using $x_2(0) = 50$ we find $c = 5/3000$. At $t = 30$, $x_2 = (300/130^2)(70^3 c + 30 \cdot 70) \approx 47.4$ lbs.

8. A model is

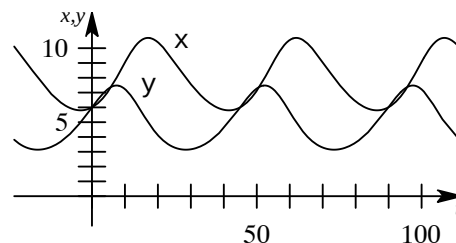
$$\begin{aligned}\frac{dx_1}{dt} &= (4 \text{ gal/min})(0 \text{ lb/gal}) - (4 \text{ gal/min})\left(\frac{1}{200}x_1 \text{ lb/gal}\right) \\ \frac{dx_2}{dt} &= (4 \text{ gal/min})\left(\frac{1}{200}x_1 \text{ lb/gal}\right) - (4 \text{ gal/min})\left(\frac{1}{150}x_2 \text{ lb/gal}\right) \\ \frac{dx_3}{dt} &= (4 \text{ gal/min})\left(\frac{1}{150}x_2 \text{ lb/gal}\right) - (4 \text{ gal/min})\left(\frac{1}{100}x_3 \text{ lb/gal}\right)\end{aligned}$$

or

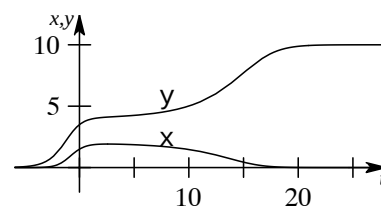
$$\begin{aligned}\frac{dx_1}{dt} &= -\frac{1}{50}x_1 \\ \frac{dx_2}{dt} &= \frac{1}{50}x_1 - \frac{2}{75}x_2 \\ \frac{dx_3}{dt} &= \frac{2}{75}x_2 - \frac{1}{25}x_3.\end{aligned}$$

Over a long period of time we would expect x_1 , x_2 , and x_3 to approach 0 because the entering pure water should flush the salt out of all three tanks.

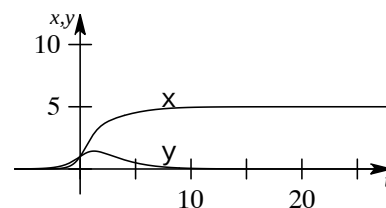
9. Zooming in on the graph it can be seen that the populations are first equal at about $t = 5.6$. The approximate periods of x and y are both 45.



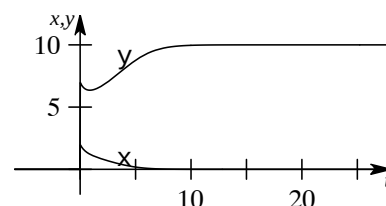
10. (a) The population $y(t)$ approaches 10,000, while the population $x(t)$ approaches extinction.



- (b) The population $x(t)$ approaches 5,000, while the population $y(t)$ approaches extinction.

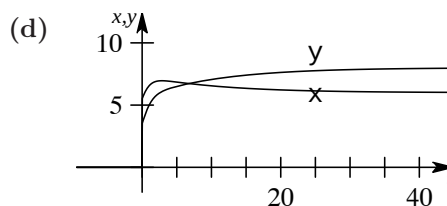
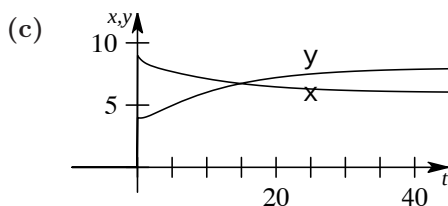
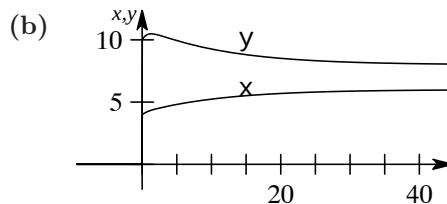
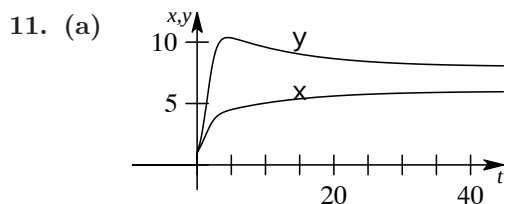
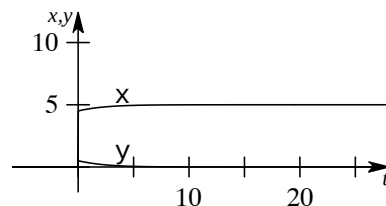


- (c) The population $y(t)$ approaches 10,000, while the population $x(t)$ approaches extinction.



2.9 Modeling with Systems of First-Order DEs

- (d) The population $x(t)$ approaches 5,000, while the population $y(t)$ approaches extinction.



In each case the population $x(t)$ approaches 6,000, while the population $y(t)$ approaches 8,000.

12. By Kirchhoff's first law we have $i_1 = i_2 + i_3$. By Kirchhoff's second law, on each loop we have $E(t) = Li_1' + R_1i_2$ and $E(t) = Li_1' + R_2i_3 + q/C$ so that $q = CR_1i_2 - CR_2i_3$. Then $i_3 = q' = CR_1i_2' - CR_2i_3$ so that the system is

$$\begin{aligned} Li_2' + Li_3' + R_1i_2 &= E(t) \\ -R_1i_2' + R_2i_3' + \frac{1}{C}i_3 &= 0. \end{aligned}$$

13. By Kirchhoff's first law we have $i_1 = i_2 + i_3$. Applying Kirchhoff's second law to each loop we obtain

$$E(t) = i_1R_1 + L_1\frac{di_2}{dt} + i_2R_2$$

and

$$E(t) = i_1R_1 + L_2\frac{di_3}{dt} + i_3R_3.$$

Combining the three equations, we obtain the system

$$\begin{aligned} L_1\frac{di_2}{dt} + (R_1 + R_2)i_2 + R_1i_3 &= E \\ L_2\frac{di_3}{dt} + R_1i_2 + (R_1 + R_3)i_3 &= E. \end{aligned}$$

14. By Kirchhoff's first law we have $i_1 = i_2 + i_3$. By Kirchhoff's second law, on each loop we have $E(t) = Li_1' + Ri_2$ and $E(t) = Li_1' + q/C$ so that $q = CRi_2$. Then $i_3 = q' = CRi_2'$ so that system is

$$\begin{aligned} Li' + Ri_2 &= E(t) \\ CRi_2' + i_2 - i_1 &= 0. \end{aligned}$$

15. We first note that $s(t) + i(t) + r(t) = n$. Now the rate of change of the number of susceptible persons, $s(t)$, is proportional to the number of contacts between the number of people infected and the number who are

2.9 Modeling with Systems of First-Order DEs

susceptible; that is, $ds/dt = -k_1 si$. We use $-k_1 < 0$ because $s(t)$ is decreasing. Next, the rate of change of the number of persons who have recovered is proportional to the number infected; that is, $dr/dt = k_2 i$ where $k_2 > 0$ since r is increasing. Finally, to obtain di/dt we use

$$\frac{d}{dt}(s + i + r) = \frac{d}{dt}n = 0.$$

This gives

$$\frac{di}{dt} = -\frac{dr}{dt} - \frac{ds}{dt} = -k_2 i + k_1 si.$$

The system of differential equations is then

$$\begin{aligned}\frac{ds}{dt} &= -k_1 si \\ \frac{di}{dt} &= -k_2 i + k_1 si \\ \frac{dr}{dt} &= k_2 i.\end{aligned}$$

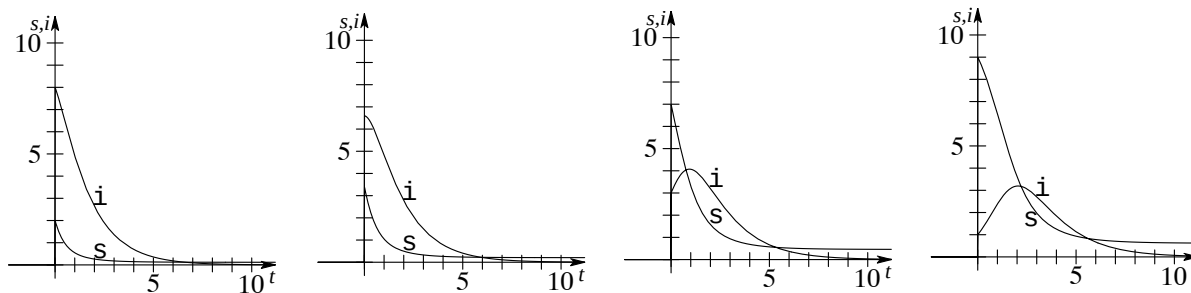
A reasonable set of initial conditions is $i(0) = i_0$, the number of infected people at time 0, $s(0) = n - i_0$, and $r(0) = 0$.

16. (a) If we know $s(t)$ and $i(t)$ then we can determine $r(t)$ from $s + i + r = n$.

- (b) In this case the system is

$$\begin{aligned}\frac{ds}{dt} &= -0.2si \\ \frac{di}{dt} &= -0.7i + 0.2si.\end{aligned}$$

We also note that when $i(0) = i_0$, $s(0) = 10 - i_0$ since $r(0) = 0$ and $i(t) + s(t) + r(t) = 0$ for all values of t . Now $k_2/k_1 = 0.7/0.2 = 3.5$, so we consider initial conditions $s(0) = 2$, $i(0) = 8$; $s(0) = 3.4$, $i(0) = 6.6$; $s(0) = 7$, $i(0) = 3$; and $s(0) = 9$, $i(0) = 1$.



We see that an initial susceptible population greater than k_2/k_1 results in an epidemic in the sense that the number of infected persons increases to a maximum before decreasing to 0. On the other hand, when $s(0) < k_2/k_1$, the number of infected persons decreases from the start and there is no epidemic.

CHAPTER 2 REVIEW EXERCISES

1. Writing the differential equation in the form $y' = k(y + A/k)$ we see that the critical point $-A/k$ is a repeller for $k > 0$ and an attractor for $k < 0$.
2. Separating variables and integrating we have

$$\frac{dy}{y} = \frac{4}{x} dx$$

$$\ln y = 4 \ln x + c = \ln x^4 + c$$

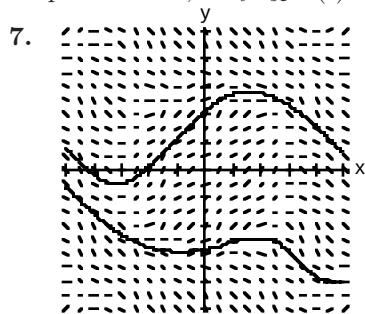
$$y = c_1 x^4.$$

We see that when $x = 0$, $y = 0$, so the initial-value problem has an infinite number of solutions for $k = 0$ and no solutions for $k \neq 0$.

3. $\frac{dy}{dx} = (y-1)^2(y-3)^2$
4. $\frac{dy}{dx} = y(y-2)^2(y-4)$
5. When n is odd, $x^n < 0$ for $x < 0$ and $x^n > 0$ for $x > 0$. In this case 0 is unstable. When n is even, $x^n > 0$ for $x < 0$ and for $x > 0$. In this case 0 is semi-stable.

When n is odd, $-x^n > 0$ for $x < 0$ and $-x^n < 0$ for $x > 0$. In this case 0 is asymptotically stable. When n is even, $-x^n < 0$ for $x < 0$ and for $x > 0$. In this case 0 is semi-stable.

6. Using a CAS we find that the zero of f occurs at approximately $P = 1.3214$. From the graph we observe that $dP/dt > 0$ for $P < 1.3214$ and $dP/dt < 0$ for $P > 1.3214$, so $P = 1.3214$ is an asymptotically stable critical point. Thus, $\lim_{t \rightarrow \infty} P(t) = 1.3214$.



8. (a) linear in y , homogeneous, exact
(b) linear in x
(c) separable, exact, linear in x and y
(d) Bernoulli in x
(e) separable
(f) separable, linear in x , Bernoulli
(g) linear in x
(h) homogeneous

(i) Bernoulli

(j) homogeneous, exact, Bernoulli

(k) linear in x and y , exact, separable, homogeneous

(l) exact, linear in y

(m) homogeneous

(n) separable

9. Separating variables and using the identity $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$, we have

$$\begin{aligned}\cos^2 x \, dx &= \frac{y}{y^2 + 1} \, dy, \\ \frac{1}{2}x + \frac{1}{4}\sin 2x &= \frac{1}{2}\ln(y^2 + 1) + c,\end{aligned}$$

and

$$2x + \sin 2x = 2\ln(y^2 + 1) + c.$$

10. Write the differential equation in the form

$$y \ln \frac{x}{y} \, dx = \left(x \ln \frac{x}{y} - y \right) \, dy.$$

This is a homogeneous equation, so let $x = uy$. Then $dx = u \, dy + y \, du$ and the differential equation becomes

$$y \ln u (u \, dy + y \, du) = (uy \ln u - y) \, dy \quad \text{or} \quad y \ln u \, du = -dy.$$

Separating variables, we obtain

$$\begin{aligned}\ln u \, du &= -\frac{dy}{y} \\ u \ln |u| - u &= -\ln |y| + c \\ \frac{x}{y} \ln \left| \frac{x}{y} \right| - \frac{x}{y} &= -\ln |y| + c \\ x(\ln x - \ln y) - x &= -y \ln |y| + cy.\end{aligned}$$

11. The differential equation

$$\frac{dy}{dx} + \frac{2}{6x+1}y = -\frac{3x^2}{6x+1}y^{-2}$$

is Bernoulli. Using $w = y^3$, we obtain the linear equation

$$\frac{dw}{dx} + \frac{6}{6x+1}w = -\frac{9x^2}{6x+1}.$$

An integrating factor is $6x + 1$, so

$$\begin{aligned}\frac{d}{dx} [(6x+1)w] &= -9x^2, \\ w &= -\frac{3x^3}{6x+1} + \frac{c}{6x+1},\end{aligned}$$

and

$$(6x+1)y^3 = -3x^3 + c.$$

(Note: The differential equation is also exact.)

12. Write the differential equation in the form $(3y^2 + 2x)dx + (4y^2 + 6xy)dy = 0$. Letting $M = 3y^2 + 2x$ and $N = 4y^2 + 6xy$ we see that $M_y = 6y = N_x$, so the differential equation is exact. From $f_x = 3y^2 + 2x$ we obtain

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$f = 3xy^2 + x^2 + h(y)$. Then $f_y = 6xy + h'(y) = 4y^2 + 6xy$ and $h'(y) = 4y^2$ so $h(y) = \frac{4}{3}y^3$. A one-parameter family of solutions is

$$3xy^2 + x^2 + \frac{4}{3}y^3 = c.$$

13. Write the equation in the form

$$\frac{dQ}{dt} + \frac{1}{t}Q = t^3 \ln t.$$

An integrating factor is $e^{\ln t} = t$, so

$$\frac{d}{dt}[tQ] = t^4 \ln t$$

$$tQ = -\frac{1}{25}t^5 + \frac{1}{5}t^5 \ln t + c$$

and

$$Q = -\frac{1}{25}t^4 + \frac{1}{5}t^4 \ln t + \frac{c}{t}.$$

14. Letting $u = 2x + y + 1$ we have

$$\frac{du}{dx} = 2 + \frac{dy}{dx},$$

and so the given differential equation is transformed into

$$u \left(\frac{du}{dx} - 2 \right) = 1 \quad \text{or} \quad \frac{du}{dx} = \frac{2u+1}{u}.$$

Separating variables and integrating we get

$$\begin{aligned} \frac{u}{2u+1} du &= dx \\ \left(\frac{1}{2} - \frac{1}{2} \frac{1}{2u+1} \right) du &= dx \\ \frac{1}{2}u - \frac{1}{4} \ln |2u+1| &= x + c \\ 2u - \ln |2u+1| &= 2x + c_1. \end{aligned}$$

Resubstituting for u gives the solution

$$4x + 2y + 2 - \ln |4x + 2y + 3| = 2x + c_1$$

or

$$2x + 2y + 2 - \ln |4x + 2y + 3| = c_1.$$

15. Write the equation in the form

$$\frac{dy}{dx} + \frac{8x}{x^2+4}y = \frac{2x}{x^2+4}.$$

An integrating factor is $(x^2+4)^4$, so

$$\begin{aligned} \frac{d}{dx} \left[(x^2+4)^4 y \right] &= 2x (x^2+4)^3 \\ (x^2+4)^4 y &= \frac{1}{4} (x^2+4)^4 + c \end{aligned}$$

and

$$y = \frac{1}{4} + c (x^2+4)^{-4}.$$

16. Letting $M = 2r^2 \cos \theta \sin \theta + r \cos \theta$ and $N = 4r + \sin \theta - 2r \cos^2 \theta$ we see that $M_r = 4r \cos \theta \sin \theta + \cos \theta = N_\theta$, so the differential equation is exact. From $f_\theta = 2r^2 \cos \theta \sin \theta + r \cos \theta$ we obtain $f = -r^2 \cos^2 \theta + r \sin \theta + h(r)$. Then $f_r = -2r \cos^2 \theta + \sin \theta + h'(r) = 4r + \sin \theta - 2r \cos^2 \theta$ and $h'(r) = 4r$ so $h(r) = 2r^2$. The solution is

$$-r^2 \cos^2 \theta + r \sin \theta + 2r^2 = c.$$

17. The differential equation has the form $(d/dx)[(\sin x)y] = 0$. Integrating, we have $(\sin x)y = c$ or $y = c/\sin x$. The initial condition implies $c = -2 \sin(7\pi/6) = 1$. Thus, $y = 1/\sin x$, where the interval $\pi < x < 2\pi$ is chosen to include $x = 7\pi/6$.

18. Separating variables and integrating we have

$$\begin{aligned}\frac{dy}{y^2} &= -2(t+1) dt \\ -\frac{1}{y} &= -(t+1)^2 + c \\ y &= \frac{1}{(t+1)^2 + c_1}, \quad \text{where } -c = c_1.\end{aligned}$$

The initial condition $y(0) = -\frac{1}{8}$ implies $c_1 = -9$, so a solution of the initial-value problem is

$$y = \frac{1}{(t+1)^2 - 9} \quad \text{or} \quad y = \frac{1}{t^2 + 2t - 8},$$

where $-4 < t < 2$.

19. (a) For $y < 0$, $\sqrt{y}$ is not a real number.

- (b) Separating variables and integrating we have

$$\frac{dy}{\sqrt{y}} = dx \quad \text{and} \quad 2\sqrt{y} = x + c.$$

Letting $y(x_0) = y_0$ we get $c = 2\sqrt{y_0} - x_0$, so that

$$2\sqrt{y} = x + 2\sqrt{y_0} - x_0 \quad \text{and} \quad y = \frac{1}{4}(x + 2\sqrt{y_0} - x_0)^2.$$

Since $\sqrt{y} > 0$ for $y \neq 0$, we see that $dy/dx = \frac{1}{2}(x + 2\sqrt{y_0} - x_0)$ must be positive. Thus, the interval on which the solution is defined is $(x_0 - 2\sqrt{y_0}, \infty)$.

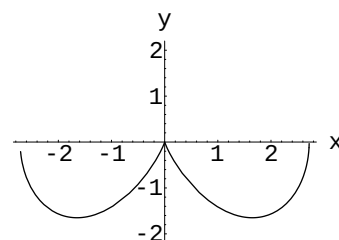
20. (a) The differential equation is homogeneous and we let $y = ux$. Then

$$\begin{aligned}(x^2 - y^2) dx + xy dy &= 0 \\ (x^2 - u^2 x^2) dx + ux^2(u dx + x du) &= 0 \\ dx + ux du &= 0 \\ u du &= -\frac{dx}{x} \\ \frac{1}{2}u^2 &= -\ln|x| + c \\ \frac{y^2}{x^2} &= -2\ln|x| + c_1.\end{aligned}$$

The initial condition gives $c_1 = 2$, so an implicit solution is $y^2 = x^2(2 - 2\ln|x|)$.

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- (b) Solving for y in part (a) and being sure that the initial condition is still satisfied, we have $y = -\sqrt{2}|x|(1 - \ln|x|)^{1/2}$, where $-e \leq x \leq e$ so that $1 - \ln|x| \geq 0$. The graph of this function indicates that the derivative is not defined at $x = 0$ and $x = e$. Thus, the solution of the initial-value problem is $y = -\sqrt{2}x(1 - \ln x)^{1/2}$, for $0 < x < e$.



21. The graph of $y_1(x)$ is the portion of the closed black curve lying in the fourth quadrant. Its interval of definition is approximately $(0.7, 4.3)$. The graph of $y_2(x)$ is the portion of the left-hand black curve lying in the third quadrant. Its interval of definition is $(-\infty, 0)$.
22. The first step of Euler's method gives $y(1.1) \approx 9 + 0.1(1 + 3) = 9.4$. Applying Euler's method one more time gives $y(1.2) \approx 9.4 + 0.1(1 + 1.1\sqrt{9.4}) \approx 9.8373$.
23. From $\frac{dP}{dt} = 0.018P$ and $P(0) = 4$ billion we obtain $P = 4e^{0.018t}$ so that $P(45) = 8.99$ billion.
24. Let $A = A(t)$ be the volume of CO_2 at time t . From $dA/dt = 1.2 - A/4$ and $A(0) = 16 \text{ ft}^3$ we obtain $A = 4.8 + 11.2e^{-t/4}$. Since $A(10) = 5.7 \text{ ft}^3$, the concentration is 0.017%. As $t \rightarrow \infty$ we have $A \rightarrow 4.8 \text{ ft}^3$ or 0.06%.
25. Separating variables, we have

$$\frac{\sqrt{s^2 - y^2}}{y} dy = -dx.$$

Substituting $y = s \sin \theta$, this becomes

$$\begin{aligned} \frac{\sqrt{s^2 - s^2 \sin^2 \theta}}{s \sin \theta} (s \cos \theta) d\theta &= -dx \\ s \int \frac{\cos^2}{\sin \theta} d\theta &= -\int dx \\ s \int \frac{1 - \sin^2 \theta}{\sin \theta} d\theta &= -x + c \\ s \int (\csc \theta - \sin \theta) d\theta &= -x + c \\ s \ln |\csc \theta - \cot \theta| + s \cos \theta &= -x + c \\ s \ln \left| \frac{s}{y} - \frac{\sqrt{s^2 - y^2}}{y} \right| + s \frac{\sqrt{s^2 - y^2}}{s} &= -x + c. \end{aligned}$$

Letting $s = 10$, this is

$$10 \ln \left| \frac{10}{y} - \frac{\sqrt{100 - y^2}}{y} \right| + \sqrt{100 - y^2} = -x + c.$$

Letting $x = 0$ and $y = 10$ we determine that $c = 0$, so the solution is

$$10 \ln \left| \frac{10}{y} - \frac{\sqrt{100 - y^2}}{y} \right| + \sqrt{100 - y^2} = -x.$$

26. From $V dC/dt = kA(C_s - C)$ and $C(0) = C_0$ we obtain $C = C_s + (C_0 - C_s)e^{-kAt/V}$.

27. (a) The differential equation

$$\begin{aligned}\frac{dT}{dt} &= k(T - T_m) = k[T - T_2 - B(T_1 - T)] \\ &= k[(1 + B)T - (BT_1 + T_2)] = k(1 + B) \left(T - \frac{BT_1 + T_2}{1 + B} \right)\end{aligned}$$

is autonomous and has the single critical point $(BT_1 + T_2)/(1 + B)$. Since $k < 0$ and $B > 0$, by phase-line analysis it is found that the critical point is an attractor and

$$\lim_{t \rightarrow \infty} T(t) = \frac{BT_1 + T_2}{1 + B}.$$

Moreover,

$$\lim_{t \rightarrow \infty} T_m(t) = \lim_{t \rightarrow \infty} [T_2 + B(T_1 - T)] = T_2 + B \left(T_1 - \frac{BT_1 + T_2}{1 + B} \right) = \frac{BT_1 + T_2}{1 + B}.$$

(b) The differential equation is

$$\frac{dT}{dt} = k(T - T_m) = k(T - T_2 - BT_1 + BT)$$

or

$$\frac{dT}{dt} - k(1 + B)T = -k(BT_1 + T_2).$$

This is linear and has integrating factor $e^{-\int k(1+B)dt} = e^{-k(1+B)t}$. Thus,

$$\begin{aligned}\frac{d}{dt}[e^{-k(1+B)t}T] &= -k(BT_1 + T_2)e^{-k(1+B)t} \\ e^{-k(1+B)t}T &= \frac{BT_1 + T_2}{1 + B}e^{-k(1+B)t} + c \\ T(t) &= \frac{BT_1 + T_2}{1 + B} + ce^{k(1+B)t}.\end{aligned}$$

Since k is negative, $\lim_{t \rightarrow \infty} T(t) = (BT_1 + T_2)/(1 + B)$.

(c) The temperature $T(t)$ decreases to $(BT_1 + T_2)/(1 + B)$, whereas $T_m(t)$ increases to $(BT_1 + T_2)/(1 + B)$ as $t \rightarrow \infty$. Thus, the temperature $(BT_1 + T_2)/(1 + B)$, (which is a weighted average,

$$\frac{B}{1+B}T_1 + \frac{1}{1+B}T_2,$$

of the two initial temperatures), can be interpreted as an equilibrium temperature. The body cannot get cooler than this value whereas the medium cannot get hotter than this value.

28. (a) By separation of variables and partial fractions,

$$\ln \left| \frac{T - T_m}{T + T_m} \right| - 2 \tan^{-1} \left(\frac{T}{T_m} \right) = 4T_m^3 kt + c.$$

Then rewrite the right-hand side of the differential equation as

$$\begin{aligned}\frac{dT}{dt} &= k(T^4 - T_m^4) = [(T_m + (T - T_m))^4 - T_m^4] \\ &= kT_m^4 \left[\left(1 + \frac{T - T_m}{T_m} \right)^4 - 1 \right] \\ &= kT_m^4 \left[\left(1 + 4 \frac{T - T_m}{T_m} + 6 \left(\frac{T - T_m}{T_m} \right)^2 + \dots \right) - 1 \right] \leftarrow \text{binomial expansion}\end{aligned}$$

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- (b) When $T - T_m$ is small compared to T_m , every term in the expansion after the first two can be ignored, giving

$$\frac{dT}{dt} \approx k_1(T - T_m), \quad \text{where} \quad k_1 = 4kT_m^3.$$

29. We first solve $(1 - t/10)di/dt + 0.2i = 4$. Separating variables we obtain $di/(40 - 2i) = dt/(10 - t)$. Then

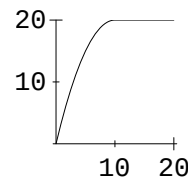
$$-\frac{1}{2} \ln |40 - 2i| = -\ln |10 - t| + c \quad \text{or} \quad \sqrt{40 - 2i} = c_1(10 - t).$$

Since $i(0) = 0$ we must have $c_1 = 2/\sqrt{10}$. Solving for i we get $i(t) = 4t - \frac{1}{5}t^2$, $0 \leq t < 10$.

For $t \geq 10$ the equation for the current becomes $0.2i = 4$ or $i = 20$. Thus

$$i(t) = \begin{cases} 4t - \frac{1}{5}t^2, & 0 \leq t < 10 \\ 20, & t \geq 10. \end{cases}$$

The graph of $i(t)$ is given in the figure.



30. From $y[1 + (y')^2] = k$ we obtain $dx = (\sqrt{y}/\sqrt{k-y})dy$. If $y = k \sin^2 \theta$ then

$$dy = 2k \sin \theta \cos \theta d\theta, \quad dx = 2k \left(\frac{1}{2} - \frac{1}{2} \cos 2\theta \right) d\theta, \quad \text{and} \quad x = k\theta - \frac{k}{2} \sin 2\theta + c.$$

If $x = 0$ when $\theta = 0$ then $c = 0$.

31. Letting $c = 0.6$, $A_h = \pi(\frac{1}{32} \cdot \frac{1}{12})^2$, $A_w = \pi \cdot 1^2 = \pi$, and $g = 32$, the differential equation becomes $dh/dt = -0.00003255\sqrt{h}$. Separating variables and integrating, we get $2\sqrt{h} = -0.00003255t + c$, so $h = (c_1 - 0.00001628t)^2$. Setting $h(0) = 2$, we find $c = \sqrt{2}$, so $h(t) = (\sqrt{2} - 0.00001628t)^2$, where h is measured in feet and t in seconds.

32. One hour is 3,600 seconds, so the hour mark should be placed at

$$h(3600) = [\sqrt{2} - 0.00001628(3600)]^2 \approx 1.838 \text{ ft} \approx 22.0525 \text{ in.}$$

up from the bottom of the tank. The remaining marks corresponding to the passage of 2, 3, 4, ..., 12 hours are placed at the values shown in the table. The marks are not evenly spaced because the water is not draining out at a uniform rate; that is, $h(t)$ is not a linear function of time.

time (seconds)	height (inches)
0	24.0000
1	22.0520
2	20.1864
3	18.4033
4	16.7026
5	15.0844
6	13.5485
7	12.0952
8	10.7242
9	9.4357
10	8.2297
11	7.1060
12	6.0648

33. In this case $A_w = \pi h^2/4$ and the differential equation is

$$\frac{dh}{dt} = -\frac{1}{7680} h^{-3/2}.$$

Separating variables and integrating, we have

$$\begin{aligned} h^{3/2} dh &= -\frac{1}{7680} dt \\ \frac{2}{5} h^{5/2} &= -\frac{1}{7680} t + c_1. \end{aligned}$$

Setting $h(0) = 2$ we find $c_1 = 8\sqrt{2}/5$, so that

$$\begin{aligned}\frac{2}{5}h^{5/2} &= -\frac{1}{7680}t + \frac{8\sqrt{2}}{5}, \\ h^{5/2} &= 4\sqrt{2} - \frac{1}{3072}t,\end{aligned}$$

and

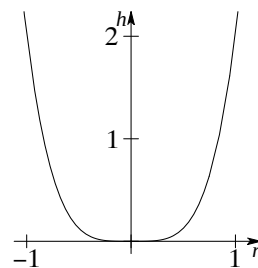
$$h = \left(4\sqrt{2} - \frac{1}{3072}t\right)^{2/5}.$$

In this case $h(4 \text{ hr}) = h(14,400 \text{ s}) = 11.8515$ inches and $h(5 \text{ hr}) = h(18,000 \text{ s})$ is not a real number. Using a CAS to solve $h(t) = 0$, we see that the tank runs dry at $t \approx 17,378 \text{ s} \approx 4.83 \text{ hr}$. Thus, this particular conical water clock can only measure time intervals of less than 4.83 hours.

34. If we let r_h denote the radius of the hole and $A_w = \pi[f(h)]^2$, then the differential equation $dh/dt = -k\sqrt{h}$, where $k = cA_h\sqrt{2g}/A_w$, becomes

$$\frac{dh}{dt} = -\frac{c\pi r_h^2 \sqrt{2g}}{\pi[f(h)]^2} \sqrt{h} = -\frac{8cr_h^2 \sqrt{h}}{[f(h)]^2}.$$

For the time marks to be equally spaced, the rate of change of the height must be a constant; that is, $dh/dt = -a$. (The constant is negative because the height is decreasing.) Thus



$$-a = -\frac{8cr_h^2 \sqrt{h}}{[f(h)]^2}, \quad [f(h)]^2 = \frac{8cr_h^2 \sqrt{h}}{a}, \quad \text{and} \quad r = f(h) = 2r_h \sqrt{\frac{2c}{a}} h^{1/4}.$$

Solving for h , we have

$$h = \frac{a^2}{64c^2r_h^4} r^4.$$

The shape of the tank with $c = 0.6$, $a = 2 \text{ ft}/12 \text{ hr} = 1 \text{ ft}/21,600 \text{ s}$, and $r_h = 1/32(12) = 1/384$ is shown in the above figure.

35. From $dx/dt = k_1x(\alpha - x)$ we obtain

$$\left(\frac{1/\alpha}{x} + \frac{1/\alpha}{\alpha - x}\right) dx = k_1 dt$$

so that $x = \alpha c_1 e^{\alpha k_1 t} / (1 + c_1 e^{\alpha k_1 t})$. From $dy/dt = k_2 xy$ we obtain

$$\ln |y| = \frac{k_2}{k_1} \ln |1 + c_1 e^{\alpha k_1 t}| + c \quad \text{or} \quad y = c_2 (1 + c_1 e^{\alpha k_1 t})^{k_2/k_1}.$$

36. In tank A the salt input is

$$\left(7 \frac{\text{gal}}{\text{min}}\right) \left(2 \frac{\text{lb}}{\text{gal}}\right) + \left(1 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_2}{100} \frac{\text{lb}}{\text{gal}}\right) = \left(14 + \frac{1}{100}x_2\right) \frac{\text{lb}}{\text{min}}.$$

The salt output is

$$\left(3 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_1}{100} \frac{\text{lb}}{\text{gal}}\right) + \left(5 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_1}{100} \frac{\text{lb}}{\text{gal}}\right) = \frac{2}{25} x_1 \frac{\text{lb}}{\text{min}}.$$

In tank B the salt input is

$$\left(5 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_1}{100} \frac{\text{lb}}{\text{gal}}\right) = \frac{1}{20} x_1 \frac{\text{lb}}{\text{min}}.$$

The salt output is

$$\left(1 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_2}{100} \frac{\text{lb}}{\text{gal}}\right) + \left(4 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_2}{100} \frac{\text{lb}}{\text{gal}}\right) = \frac{1}{20} x_2 \frac{\text{lb}}{\text{min}}.$$

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The system of differential equations is then

$$\frac{dx_1}{dt} = 14 + \frac{1}{100}x_2 - \frac{2}{25}x_1$$

$$\frac{dx_2}{dt} = \frac{1}{20}x_1 - \frac{1}{20}x_2.$$

37. From $y = -x - 1 + c_1 e^x$ we obtain $y' = y + x$ so that the differential equation of the orthogonal family is

$$\frac{dy}{dx} = -\frac{1}{y+x} \quad \text{or} \quad \frac{dx}{dy} + x = -y.$$

This is a linear differential equation and has integrating factor $e^{\int dy} = e^y$, so

$$\frac{d}{dy}[e^y x] = -ye^y$$

$$e^y x = -ye^y + e^y + c_2$$

$$x = -y + 1 + c_2 e^{-y}.$$

38. Differentiating the family of curves, we have

$$y' = -\frac{1}{(x+c_1)^2} = -\frac{1}{y^2}.$$

The differential equation for the family of orthogonal trajectories is then $y' = y^2$. Separating variables and integrating we get

$$\frac{dy}{y^2} = dx$$

$$-\frac{1}{y} = x + c_1$$

$$y = -\frac{1}{x+c_1}.$$

